

# Forecasting with a Dutch Agent-Based Model and Optimally Behaving Households

Steven Hoekstra<sup>\*1,2</sup>, Cars Hommes<sup>1,2,3</sup>, and Frank Pijpers<sup>4,5</sup>

<sup>1</sup>*CeNDEF, University of Amsterdam, Amsterdam, Netherlands*

<sup>2</sup>*Tinbergen Institute, Amsterdam, Netherlands*

<sup>3</sup>*Bank of Canada, Ottawa, Canada*

<sup>4</sup>*Statistics Netherlands, The Hague, Netherlands*

<sup>5</sup>*Korteweg de Vries institute for Mathematics, University of Amsterdam, Amsterdam, Netherlands*

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## Abstract

We develop an agent-based model of the Dutch economy that embeds a HANK-style household sector with heterogeneous marginal propensities to consume. Building on the stock-flow consistent framework of Poledna et al. (2023), we replace the constant marginal propensity to consume heuristic with forward-looking households who optimise consumption, savings and investments over time as in Kaplan et al. (2018). Policy functions are computed using an amortised neural-network method that enables efficient recalculation across time periods and Monte Carlo simulations without re-solving the household problem. Following Adam and Marcet (2011), households are internally rational, optimising given beliefs, but form expectations using adaptive AR(1) learning rules rather than model-consistent forecasts. We calibrate the model to Eurostat national accounts and administrative microdata from Statistics Netherlands, targeting empirically realistic MPC heterogeneity driven by the joint distribution of income, illiquid wealth, and liquid wealth. In rolling out-of-sample forecasting exercises over 2010–2019, we compare the optimal-consumption ABM against the original model, ten heuristic variants spanning static rules, habit formation, buffer-stock target wealth, capital-return MPC asymmetry, and Merton portfolio choice, and benchmark time-series models. The optimal-consumption ABM achieves household consumption forecast improvements of 10–22% at the 12-quarter horizon relative to univariate benchmarks depending on the expectation mechanism (Diebold–Mariano  $p < 0.01$ ), and up to 28% relative to the VAR(1) under AR expectations. A channel decomposition isolating the portfolio-choice and capital-return channels reveals that portfolio choice accounts for the majority of the OC-ABM’s residual advantage over a buffer-stock heuristic (+11.7 pp on consumption RMSFE), while capital-return MPC asymmetry contributes negligibly; the OC-ABM remains the only variant whose consumption gain is statistically significant under every expectation mechanism, consistent with Euler-equation discipline operating through reduced forecast variance rather than higher central tendency. GDP improvements of 11–15% are directionally positive across all mechanisms but not statistically significant. These results demonstrate the value of combining microfounded household behaviour with the flexibility of agent-based modelling for macroeconomic forecasting.

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\*Corresponding author: s.g.hoekstra@uva.nl

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## 1 Introduction

Accurate macroeconomic forecasting of household consumption is central to monetary policy, fiscal planning, and financial stability assessment. Yet no single modelling tradition captures all relevant dimensions. Representative-agent DSGE models commonly deployed for operational forecasting provide disciplined structural frameworks but, by construction, abstract from the distributional dynamics that drive aggregate consumption at the household level; reduced-form time-series models complement these with strong short-run fit but lack structural content for longer horizons. The Heterogeneous Agent New Keynesian (HANK) literature has demonstrated that the joint distribution of income and wealth, and the marginal propensity to consume (MPC) heterogeneity it implies, is a first-order determinant of aggregate consumption dynamics (Kaplan et al., 2018, 2014; Kaplan and Violante, 2022). This is confirmed empirically by Fagereng et al. (2021), who show that marginal propensities to consume vary systematically with liquid assets. Yet incorporating household heterogeneity does not automatically translate into forecasting gains: Acharya et al. (2023) show that an estimated two-asset HANK model does not improve consumption forecasts relative to the representative-agent benchmark of Smets and Wouters (2007), suggesting that the modelling environment in which heterogeneity is embedded matters as much as heterogeneity itself. No forecasting framework to date embeds microfounded consumption heterogeneity in an environment that also features out-of-equilibrium adjustment, decentralised markets, and stock–flow consistent accounting.

Agent-based models (ABMs) offer a complementary source of structural richness. In an ABM, aggregate outcomes emerge from interactions among heterogeneous, boundedly rational agents operating within explicit institutional settings (Axtell and Farmer, 2025; Dawid and Delli Gatti, 2018a). Rather than enforcing market clearing and solving for equilibrium, ABMs specify micro-level decision rules and interaction protocols, including search-and-matching in goods, labour, and credit markets, production networks, and credit networks, and simulate the economy forward under stochastic shocks. This makes them well suited to study non-linear amplification, path dependence, and the joint propagation of real and financial shocks. Recent large-scale ABMs calibrated to national accounts have demonstrated competitive or better forecasting performance relative to standard benchmarks (Poledna et al., 2023; Hommes et al., 2025; Wiese et al., 2024). Yet a common limitation persists across these models: household consumption is governed by simple heuristics, typically a constant marginal propensity to consume, that cannot capture the forward-looking, portfolio-dependent behaviour the HANK literature identifies as essential.<sup>1</sup>

This paper asks whether replacing heuristic consumption rules with microfounded, optimising household behaviour improves ABM forecasting performance. We develop an optimal-consumption ABM (OC-ABM) of the Dutch economy, chosen for its high-quality administrative microdata (Statistics Netherlands), well-documented national accounts (Eurostat), and comparability to the Austrian ABM of Poledna et al. (2023), in which households make forward-looking consumption–saving decisions over liquid and illiquid assets in the spirit

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<sup>1</sup>Some ABM frameworks include richer consumption specifications with liquidity-constrained households or rule-of-thumb heterogeneity (Dawid and Delli Gatti, 2018b), but none embeds the microfounded two-asset intertemporal optimisation that Kaplan and Violante (2022) identify as necessary to match the empirical MPC hierarchy.

of Kaplan et al. (2018). Households operate under imperfect market knowledge. They are internally rational in the sense of Adam and Marcet (2011), solving their optimisation problem correctly given their beliefs. However, departing from the Bayesian updating in Adam and Marcet (2011), and motivated by the growing recognition that rational expectations place implausible computational demands on agents in heterogeneous-agent settings (Moll, 2025), they form expectations using OLS-estimated AR(1) rules in the tradition of adaptive learning (Evans and Honkapohja, 2001). Expectations are parsimonious in the spirit of the behavioral learning equilibria of Hommes and Zhu (2014): households condition on a small number of aggregates each period rather than the full state of the economy. Recent work has begun to explore this intersection from the HANK side: Pfäuti and Seyrich (2022) embed cognitive discounting in a HANK framework. Our approach differs by situating optimising households in a fully decentralised ABM environment with network interactions and stock–flow consistent accounting. This bridges two literatures: from HANK, we retain rigorous microfoundations rooted in the incomplete-markets tradition (Bewley, 1977; Huggett, 1993; Aiyagari, 1994) and distributional transmission channels; from ABMs, we preserve decentralised dynamics, network propagation, and explicit stock–flow consistency across sectors.

Three contributions support this synthesis. First, the OC-ABM embeds HANK-style two-asset households in a stock–flow consistent agent-based model with decentralised markets, augmented with a simple equity–fund market that pays firm and bank dividends. This provides a structural environment in which distributional dynamics and out-of-equilibrium adjustment interact with forward-looking household behaviour. Second, we compute household policy functions using an amortised neural-network method adapted from Maliar et al. (2021), trained via Euler-equation residual minimisation and generalised to a continuous parameter space. This allows preference and behavioural parameters to vary across calibration periods without re-solving the household problem, enabling efficient, data-driven calibration to quarterly data. Third, we calibrate household behaviour using anonymised administrative micro-data from Statistics Netherlands (CBS), matching the joint distribution of income and wealth rather than aggregated moments alone. We target a quarterly MPC out of transitory income of 0.20, MPCs out of capital and deposits of 0.02, and an average propensity to consume of approximately 0.86, consistent with European household survey evidence (Drescher et al., 2020) and Eurostat data.

We evaluate forecasting performance using rolling out-of-sample exercises over 2010Q1–2019Q4, a pre-COVID period of relative macroeconomic stability. We compare the OC-ABM against the base ABM of Poledna et al. (2023), seven heuristic extensions spanning static, habit-formation, and buffer-stock consumption rules (H1–H7) under a common borrowing constraint, and four benchmark time-series models: AR(1), ARX(1), VAR(1), and VARX(1). Our headline finding is that the OC-ABM substantially improves household consumption forecasts at the 12-quarter horizon, as measured by root mean squared forecast error (RMSFE). Relative to the corresponding univariate benchmark, the OC-ABM achieves consumption RMSFE reductions of 10–22% across all four expectation mechanisms (Diebold–Mariano  $p < 0.01$ ; Diebold and Mariano, 1995). The largest gains arise under mechanisms without exogenous predictors (21.8% under AR, 18.6% under VAR); smaller but significant gains obtain when exogenous variables are included (10.7% under ARX, 9.9% under VARX). Against multivariate benchmarks, improvements reach up to 28%. The base ABM, by contrast, achieves a statistically insignificant 6.9% ( $p = 0.22$ ) under AR expectations. These gains are concentrated at horizons beyond four quarters; at shorter horizons, ABMs underperform time-series benchmarks, consistent with the latter’s advantage in capturing short-run serial correlation. Results are also variable-specific: GDP improvements of 11–15% are directionally positive across all mechanisms but not statistically significant, inflation forecasts show a small but significant

deterioration under AR expectations, and government consumption forecasts are consistently poor across all model variants.

MPC heterogeneity may contribute to forecasting gains. The H2-ABM improves consumption forecasts by 5.2% under AR expectations, though this gain is not statistically significant ( $p = 0.70$ ), and its performance is inconsistent across expectation mechanisms. The OC-ABM's more robust advantage suggests that the intertemporal consumption-smoothing channel, through which forward-looking households absorb transitory shocks via portfolio adjustment rather than passing them through to spending one-for-one, provides an additional stabilising mechanism that heuristic rules cannot replicate.

The remainder of the paper proceeds as follows. Section 2 introduces the agent-based model and the extended household sector. Section 3 details the calibration strategy, including the neural-network solution method and microdata matching. Section 5 presents the forecast comparison. Section 6 assesses internal validity through auto- and cross-correlation analysis. Section 7 discusses limitations and concludes.

## 2 Model

This section describes the model, which builds on the macroeconomic agent-based model of Poledna et al. (2023). We make two principal extensions. First, we replace the constant marginal propensity to consume heuristic with a HANK-style optimal consumption block inspired by the two-asset framework of Kaplan et al. (2018), allowing consumption to depend on liquid and illiquid wealth, asset returns, and inflation in addition to income. Second, we introduce an equity-fund market through which households hold claims on firm and bank profits, earning dividend income. We also compare the optimal consumption model against a suite of increasingly flexible heuristic consumption rules (Section 2.4) and a baseline constant-consumption heuristic (Section 2.5).

### 2.1 Overview

Figure 1 shows the overall structure of the model. The economy consists of six sectors: households, firms, a commercial bank, a central bank, the government, and the rest of the world.

- **Households** sell labour to firms in return for wages and invest in the equity fund to claim a share of profits. They purchase consumption goods from firms, deposit money in the bank or borrow from it, and receive dividends from the equity fund.
- **Firms** produce goods that serve as consumption goods, investment goods, intermediate goods, or material inputs within an input–output network. Firms borrow from the bank, deposit cash reserves, and pay dividends to equity-fund holders.
- **The bank** takes deposits and extends loans to firms and households, subject to capital requirements and firm-level risk assessments. Bank profits are paid as dividends to equity-fund holders.
- **The government** levies taxes and social contributions from households, provides subsidies and social benefits, and purchases government consumption goods from firms.

- **The central bank** provides liquidity to the commercial bank through advances or accepts reserve deposits, paying or receiving interest accordingly. It sets the policy interest rate according to a Taylor (1993) rule.
- **The rest of the world** imports and exports goods on the goods market.

The model represents heterogeneous agents aligned with European System of Accounts (ESA) classifications and is calibrated to Eurostat national accounts and CBS microdata. Markets are fully decentralised: all transactions occur through search-and-matching processes. The remainder of the model not described in this section, including firm production, pricing, the banking sector, and government fiscal rules, follows Poledna et al. (2023) and is detailed in Appendix A.

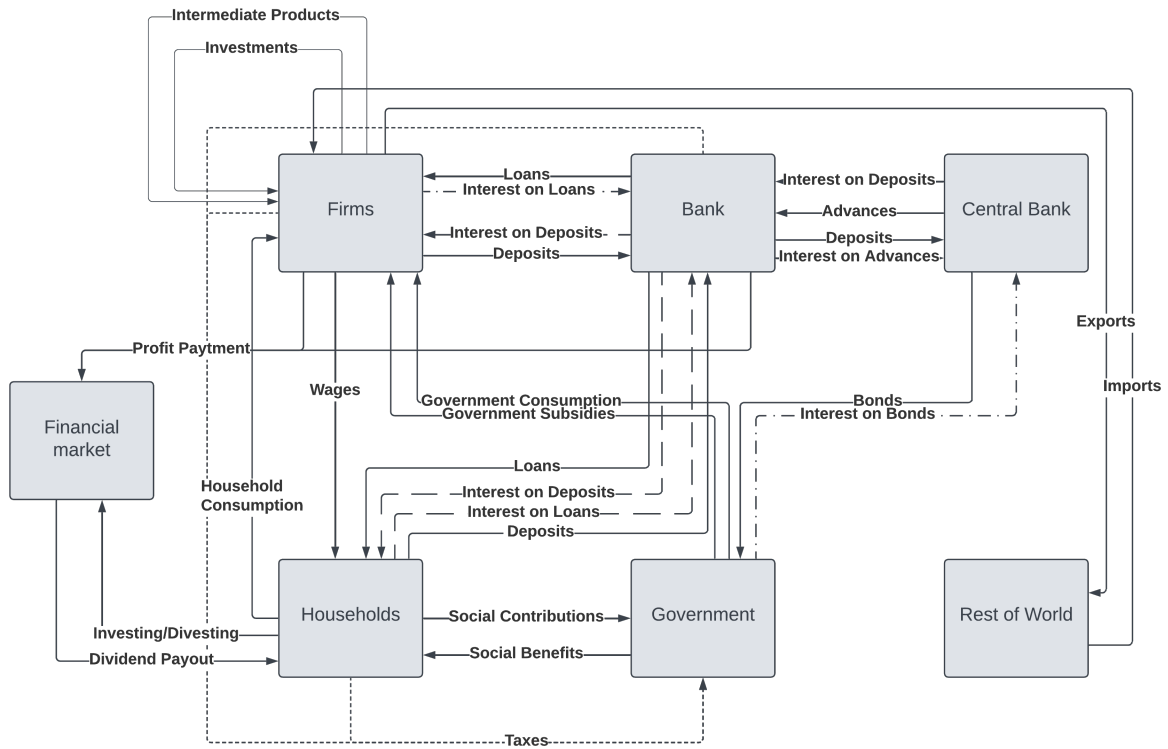


Figure 1: Flow diagram of the model. Arrow direction indicates flow direction.

## 2.2 Households

As Statistics Netherlands (CBS) registers wealth at the household level, we model the household sector accordingly. The sector consists of  $H$  households, each of which receives the sum of the incomes of all individuals in the household.

### 2.2.1 Labour Income and Search

Household income depends on activity status. If household  $h$  is employed by firm  $i$  in period  $t$ , it receives the wage  $w_{h,t} = w_{i,t}$ . If unemployed and economically active, the household

searches for work through the search-and-matching mechanism on the labour market: it visits firms with open vacancies in random order and accepts the first available position. There are no hiring or firing costs; fired employees begin searching in the same period. If no vacancy is found, the household remains unemployed and retains its previous wage as a reference:

$$w_{h,t} = \begin{cases} w_{i,t} & \text{if employed by firm } i, \\ w_{h,t-1} & \text{if unemployed.} \end{cases} \quad (1)$$

An economically inactive household receives benefits  $sb_t^{\text{inact}}$  and does not search for work:

$$sb_t^{\text{inact}} = sb_{t-1}^{\text{inact}}(1 + \gamma_t^e), \quad (2)$$

where  $\gamma_t^e$  is the expected aggregate growth rate. Each household additionally receives other social transfers (related to family, children, etc.), assumed uniform across households:

$$sb_t^{\text{other}} = sb_{t-1}^{\text{other}}(1 + \gamma_t^e). \quad (3)$$

Disposable income from labour and social benefits is:

$$Y_{h,t} = \begin{cases} (w_{h,t}(1 - \tau^{\text{SIW}} - \tau^{\text{INC}}(1 - \tau^{\text{SIW}})) + sb_t^{\text{other}}) \bar{P}_{t-1}^{\text{HH}} & \text{if employed,} \\ (\theta^{\text{UB}} w_{h,t} + sb_t^{\text{other}}) \bar{P}_{t-1}^{\text{HH}} & \text{if unemployed,} \\ (sb_t^{\text{inact}} + sb_t^{\text{other}}) \bar{P}_{t-1}^{\text{HH}} & \text{if not economically active,} \end{cases} \quad (4)$$

where  $\tau^{\text{INC}}$  is the income tax rate,  $\tau^{\text{SIW}}$  is the rate of social insurance contributions paid by the employee, and  $\theta^{\text{UB}}$  is the unemployment benefit replacement rate. We denote deflated disposable income by  $y_{h,t} = Y_{h,t} / \bar{P}_{t-1}^{\text{HH}}$ , which enters the household budget constraint below.<sup>2</sup>

### 2.2.2 Household Optimisation Problem

Having described household income, we now turn to the consumption–saving decision. Households also receive interest on liquid assets and a return on illiquid capital (described in Section 2.2.3). We adopt the two-asset formulation of Kaplan et al. (2018). Kaplan and Violante (2022) show that a two-asset structure, with sufficiently different returns on liquid and illiquid assets, is necessary for theoretical MPCs to match their empirical counterparts.<sup>3</sup>

Households solve the following maximisation problem, a variation of the household problem in Kaplan et al. (2018). We depart from their specification in three ways. First, we remove labour supply from the consumption problem, as labour hours are not available in the CBS microdata whereas wages are. Second, we add utility from holding liquid and illiquid assets. While these terms can be motivated by money-in-the-utility and wealth-in-the-utility formulations (Michaillat and Saez, 2021), their primary role in our specification is empirical: the baseline CRRA utility with adjustment costs alone cannot simultaneously match the MPC hierarchy ( $\text{MPC} > \text{MPC}_k \approx \text{MPC}_b$ ) and the aggregate consumption rate for our calibrated parameter values (Section 3.4.3). The additional terms provide the degrees of freedom needed

<sup>2</sup>We deflate by the lagged price index  $\bar{P}_{t-1}^{\text{HH}}$  rather than the current level because households make their consumption–saving decisions before observing the current-period price realisation.

<sup>3</sup>A necessary condition is that returns on the liquid and illiquid assets differ substantially. Kaplan and Violante (2022) argue that both theory and data provide sufficient evidence that this holds.

to replicate this moment hierarchy. Third, households have imperfect market knowledge: they are internally rational in the sense of Adam and Marcet (2011), solving their optimisation problem given beliefs, but form expectations using a parsimonious perceived law of motion rather than conditioning on the full state of the economy. This follows the approach of Krusell and Smith (1998), who solve an equilibrium model with wealth and income distributions by letting agents condition only on a small set of moments.

The household problem (in real terms) is:

$$\max_{\{c_{h,t}, d_{h,t}, b_{h,t+1}\}_{t \in \mathbb{N}}} \mathbb{E}_0^* \sum_{t=0}^{\infty} \beta^t (u(c_{h,t}) + \mathbb{I}_b u_b(b_{h,t}) + \mathbb{I}_k u_k(k_{h,t})), \quad (5)$$

subject to

$$\begin{aligned} (1 + \tau^{\text{VAT}})c_{h,t} &= y_{h,t} + (1 + r_t^b + \mathbb{I}_{\{b_{h,t+1} < 0\}} r_{cc})b_{h,t} - d_{h,t} - \chi(d_{h,t}) - b_{h,t+1}, \\ k_{h,t+1} &= (1 + r_t^k - \tau^K)k_{h,t} + d_{h,t}, \\ b_{h,t+1} &\geq -\underline{b}, \quad k_{h,t} \geq 0, \end{aligned} \quad (6)$$

with adjustment cost

$$\chi(d_{h,t}) = x_0 |d_{h,t}|^{x_1}, \quad x_0, x_1 > 0. \quad (7)$$

Here  $\beta \in (0, 1)$  is the discount factor,  $c_{h,t}$  is consumption,  $b_{h,t}$  denotes liquid assets with borrowing limit  $\underline{b}$ ,  $k_{h,t}$  denotes illiquid asset holdings, and  $d_{h,t}$  is net deposits into the illiquid asset. Note that  $k_{h,t+1}$  is pinned by the illiquid asset accumulation equation given  $d_{h,t}$ , so the independent choice variables are  $(c_{h,t}, d_{h,t}, b_{h,t+1})$ . The prices  $r_t^b$  and  $r_t^k$  are the real returns on liquid and illiquid assets respectively,  $\bar{p}_t^{\text{HH}}$  is the consumer price index,  $r_{cc}$  is a credit surcharge on negative liquid balances,  $\tau^{\text{VAT}}$  is value-added tax, and  $\tau^K$  is the capital tax rate levied on the stock of illiquid assets, consistent with the Dutch Box 3 imputed-return regime. The parameters  $x_0$  and  $x_1$  govern the size and convexity of adjustment costs for reallocating between liquid and illiquid assets.

The utility function takes the Stone–Geary form:

$$u(c_{h,t}) = \frac{(c_{h,t} - \bar{c})^{1-\sigma}}{1-\sigma}, \quad (8)$$

where  $\sigma > 0$  is the coefficient of relative risk aversion (so that  $1/\sigma$  is the intertemporal elasticity of substitution) and  $\bar{c} \geq 0$  is subsistence consumption, below which utility is undefined. The Stone–Geary subsistence level  $\bar{c}(W_{h,t}) = c_{\text{int}} + c^* \cdot W_{h,t}$  enters the household budget as a mandatory expenditure and is passed to the neural network as a conditioning input, so the policy function adapts to the subsistence specification. Together,  $c^*$  and  $c_{\text{int}}$  are calibrated jointly with  $\sigma$  and the discount factor  $\beta$  as part of the preference parameter vector  $\Gamma = (\sigma, \beta, c^*, c_{\text{int}})$  (Section 3.4.3). The auxiliary utility functions are  $u_b(b_{h,t}) = b_{h,t}^{1-\sigma_b} / (1-\sigma_b)$  and  $u_k(k_{h,t}) = k_{h,t}^{1-\sigma_k} / (1-\sigma_k)$ . Since households can hold negative liquid balances (borrowing up to  $\underline{b}$ ), the liquidity preference term is evaluated at  $\max(0, b_{h,t})$ , so it rewards positive balances but does not penalise borrowing. The indicators  $\mathbb{I}_b$  and  $\mathbb{I}_k$  select which terms are active, yielding four specifications. We find that the bond-liquidity specification ( $\mathbb{I}_b = 1, \mathbb{I}_k = 0$ ) fits the data best, as discussed in Section 3.4.3.

Each period, agents update the parameters of their perceived law of motion as new data arrive and recalculate their optimal policy accordingly. Wage expectations  $\mathbb{E}_t^*[w_{t+1}]$  enter the Euler

equation through the continuation value: households forecast future income to evaluate the expected marginal utility of future consumption, so that anticipated wage changes affect current consumption–saving decisions. The perceived law of motion and the actual law of motion cannot exactly coincide, since the economy is too complex for an AR(1) rule to capture fully. Expectations can, however, be consistent with rational expectations if the perceived law of motion happens to coincide with the actual law of motion, a condition that will not hold exactly in our setting. We describe the household-level perceived laws of motion in Section 2.2.5 and the aggregate expectation formation mechanisms in Section 2.6.

### 2.2.3 Capital Market

Households hold capital  $k_{h,t}$  representing equity stakes in the productive sector, which comprises all firms and the commercial bank. Total household capital is  $K_t = \sum_{h=1}^H k_{h,t}$ . Holdings are initially allocated in proportion to household income and subsequently evolve through investment decisions:

$$k_{h,t+1} = (1 + r_t^k - \tau^K)k_{h,t} + d_{h,t}, \quad (9)$$

where  $d_{h,t}$  is net deposits into the illiquid asset (positive for purchases, negative for sales), consistent with the budget constraint in Section 2.2.2.

Capital earns a dividend return from profits distributed by firms and the bank. Both pay out a fraction  $\theta^{\text{DIV}}$  of their after-tax profits as dividends:

$$D_t = \underbrace{\sum_{i=1}^I \theta^{\text{DIV}} (1 - \tau^{\text{INC}}) (1 - \tau^{\text{FIRM}}) \max(0, \Pi_{i,t})}_{\text{Firm dividends}} + \underbrace{\theta^{\text{DIV}} (1 - \tau^{\text{INC}}) (1 - \tau^{\text{FIRM}}) \max(0, \Pi_t^{\text{bank}})}_{\text{Bank dividends}}, \quad (10)$$

where  $\tau^{\text{FIRM}}$  is the corporate tax rate,  $\tau^{\text{INC}}$  is the income tax rate on dividends,  $\Pi_{i,t}$  is the profit of firm  $i$ , and  $\Pi_t^{\text{bank}}$  is bank profit. The return on capital is the dividend yield:

$$r_t^k = \frac{D_t}{K_t}. \quad (11)$$

We abstract from secondary-market equity pricing: households deposit into or withdraw from the common capital stock at par, and net purchases increase  $K_t$ , diluting the dividend yield for all holders. Returns therefore adjust through quantity rather than price, and  $r_t^k$  captures dividend yield only.

### 2.2.4 Demanded and Realised Consumption

As in Poledna et al. (2023), households purchase goods from multiple types of firms through search-and-matching. The household calculates a demanded consumption budget for good  $g$ :

$$c_{g,t}^d = \frac{b_g^{\text{HH}} \bar{P}_{g,t}}{\bar{P}_{t-1}^{\text{HH}}} \frac{c_{h,t}^d}{1 + \tau^{\text{VAT}}}, \quad (12)$$

where  $b_g^{\text{HH}}$  is the consumption coefficient for product  $g$ ,  $\bar{P}_t^{\text{HH}}$  is the consumer price index, and  $\bar{P}_{g,t}$  is the producer price index for good  $g$  (both defined in Section A.1.8). The household then

visits firms to purchase goods. Depending on firm inventory and production capacity, demand may or may not be fully satisfied:

$$c_{h,t} \begin{cases} = \sum_g c_{g,t}^d & \text{if demand is fully met,} \\ < \sum_g c_{g,t}^d & \text{if firms visited cannot satisfy demand.} \end{cases} \quad (13)$$

Once realised consumption is determined, realised savings follow from the budget constraint:

$$b_{h,t+1} = y_{h,t} + (1 + r_t^b + \mathbb{I}_{\{b_{h,t+1} < 0\}} r_{cc}) b_{h,t} - d_{h,t} - \chi(d_{h,t}) - \bar{P}_t^{\text{HH}}(1 + \tau^{\text{VAT}}) c_{h,t}, \quad (14)$$

$$k_{h,t+1} = (1 + r_t^k - \tau^K) k_{h,t} + d_{h,t}. \quad (15)$$

### 2.2.5 Household-Level Expectations

The economy is too complex for households to learn the true data-generating process. Following Hommes and Zhu (2014), households learn parsimonious AR(1) forecasting rules for the variables that enter their optimisation problem. They observe past realisations and estimate the parameters by ordinary least squares in the tradition of Marcet and Sargent (1989) and Evans and Honkapohja (2001). Households estimate perceived laws of motion (PLMs) by OLS:

$$r_{t+1}^k = \alpha_t^k + \rho_t^k r_t^k + \epsilon_{t+1}^k, \quad (16)$$

$$r_{t+1}^b = \alpha_t^b + \rho_t^b r_t^b + \epsilon_{t+1}^b, \quad (17)$$

$$\pi_{t+1} = \alpha_t^\pi + \rho_t^\pi \pi_t + \epsilon_{t+1}^\pi, \quad (18)$$

$$w_{t+1} = \alpha_t^w + \rho_t^w w_t + \epsilon_{t+1}^w, \quad (19)$$

where  $\epsilon_{t+1}^j$  are zero-mean disturbances. Parameters  $\alpha_t^j$  and  $\rho_t^j$  for  $j \in \{k, b, \pi, w\}$  are re-estimated each period by OLS on an expanding window of observations  $\{1, \dots, t-1\}$ . Point forecasts are formed as conditional means:  $\hat{r}_{t+1|t}^k = \alpha_t^k + \rho_t^k r_t^k$ , and analogously for the other variables. All households share the same PLM parameter estimates. These household-level expectations feed into the optimal consumption problem. The aggregate expectation formation mechanisms used by all other agents in the model (firms, the bank, and the government) are described separately in Section 2.6.

### 2.3 Policy Function Computation

We compute household policy functions using an amortised deep-learning approach adapted from Maliar et al. (2021). Neural networks have become a standard tool for solving heterogeneous-agent models (see Fernández-Villaverde et al., 2024, for a survey), including deep equilibrium nets (Azinovic et al., 2022) and models with financial frictions (Fernández-Villaverde et al., 2023). We follow Maliar et al. (2021) for two reasons: their all-in-one expectation operator avoids costly numerical integration over future states, and their KKT residual formulation handles the borrowing constraints in our household problem without requiring separate case distinctions.

The key idea, known as amortisation in the machine-learning literature, is to separate *solving* the household problem from *calibrating* it. Rather than solving the Karush–Kuhn–Tucker

(KKT) system for a single parameter vector, the neural network learns to solve it for any parameter combination within specified bounds. The preference parameter vector  $\Gamma$ , comprising  $(\sigma, \beta, c^*, c_{\text{int}})$  in the baseline specification, extended with  $\sigma_k$  and/or  $\sigma_b$  in the liquidity variants, is treated as an additional input, so the network maps

$$(\mathbf{s}, \Gamma) \mapsto (c_{h,t}, k_{h,t+1}, b_{h,t+1}, \lambda_{h,t}^b, \lambda_{h,t}^k),$$

where  $\mathbf{s}$  collects current states and  $\lambda_{h,t}^b, \lambda_{h,t}^k$  are Lagrange multipliers on the asset constraints. This allows us to search over preference parameters to match empirical moments without resolving the optimisation problem, a property essential for embedding HANK-style households in an ABM, where the perceived law of motion evolves each period and recalibration must be computationally cheap.

Our approach relates to the ‘extended policy functions’ of Kase et al. (2025), who treat structural parameters as pseudo-state variables when estimating nonlinear HANK models via likelihood methods. While their method targets likelihood estimation, ours targets moment matching within an agent-based framework. A key difference is that during simulation we employ online learning, where the network receives incremental updates each period using KKT residual minimisation with small batch updates, allowing it to adapt to evolving conditions such as changing interest rates and shifting wealth distributions. This contrasts with Kase et al. (2025), who solve once and hold the network fixed during simulation.

This approach offers several advantages over traditional grid-based or basis-function methods. First, neural networks achieve faster convergence while maintaining accuracy (Maliar et al., 2021). We experimented with classical grid methods, projection methods, and sparse adaptive grids (Brumm and Scheidegger, 2017), but found them prohibitively slow for our application, which requires solving the household problem at each quarterly calibration period across multiple Monte Carlo simulations. Second, our sequential training exploits temporal continuity: each period’s network is warm-started from the previous period’s parameters. Because the perceived law of motion evolves gradually, consecutive solutions lie close together in parameter space, substantially reducing training time. Third, unlike discretised grids, neural networks naturally accommodate time-varying state-space bounds, because re-discretising the state space each period discards previously computed policy values, which is important for handling the potential nonstationarity inherent in agent-based models.

The network is trained to minimise expected residuals of the KKT system. The theoretical loss is  $\Xi(\theta) = \mathbb{E}_{\mathbf{s}}[\xi(\mathbf{s}; \Gamma, \theta)]$ , where  $\theta$  denotes the neural network weight vector<sup>4</sup> and  $\xi$  stacks five KKT residuals: two Euler equations (for bonds and capital), two complementarity conditions enforcing the inequality constraints on asset holdings via Fischer–Burmeister functions, and the budget constraint. We minimise an empirical counterpart using the cross-product estimator of Maliar et al. (2021), which provides an unbiased estimate of expected squared residuals with reduced variance: the cross-product of two independent shock realisations  $\xi_m^{(1)} \cdot \xi_m^{(2)}$  estimates  $\mathbb{E}[\xi_m]^2$  rather than  $\mathbb{E}[\xi_m^2]$ , since  $\mathbb{E}[\xi_m^{(1)} \cdot \xi_m^{(2)}] = \mathbb{E}[\xi_m]^2$  by independence, thereby avoiding the positive bias from the variance term that contaminates single-draw squared residuals. For each sampled state  $\mathbf{s}_i$ , we draw two independent shock realisations and compute:

$$\Xi^n(\theta) = \frac{1}{n} \sum_{i=1}^n \sum_{m=1}^5 w_m \cdot \xi_m^{(1)}(\mathbf{s}_i; \theta) \cdot \xi_m^{(2)}(\mathbf{s}_i; \theta), \quad (20)$$

where  $m$  indexes residual types and  $w_m = 1$  are uniform residual-specific weights (so the total loss is the sum, not the average, across residual types). Gradients are computed via automatic differentiation.

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<sup>4</sup>Not to be confused with  $\theta^{\text{UB}}$  (replacement rate) or  $\theta^{\text{DIV}}$  (dividend payout ratio) used elsewhere.

We implement the separation of solving and calibrating through a three-stage training procedure, joint training of network and parameters, KKT-focused refinement with frozen parameters, and moment calibration with a frozen network, detailed in Appendix C together with the network architecture and hyperparameters. We consider four model variants with different liquidity preference specifications (baseline, capital liquidity, bond liquidity, and full liquidity), which differ in whether  $\sigma_k$  and/or  $\sigma_b$  enter the preference vector  $\Gamma$ . Since we train a separate network for each quarterly period, the complete procedure is repeated 40 times (2010Q1–2019Q4). Calibration targets and results are presented in Section 3.4.3.

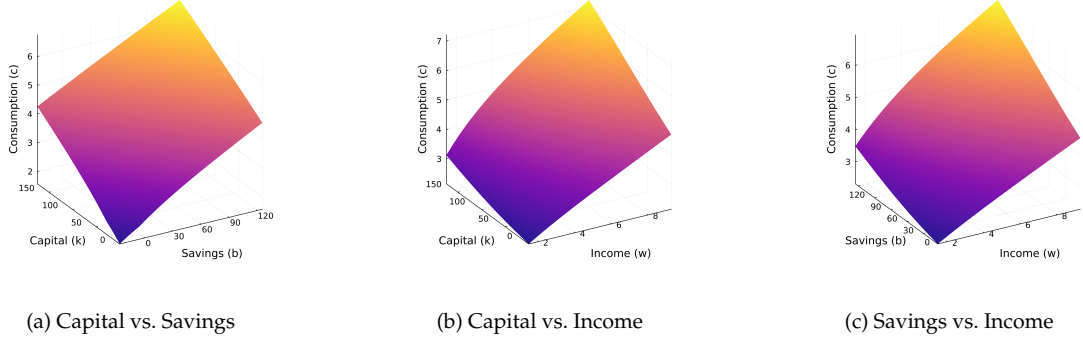


Figure 2: Policy function slices from the optimal consumption model, showing consumption as a function of two state variables at a time while holding the remaining states at their empirical sample medians.

The consumption surfaces in Figure 2 display the concavity in wealth expected under CRRA preferences with Stone–Geary subsistence: consumption increases in both capital and bond holdings but at a diminishing rate. Panel (a) shows a visible flattening along the capital axis at low savings, where the non-negativity constraint on deposits begins to bind and the Lagrange multiplier  $\lambda^b$  becomes active. The steeper gradient with respect to income relative to either asset in panels (b) and (c) reflects the calibrated moment hierarchy  $MPC > MPC_k \approx MPC_b$ .

## 2.4 Heuristic Consumption Models

We compare the optimal consumption model against a suite of ten heuristic consumption rules, denoted H1–H10, designed to span the space of reduced-form consumption functions used in applied macroeconomics and to progressively close the structural channels that distinguish reduced-form rules from the optimal policy. H1–H4 are static rules combining linear or concave income response with linear wealth dependence; H5–H6 add habit formation on the income component (Abel, 1990; Campbell, 1999); H7 implements the buffer-stock rule of Deaton (1991) and Carroll (1997), in which consumption is a smooth function of distance from a target wealth level. The remaining three variants close the additional structural channels identified in Table 2: H8 partitions the income aggregate to allow a separate marginal propensity to consume out of capital returns (Christelis et al., 2019); H9 replaces the fixed liquid/illiquid savings split with a state-contingent Merton (1969) portfolio rule; H10 combines both. The progression is not motivated by nesting — H5–H7 are not strict extensions of H1–H4 — but by spanning the families of reduced-form behaviour (income response, habit, target-wealth) and by sequentially equalising the structural channels separating the heuristic class from the OC-ABM, so that the residual OC-ABM advantage attributes more sharply to Euler-equation discipline.

Before stating the heuristic equations, we define the income variable used by all heuristic models. Unlike the optimal model, where capital returns enter through a separate portfolio chan-

nel, heuristic households fold capital returns into income. This means heuristic households respond to asset returns only through the income–consumption relationship, whereas optimal households respond through both intertemporal and portfolio channels, a structural difference that matters for interpreting the forecasting comparison. Expected income is:

$$Y_{h,t}^e = \begin{cases} (w_{h,t}(1 - \tau^{\text{SIW}} - \tau^{\text{INC}}(1 - \tau^{\text{SIW}})) + sb_t^{\text{other}}) \bar{P}_{t-1}^{\text{HH}}(1 + \pi_t^e) + r_t^k k_{h,t} & \text{if employed,} \\ (\theta^{\text{UB}} w_{h,t} + sb_t^{\text{other}}) \bar{P}_{t-1}^{\text{HH}}(1 + \pi_t^e) + r_t^k k_{h,t} & \text{if unemployed,} \\ (sb_t^{\text{inact}} + sb_t^{\text{other}}) \bar{P}_{t-1}^{\text{HH}}(1 + \pi_t^e) + r_t^k k_{h,t} & \text{if not economically active.} \end{cases} \quad (21)$$

We denote expected income by  $y_{h,t}^e$ , total wealth by  $W_{h,t} = k_{h,t} + b_{h,t}$ , and use  $\kappa$  for the intercept,  $\hat{\gamma}$  for the linear MPC out of income,  $\hat{\delta}$  for the linear MPC out of wealth,  $\hat{\alpha}$  and  $\hat{\beta}$  for the concave income parameters,  $\hat{\lambda}$  for the habit-persistence coefficient,  $\hat{\phi}^{\text{bs}}$  for the buffer-stock adjustment speed, and  $\bar{W}$  for the target wealth-to-income level.<sup>5</sup> Let  $a_{h,t} = y_{h,t} + \max(0, b_{h,t} + \underline{b})$  denote available resources — realised income plus liquid dissaving capacity down to the borrowing limit. The seven heuristics are:

$$\begin{aligned} \text{H1: } c_{h,t} &= \frac{\min(\kappa + \hat{\gamma} y_{h,t}^e, a_{h,t})}{1 + \tau^{\text{VAT}}} & \text{H5: } c_{h,t} &= \frac{\min(\hat{\lambda} \tilde{c}_{h,t-1} + (1 - \hat{\lambda})(\kappa + \hat{\alpha} y_{h,t}^e \hat{\beta}), a_{h,t})}{1 + \tau^{\text{VAT}}} \\ \text{H2: } c_{h,t} &= \frac{\min(\kappa + \hat{\alpha} y_{h,t}^e \hat{\beta}, a_{h,t})}{1 + \tau^{\text{VAT}}} & \text{H6: } c_{h,t} &= \frac{\min(\hat{\lambda} \tilde{c}_{h,t-1}^y + (1 - \hat{\lambda})(\kappa + \hat{\alpha} y_{h,t}^e \hat{\beta}) + \hat{\delta} W_{h,t}, a_{h,t})}{1 + \tau^{\text{VAT}}} \\ \text{H3: } c_{h,t} &= \frac{\min(\kappa + \hat{\gamma} y_{h,t}^e + \hat{\delta} W_{h,t}, a_{h,t})}{1 + \tau^{\text{VAT}}} & \text{H7: } c_{h,t} &= \frac{\min(\kappa + \hat{\alpha} y_{h,t}^e \hat{\beta} + \hat{\phi}^{\text{bs}}(W_{h,t} - \bar{W}), a_{h,t})}{1 + \tau^{\text{VAT}}} \\ \text{H4: } c_{h,t} &= \frac{\min(\kappa + \hat{\alpha} y_{h,t}^e \hat{\beta} + \hat{\delta} W_{h,t}, a_{h,t})}{1 + \tau^{\text{VAT}}} \end{aligned}$$

where  $\tilde{c}_{h,t-1}$  is realised consumption in the previous quarter and  $\tilde{c}_{h,t-1}^y$  is its income component (consumption net of the additive wealth contribution). All ten heuristics share the same borrowing constraint as the OC-ABM: by capping consumption at  $a_{h,t}$ , households can dissave their liquid balances down to the limit  $\underline{b}$ , mirroring the constraint  $b_{h,t+1} \geq -\underline{b}$  in the optimisation problem of Section 2.2.2. This equalises the budget set across heuristic and optimal variants, isolating differences in the consumption rule itself rather than in the feasible action space.

The remaining three variants build on H7 by closing the additional structural channels distinguishing the heuristic class from the OC-ABM (Table 2). They share the H7 functional form and differ only in the treatment of capital-return income and portfolio composition.

**H8: Capital-return MPC asymmetry.** To equalise the capital-return channel, we partition the income aggregate in Equation (21) into labour-and-transfer income  $y_{h,t}^w$  and capital-return income  $r_t^k k_{h,t}$ , attenuating the latter by a Christelis factor  $\alpha_{\text{cap}} = 0.5$ :

$$y_{h,t}^{\text{e,H8}} = y_{h,t}^w + \alpha_{\text{cap}} \cdot r_t^k k_{h,t}. \quad (22)$$

Consumption then follows the H7 rule with  $y_{h,t}^e$  replaced by  $y_{h,t}^{\text{e,H8}}$ . The factor  $\alpha_{\text{cap}}$  scales the contribution of capital returns to the income aggregate so that the implied marginal propensity to consume out of capital returns is half of that out of labour income, consistent with the asymmetry documented by Christelis et al. (2019).

<sup>5</sup>We use hatted symbols for heuristic parameters to distinguish them from the discount factor  $\beta$ , growth rate  $\gamma_t^e$ , and AR coefficients  $\rho_t^j$  used elsewhere.

**H9: Merton portfolio rule.** H7 splits new savings between the liquid and illiquid asset in the proportion observed in the calibration quarter. H9 replaces this rule with a state-contingent target. Following Merton (1969), the target portfolio share is

$$\omega_{h,t} = \text{clip}\left(\frac{\mu_t^k - r_t^b}{\hat{\gamma}_M (\sigma_t^k)^2}, 0.3, 0.85\right), \quad (23)$$

where  $\mu_t^k$  and  $\sigma_t^k$  are perceived expected return and volatility on capital, and the clipping bounds avoid pathological allocations during transitions. The Merton risk-aversion coefficient  $\hat{\gamma}_M$  is inverted at each forecast origin so that the unconstrained share equals the empirical aggregate liquid-illiquid ratio at the prevailing risk-free rate  $r_0^b$ . The target illiquid stock is  $K_{h,t}^{\text{target}} = \text{clip}(\omega_{h,t} W_{h,t}, 0, W_{h,t} - \underline{b})$ , and the household closes a fraction of the gap each period:

$$d_{h,t} = 0.1 \cdot (K_{h,t}^{\text{target}} - k_{h,t}). \quad (24)$$

The smoothing rate 0.1 mirrors the gradual portfolio adjustment of the OC-ABM's neural-network policy. The bond balance is then determined by the household budget constraint exactly as in H7. To preserve comparability with the OC-ABM treatment of equity, deposits adjust  $k_{h,t}$  at par without entering aggregate investment, matching the OC-ABM convention.

**H10: Combined.** H10 combines the capital-return MPC asymmetry of H8 with the Merton portfolio rule of H9, applied jointly to the H7 buffer-stock specification. H10 is the channel-equalised heuristic against which the OC-ABM's residual advantage attributes to Euler-equation discipline alone.

**Heuristic calibration.** We pin the structural MPC-shape parameters  $(\hat{\gamma}, \hat{\alpha}, \hat{\beta}, \hat{\delta}, \hat{\varphi}^{\text{bs}})$  to match the empirical MPC targets from the DNB Household Survey (Christelis et al., 2019) —  $\text{MPC}_y = 0.20$  for income and  $\text{MPC}_W = 0.03$  for wealth — and then solve numerically for the intercept  $\kappa$  by bisection so that aggregate consumption equals the average propensity to consume from national accounts. The habit coefficient  $\hat{\lambda} = 0.7$  matches the consumption-persistence estimate of Christiano et al. (2005), and the buffer-stock target  $\bar{W}$  is set to the cross-sectional median wealth at each forecast origin. The capital-return scaling factor  $\alpha_{\text{cap}} = 0.5$  (H8, H10) is fixed at the Christelis asymmetry value and not iterated. The Merton risk-aversion coefficient  $\hat{\gamma}_M$  (H9, H10) is inverted at each forecast origin so that the unconstrained portfolio share  $\omega = (\mu^k - r^b) / (\hat{\gamma}_M (\sigma^k)^2)$  equals the empirical aggregate liquid-illiquid ratio at the prevailing risk-free rate; the perceived first two moments  $\mu^k$  and  $\sigma^k$  are estimated by OLS on the dividend-yield series available at the forecast origin. Because the capped consumption rule is non-linear in the structural parameters, we iterate this two-step calibration until the effective cross-sectional mean MPCs (measured by numerical perturbation of  $y$  and  $W$  through the capped rule) match their targets to within  $5 \times 10^{-3}$ . Calibration is performed at every forecast origin, mirroring the OC-ABM's quarterly retraining schedule.

Table 1 summarises each specification.

**Remaining structural asymmetries.** The borrowing constraint equalised across all ten heuristics removes one of four structural differences between the heuristic and optimal models. The H1–H7 ladder leaves three channels open. First, heuristic households fold capital returns into income (Equation 21) and therefore respond to  $r_t^k$  through the same MPC as labour

Table 1: Heuristic Consumption Model Specifications

| Model | Income response                              | Wealth response                        | Dynamics / Channel extension                         |
|-------|----------------------------------------------|----------------------------------------|------------------------------------------------------|
| H1    | Linear ( $\hat{\gamma}$ )                    | —                                      | Static                                               |
| H2    | Concave ( $\hat{\alpha}, \hat{\beta}$ )      | —                                      | Static                                               |
| H3    | Linear                                       | Linear ( $\hat{\delta}$ )              | Static                                               |
| H4    | Concave                                      | Linear                                 | Static                                               |
| H5    | Concave                                      | —                                      | Habit ( $\hat{\lambda}$ )                            |
| H6    | Concave (habit-smoothed)                     | Linear (additive)                      | Habit on income only                                 |
| H7    | Concave                                      | Target deviation ( $\hat{\phi}^{bs}$ ) | Buffer-stock                                         |
| H8    | Concave (capital MPC scaled $\alpha_{cap}$ ) | Target deviation                       | Buffer-stock + Christelis asymmetry                  |
| H9    | Concave                                      | Target deviation                       | Buffer-stock + Merton portfolio ( $\hat{\gamma}_M$ ) |
| H10   | Concave (capital MPC scaled)                 | Target deviation                       | Buffer-stock + Christelis asymmetry                  |

Note: All ten heuristics share the borrowing constraint  $c_{h,t} \leq a_{h,t} / (1 + \tau^{VAT})$  where  $a_{h,t} = y_{h,t} + \max(0, b_{h,t} + \underline{b})$ , matching the OC-ABM’s liquid-asset floor. Free parameter counts include the intercept  $\kappa$ , which is common to all specifications. H5 and H6 introduce habit through  $\hat{\lambda} \in [0, 1]$  applied to the income component only; the wealth term in H6 enters additively. H7 follows Deaton (1991), with  $\hat{\phi}^{bs}$  governing the speed of adjustment toward the target wealth  $\bar{W}$ . H8 fixes  $\alpha_{cap} = 0.5$  from Christelis et al. (2019) and is not counted as a free parameter; H9’s  $\hat{\gamma}_M$  is inverted at calibration to match the empirical aggregate portfolio share. H10 combines H8 and H9 with no additional free parameters beyond those of H9.

income, whereas the OC-ABM allows capital returns to enter through a separate continuation-value channel and at a distinct MPC. Second, H1–H7 households split new savings between liquid and illiquid assets in the proportion observed in the calibration quarter, while the OC-ABM determines portfolio shares endogenously through the neural-network policy.<sup>6</sup> Third, H1–H7 rules respond only to current state, while the optimal model incorporates an explicit continuation value through the Euler equation. H8 closes the first channel by partitioning the income aggregate and scaling the capital-return contribution by  $\alpha_{cap}$  (Equation 22); H9 closes the second by replacing the fixed split with a Merton-targeted portfolio rule (Equations 23–24); H10 closes both jointly. The optimisation/Euler channel remains by construction — it cannot be replicated by any reduced-form rule. Table 2 summarises the treatment of each channel across model classes. Section 5.3 reports the consumption-forecast attribution implied by this decomposition.

Table 2: Structural channels: OC-ABM versus heuristic ladder

| Channel              | OC-ABM                                     | H1–H7           | H8                       | H9              | H10                      |
|----------------------|--------------------------------------------|-----------------|--------------------------|-----------------|--------------------------|
| Borrowing constraint | $b_{h,t+1} \geq -\underline{b}$            | equalised       | equalised                | equalised       | equalised                |
| Capital-return MPC   | Separate continuation-value channel        | Folded into $y$ | Split ( $\alpha_{cap}$ ) | Folded into $y$ | Split ( $\alpha_{cap}$ ) |
| Portfolio choice     | Endogenous (NN policy on $d_{h,t}$ )       | Fixed split     | Fixed split              | Merton target   | Merton target            |
| Optimisation / Euler | Euler-equation-disciplined (KKT residuals) | Reduced form    | Reduced form             | Reduced form    | Reduced form             |

Note:  $a_{h,t} = y_{h,t} + \max(0, b_{h,t} + \underline{b})$ . The borrowing constraint is implemented by capping consumption at the same effective resource limit as the OC-ABM’s liquid-asset floor across all ten heuristics. H8 partitions the income aggregate into labour and capital components with separate MPC scaling. H9 replaces the fixed liquid/illiquid split with a state-contingent Merton target. H10 combines both. The optimisation/Euler channel is closed only by the OC-ABM and constitutes the residual identification gap once the other three channels are equalised.

<sup>6</sup>This is necessary because Poledna et al. (2023) assign each firm a single household as owner, producing very thick tails in the wealth distribution. With a linear consumption function this is immaterial, but with heterogeneous MPCs and endogenous portfolio choice it would distort forecasting performance.

## 2.5 Baseline Heuristic

The baseline consumption heuristic of Poledna et al. (2023) sets consumption as a fixed fraction of income:  $c_{h,t} = \phi^C y_{h,t}$ , where  $\phi^C$  is calibrated from national accounts as the ratio of household consumption to disposable income. This model features no MPC heterogeneity and does not account for wealth differences. It uses the same income process as the heuristic models above. The remainder of the model is unchanged from Poledna et al. (2023) and is described in Appendix A.

## 2.6 Aggregate Expectation Formation Mechanisms

The household-level expectations described in Section 2.2.5 govern the optimal consumption problem. All other agents in the model, namely firms, the bank, and the government, form expectations about aggregate output growth and inflation using one of four mechanisms, which we also use to generate the aggregate forecasts evaluated in Section 5. For the heuristic ABM variants, all agents (including households) share the same aggregate expectation mechanism. For the optimal-consumption ABM, households retain their own perceived laws of motion while all other agents use the aggregate mechanism; this separation is necessitated by computational cost, since the household optimisation problem requires training a neural network for each period's perceived law of motion, whereas the aggregate mechanism is a simple OLS regression.

Throughout, we express output growth in log-differences for stationarity. The expected discrete growth rate and expected inflation are recovered as:

$$\gamma_t^e = \exp\left(\mathbb{E}_t^* \left[ \log\left(\frac{Y_t}{Y_{t-1}}\right) \right]\right) - 1, \quad (25)$$

$$\pi_t^e = \mathbb{E}_t^*(1 + \pi_t) - 1. \quad (26)$$

These recovery formulae apply identically across all four specifications below.<sup>7</sup>

### 2.6.1 AR(1) Expectations

Following Poledna et al. (2023), the simplest specification models output growth and inflation as independent AR(1) processes. This serves as the parsimony benchmark: a minimal adaptive learning model that is progressively relaxed in the specifications that follow. Output growth expectations are:

$$\log\left(\frac{Y_t}{Y_{t-1}}\right) = \rho_t^Y \log\left(\frac{Y_{t-1}}{Y_{t-2}}\right) + \alpha_t^Y + \epsilon_t^Y, \quad (27)$$

and inflation perceived law of motion is:

$$(1 + \pi_t) = \rho_t^\pi (1 + \pi_{t-1}) + \alpha_t^\pi + \epsilon_t^\pi, \quad (28)$$

where all coefficients are re-estimated each period by OLS on observations  $t' \in \{-T', \dots, t-1\}$ . Expected growth and inflation are recovered via equations (25)–(26).

<sup>7</sup>The transformation  $\gamma_t^e = \exp(\mathbb{E}_t^*[\log(Y_t/Y_{t-1})]) - 1$  recovers the geometric rather than arithmetic expected growth rate. By Jensen's inequality, this systematically understates arithmetic expected growth for symmetric log-growth shocks. The bias is negligible for the magnitude of quarterly growth rates in our sample.

### 2.6.2 VAR Expectations

The VAR specification models output growth and inflation jointly, capturing their interdependencies:

$$\begin{bmatrix} \log\left(\frac{Y_t}{Y_{t-1}}\right) \\ 1 + \pi_t \end{bmatrix} = \begin{bmatrix} \rho_t^{YY} & \rho_t^{Y\pi} \\ \rho_t^{\pi Y} & \rho_t^{\pi\pi} \end{bmatrix} \begin{bmatrix} \log\left(\frac{Y_{t-1}}{Y_{t-2}}\right) \\ 1 + \pi_{t-1} \end{bmatrix} + \begin{bmatrix} \alpha_t^Y \\ \alpha_t^\pi \end{bmatrix} + \begin{bmatrix} \epsilon_t^Y \\ \epsilon_t^\pi \end{bmatrix}, \quad (29)$$

with all coefficients re-estimated each period by OLS. Expected growth and inflation are recovered via equations (25)–(26).

### 2.6.3 ARX Expectations

Following Poledna et al. (2023), we incorporate predetermined paths for imports ( $M_t$ ), exports ( $X_t$ ), and government consumption ( $G_t$ ) as exogenous predictors, consistent with a small open economy framework. Output growth expectations are:

$$\log\left(\frac{Y_t}{Y_{t-1}}\right) = \rho_t^Y \log\left(\frac{Y_{t-1}}{Y_{t-2}}\right) + \phi_t^M \Delta \log M_{t-1} + \phi_t^G \Delta \log G_{t-1} + \phi_t^X \Delta \log X_{t-1} + \alpha_t^Y + \epsilon_t^Y, \quad (30)$$

and inflation perceived law of motion is:

$$(1 + \pi_t) = \rho_t^\pi (1 + \pi_{t-1}) + \phi_t^M \Delta \log M_{t-1} + \phi_t^G \Delta \log G_{t-1} + \phi_t^X \Delta \log X_{t-1} + \alpha_t^\pi + \epsilon_t^\pi, \quad (31)$$

where  $\Delta \log X_t = \log X_t - \log X_{t-1}$ . All coefficients are re-estimated each period. Expected growth and inflation follow from equations (25)–(26).

### 2.6.4 VARX Expectations

The VARX specification combines the VAR structure with exogenous predictors:

$$\begin{bmatrix} \log\left(\frac{Y_t}{Y_{t-1}}\right) \\ 1 + \pi_t \end{bmatrix} = \begin{bmatrix} \rho_t^{YY} & \rho_t^{Y\pi} \\ \rho_t^{\pi Y} & \rho_t^{\pi\pi} \end{bmatrix} \begin{bmatrix} \log\left(\frac{Y_{t-1}}{Y_{t-2}}\right) \\ 1 + \pi_{t-1} \end{bmatrix} + \begin{bmatrix} \phi_t^{YM} & \phi_t^{YG} & \phi_t^{YX} \\ \phi_t^{\pi M} & \phi_t^{\pi G} & \phi_t^{\pi X} \end{bmatrix} \begin{bmatrix} \Delta \log M_{t-1} \\ \Delta \log G_{t-1} \\ \Delta \log X_{t-1} \end{bmatrix} + \begin{bmatrix} \alpha_t^Y \\ \alpha_t^\pi \end{bmatrix} + \begin{bmatrix} \epsilon_t^Y \\ \epsilon_t^\pi \end{bmatrix}, \quad (32)$$

with all coefficients re-estimated each period. Expected growth and inflation follow from equations (25)–(26).

## 3 Calibration

This section describes the calibration of the agent-based model. The calibration strategy follows the approach of Poledna et al. (2023) for the macroeconomic structure and extends it with microdata-based calibration of the household sector. We proceed in four steps. Section 3.1 describes the data sources. Section 3.2 explains how macroeconomic parameters are calibrated from national accounts, input–output tables, and institutional guidelines. Section 3.3 sets out the initial conditions for all sectors. Section 3.4 presents the microdata-based calibration of the household sector, including the empirical wealth–income distribution, the neural-network solution method, and the calibration results.

### 3.1 Data Sources

We draw on two primary data sources: Eurostat national accounts and Statistics Netherlands (CBS) microdata. The model is calibrated at quarterly frequency, with each agent representing a natural or legal entity (household, firm, government body, or other institution) in the Netherlands. The Netherlands is a small open economy with approximately 17 million inhabitants.<sup>8</sup> Government expenditures amount to 44.5% of GDP, exports to 51% and imports to 45%, and the labour income share is 77.3%. Contrary to Poledna et al. (2023), we scale the model to 1:100 due to computational and data privacy constraints, yielding 172,699 individuals and 13,814 firms.

Table 3 reports the Eurostat datasets used for macroeconomic calibration. For the household sector, we use CBS Microdata (Table 4), which provides individual-level information on personal income, household income, and household wealth, enabling calibration of the joint income–wealth distribution that drives MPC heterogeneity in the model.

Table 3: Eurostat datasets used for model calibration

| Name                                                                          | Code             |
|-------------------------------------------------------------------------------|------------------|
| Population by current activity status, NACE Rev. 2 activity and NUTS 2 region | cens_11an_r2     |
| Business demography by legal form (from 2004 onwards, NACE Rev. 2)            | bd_9ac_1_form_r2 |
| Symmetric input–output table at basic prices (product by product)             | naio_10_cp1700   |
| Cross-classification of fixed assets by industry and by asset (stocks)        | nama_10_nfa_st   |
| Government revenue, expenditure and main aggregates                           | gov_10a_main     |
| General government expenditure by function (COFOG)                            | gov_10a_exp      |
| Quarterly non-financial accounts for general government                       | gov_10q_ggnfa    |
| Quarterly government debt                                                     | gov_10q_ggdebt   |
| Financial balance sheets (annual)                                             | nasa_10_f_bs     |
| Financial balance sheets (quarterly)                                          | nasq_10_f_bs     |
| Non-financial transactions (annual)                                           | nasa_10_nf_tr    |
| Non-financial transactions (quarterly)                                        | nasq_10_nf_tr    |
| GDP and main components (output, expenditure and income)                      | namq_10_gdp      |
| Money market interest rates (quarterly)                                       | irt_st_q         |

Table 4: CBS Microdata tables for household sector calibration

| Name                                                 | Code    |
|------------------------------------------------------|---------|
| Income of persons (annual, from tax declarations)    | INPATAB |
| Income of households (annual, from tax declarations) | INHATAB |
| Wealth of households (annual, from tax declarations) | VEHTAB  |

INPATAB contains income data for all individuals in the Dutch population (available from 2011 onwards), INHATAB contains income data at the household level (available from 2011 onwards), and VEHTAB contains wealth data at the household level (available from 2006 onwards).

### 3.2 Macroeconomic Parameters

#### 3.2.1 Census and Demography

The number of agents is calibrated from census and business demography data. We use Eurostat table cens\_11an\_r2 to set the number of economically active persons  $H^{\text{act}}$  to the 2011

<sup>8</sup>Data retrieved from Statistics Netherlands, <https://opendata.cbs.nl/#/CBS/en/>, accessed 22 March 2023.

value. The classification of industries follows the Statistical Classification of Economic Activities in the European Community (NACE Rev. 2). Eurostat compiles tables, including input–output tables, demographic data, and cross-classification tables, at a breakdown of 64 activities. We exclude sectors T (“Services of households as employers”) and U (“Services provided by extraterritorial organisations”), both of which have zero output in the Netherlands, giving  $S = G = 62$  industries and products. The number of firms per industry  $I_s$  is set using business demography data (bd\_9ac\_1\_form\_r2). Following Poledna et al. (2023), we set the number of foreign firms importing domestic goods to 50% and the number of government entities to 25%, corresponding approximately to the shares of exports and government consumption in total value added.

### 3.2.2 Input–Output Tables

Parameters related to productivity, technology, capital formation, and consumption are calculated from input–output tables and updated annually. To calibrate the technology, consumption, and capital formation coefficients ( $a_{sg}, b_g^{CF}, b_g^{HH}, b_g^{CFH}, c_g^G, c_g^E, c_g^I$ ) and the productivity coefficient of intermediate inputs and net tax rates ( $\beta_i^M, \tau_i^Y, \tau_i^{PROD}$ ), we use the symmetric input–output table at basic prices (naio\_10\_cp1700). Parameters are computed by normalising the table columns. Figure 3 illustrates the Dutch input–output network aggregated to 19 sectors for readability; the model operates at the full 62-sector NACE Rev. 2 disaggregation.

Some parameters require both input–output tables and cross-classification data. The productivity coefficients for labour and capital ( $\bar{\alpha}_i, \kappa_i$ ), the depreciation rate ( $\delta_i$ ), and the average wage rate ( $\bar{w}_i$ ) are calibrated by combining these sources.

### 3.2.3 Tax Rates and Sector Accounts

Tax rates are calculated from national accounting identities to match financial flows in the input–output tables, government statistics, and sector accounts. An average tax rate for an institutional sector (firms, households, etc.) is defined as the aggregate tax flow divided by the aggregate monetary flow constituting the tax base. Tax rates are computed per calendar year and applied uniformly across agents within each sector.<sup>9</sup>

The corporate profit tax rate  $\tau^{FIRM}$  equals total corporate tax flows divided by total operating surplus and mixed income, sourced from sector accounts (nasa\_10\_nf\_tr) and input–output tables respectively. Social security contribution rates for employers ( $\tau^{SIF}$ ) and employees ( $\tau^{SIW}$ ) are calibrated similarly. Value-added tax rates ( $\tau^{VAT}, \tau^{CF}, \tau^G, \tau^{EXPORT}$ ) are obtained from input–output tables.

The dividend payout ratio  $\theta^{DIV}$  equals interest and dividend receipts plus mixed income received by the household sector (nasa\_10\_nf\_tr) divided by total net operating surplus and mixed income from input–output tables. The risk premium  $\mu$  on firms’ outstanding debt is calculated as the difference between the 3-month Euribor rate (irt\_st\_q) and the ratio of observed interest payments to total liabilities of non-financial corporations (nasq\_10\_f\_bs). The interest rate on government debt  $r^G$  is calibrated as interest due per quarter (gov\_10q\_ggnfa) divided by total government debt (gov\_10q\_ggdebt).

<sup>9</sup>The Netherlands has a progressive marginal tax system, which we abstract from for model parsimony.

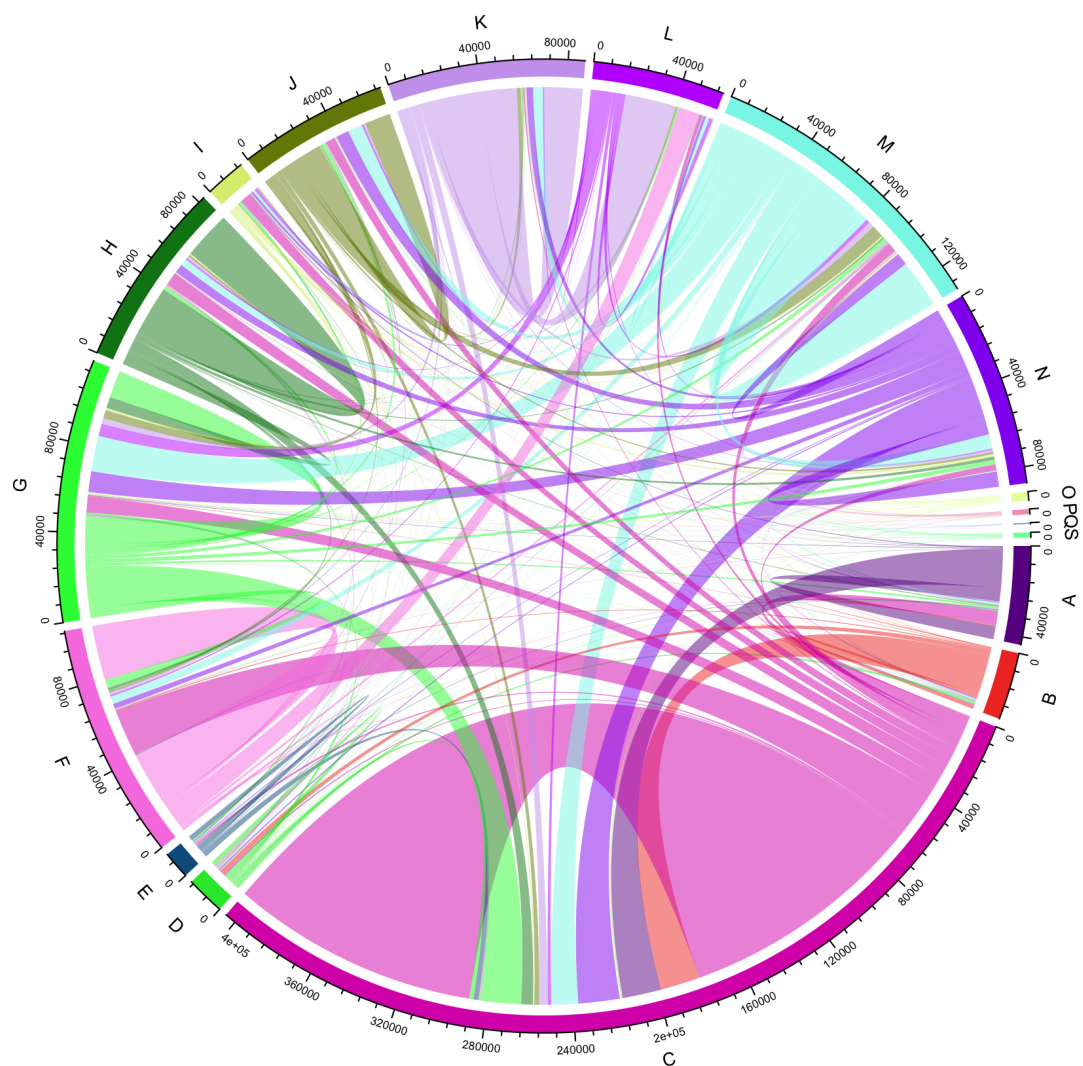


Figure 3: Intermediate consumption between 19 industries derived from Eurostat input-output tables, aggregated from the 62-sector NACE Rev. 2 classification used in the model.

|   |                                                     |   |                                                   |   |                                         |
|---|-----------------------------------------------------|---|---------------------------------------------------|---|-----------------------------------------|
| A | Agriculture, forestry and fishing                   | H | Transportation and storage                        | O | Public administration and defence       |
| B | Mining and quarrying                                | I | Accommodation and food service activities         | P | Education                               |
| C | Manufacturing                                       | J | Information and communication                     | Q | Human health and social work activities |
| D | Electricity, gas, steam and air conditioning supply | K | Financial and insurance activities                | R | Arts, entertainment and recreation      |
| E | Water supply                                        | L | Real estate activities                            | S | Other service activities                |
| F | Construction                                        | M | Professional, scientific and technical activities |   |                                         |
| G | Wholesale and retail trade                          | N | Administrative activities                         |   |                                         |

### 3.2.4 Institutional Guidelines and Banking Practices

Several parameters are set according to statutory guidelines and financial regulation. The capital ratio  $\zeta$  follows Basel III requirements. The central bank inflation target  $\pi^*$  is 2%, per ECB statutes. The rate of debt instalment  $\theta$  is 5% quarterly. The maximum loan-to-value ratio  $\zeta^{\text{LTV}}$  is 60%, and the initial loan-to-value ratio for replacement firms  $\zeta^b$  is 50%.

### 3.2.5 Exogenous Processes

Exogenous processes, including imports, exports, government consumption, euro area inflation, and euro area GDP growth, are estimated from national accounts (namq\_10\_gdp) and money market interest rates (irt\_st\_q) using data from 1997Q1 through the calibration quarter.

## 3.3 Initial Conditions

This section describes the initial conditions for all sectors. Table 6 reports per-agent values and Table 7 reports sector-level aggregates, both for 2010Q1.

### 3.3.1 Firms

The size distribution of firms in industrialised economies is highly skewed, with many small firms and few large ones (Axtell, 2001). Initial employment  $N_{i,0}$  for firm  $i$  in industry  $s$  is sampled from a power-law distribution with exponent  $-2$  (subject to  $\sum_{i \in I_s} N_{i,0} = N_s$  and  $N_{i,0} > 0$ ), following the parameterisation of Poledna et al. (2023).<sup>10</sup>

Initial production is determined by labour productivity:

$$Y_{i,0} = Q_{i,0}^d = \bar{\alpha}_i N_{i,0}. \quad (33)$$

The initial capital stock, where  $K_{i,0}$  denotes the physical capital stock of firm  $i$  (distinct from household equity holdings  $k_{h,t}$ ), is set to provide the firm's target reserve capacity:

$$K_{i,0} = \frac{Y_{i,0}}{\kappa_i \omega}. \quad (34)$$

Initial intermediate inputs are sized to meet expected usage plus a buffer stock:

$$M_{i,0} = \frac{Y_{i,0}}{\omega \beta_i^M}. \quad (35)$$

Cross-classification tables for financial assets by industry are not available at the NACE 64-sector level. We therefore allocate initial firm debt  $L_{i,0}$  in proportion to capital:

$$L_{i,0} = L^I \frac{K_{i,0}}{\sum_{j=1}^I K_{j,0}}, \quad (36)$$

<sup>10</sup>As we currently lack access to data on the firm-size distribution in the Netherlands, we follow Poledna et al. (2023), who calibrate to Austrian data. A sensitivity analysis using alternative distributional assumptions is left for future work.

Table 5: Model parameters

| Parameter                  | Description                                                                           | Value                                                | Source                                                             |
|----------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------|
| $G/S$                      | Number of products/industries                                                         | 62                                                   | census data,<br>business<br>demography<br>data                     |
| $H^{\text{act}}$           | Number of economically active persons                                                 | 7860391                                              |                                                                    |
| $H^{\text{inact}}$         | Number of economically inactive persons                                               | 3921200                                              |                                                                    |
| $J$                        | Number of government entities                                                         | 152820                                               |                                                                    |
| $L$                        | Number of foreign consumers                                                           | 305639                                               |                                                                    |
| $I$                        | Number of firms                                                                       | 1536974                                              |                                                                    |
| $\beta$                    | Household discount rate                                                               | 0.94                                                 | Internal<br>calibra-<br>tion                                       |
| $1/\sigma$                 | Intertemporal elasticity of substitution                                              | 0.7                                                  |                                                                    |
| $x_0$                      | Size of transaction cost of investment                                                | 0.1                                                  |                                                                    |
| $x_1$                      | Convexity of transaction cost of investment                                           | 1.5                                                  |                                                                    |
| $\bar{a}_i$                | Average productivity of labour of the $i^{\text{th}}$ firm                            | parameters are firm/sector<br>specific; see table 13 | input-output<br>tables                                             |
| $\kappa_i$                 | Productivity of capital of the $i^{\text{th}}$ firm                                   |                                                      |                                                                    |
| $\beta_i$                  | Productivity of intermediate consumption of the $i^{\text{th}}$ firm                  |                                                      |                                                                    |
| $\delta_i$                 | Depreciation rate for capital of the $i^{\text{th}}$ firm                             |                                                      |                                                                    |
| $\bar{w}_i$                | Average wage rate of firm $i$                                                         |                                                      |                                                                    |
| $a_{sg}$                   | Technology coefficient of the $g^{\text{th}}$ product in the $s^{\text{th}}$ industry |                                                      |                                                                    |
| $b_g^{\text{CF}}$          | Capital formation coefficient of the $g^{\text{th}}$ product (firm investment)        |                                                      |                                                                    |
| $b_g^{\text{CFH}}$         | Household investment coefficient of the $g^{\text{th}}$ product                       |                                                      |                                                                    |
| $b_g^{\text{HH}}$          | Consumption coefficient of the $g^{\text{th}}$ product of households                  |                                                      |                                                                    |
| $c_g^{\text{G}}$           | Consumption of the $g^{\text{th}}$ product of the government in mln. Euro             |                                                      |                                                                    |
| $c_g^{\text{E}}$           | Exports of the $g^{\text{th}}$ product in mln. Euro                                   |                                                      |                                                                    |
| $c_g^{\text{I}}$           | Imports of the $g^{\text{th}}$ product in mln. Euro                                   |                                                      |                                                                    |
| $\tau_i^{\text{Y}}$        | Net tax rate on products of the $i^{\text{th}}$ firm                                  |                                                      |                                                                    |
| $\tau_i^{\text{K}}$        | Net tax rate on production of the $i^{\text{th}}$ firm                                |                                                      |                                                                    |
| $\tau^{\text{INC}}$        | Income tax rate                                                                       | 0.1715                                               | government statistics,<br>sector accounts                          |
| $\tau^{\text{FIRM}}$       | Corporate tax rate                                                                    | 0.0805                                               |                                                                    |
| $\tau^{\text{VAT}}$        | Value-added tax rate                                                                  | 0.1381                                               |                                                                    |
| $\tau^{\text{SIF}}$        | Social insurance rate (employers' contributions)                                      | 0.2922                                               |                                                                    |
| $\tau^{\text{SIW}}$        | Social insurance rate (employees' contributions)                                      | 0.1103                                               |                                                                    |
| $\tau^{\text{EXPORT}}$     | Export tax rate                                                                       | 0.0024                                               |                                                                    |
| $\tau^{\text{CF}}$         | Tax rate on capital formation                                                         | 0.0969                                               |                                                                    |
| $\tau^{\text{G}}$          | Tax rate on government consumption                                                    | 0.0019                                               |                                                                    |
| $r^{\text{G}}$             | Interest rate on government bonds                                                     | 0.0057                                               |                                                                    |
| $\mu$                      | Risk premium on policy rate                                                           | 0.0243                                               |                                                                    |
| $\psi$                     | Fraction of income devoted to consumption                                             | 0.8487                                               |                                                                    |
| $\psi^{\text{H}}$          | Fraction of income devoted to investment in housing                                   | 0.0936                                               |                                                                    |
| $\theta^{\text{DIV}}$      | Dividend payout ratio                                                                 | 0.5354                                               |                                                                    |
| $\theta^{\text{UB}}$       | Unemployment benefit replacement rate                                                 | 0.4054                                               | Basel III, ECB<br>statutes,<br>banking<br>practices,<br>literature |
| $\theta$                   | Rate of instalment on debt                                                            | 0.05                                                 |                                                                    |
| $\zeta$                    | Banks' capital ratio                                                                  | 0.03                                                 |                                                                    |
| $\zeta^{\text{LTV}}$       | Loan-to-value (LTV) ratio                                                             | 0.6                                                  |                                                                    |
| $\zeta^{\text{b}}$         | Loan-to-capital ratio for new firms after bankruptcy                                  | 0.5                                                  |                                                                    |
| $\pi^*$                    | Inflation target of the monetary authority                                            | 0.005                                                | national accounts<br>(exogenously estimated)                       |
| $\alpha^{\text{G}}$        | Autoregressive coefficient for government consumption                                 | 0.9821                                               |                                                                    |
| $\beta^{\text{G}}$         | Scalar constant for government consumption                                            | 0.1943                                               |                                                                    |
| $\sigma^{\text{G}}$        | Standard deviation of government consumption                                          | 0.0095                                               |                                                                    |
| $\alpha^{\text{E}}$        | Autoregressive coefficient for exports                                                | 0.9747                                               |                                                                    |
| $\beta^{\text{E}}$         | Scalar constant for exports                                                           | 0.3029                                               |                                                                    |
| $\alpha^{\text{I}}$        | Autoregressive coefficient for imports                                                | 0.9664                                               |                                                                    |
| $\beta^{\text{I}}$         | Scalar constant for imports                                                           | 0.3963                                               |                                                                    |
| $\alpha^{\text{YEA}}$      | Autoregressive coefficient for euro area GDP                                          | 0.9650                                               |                                                                    |
| $\beta^{\text{YEA}}$       | Scalar constant for euro area GDP                                                     | 0.5152                                               |                                                                    |
| $\alpha^{\pi^{\text{EA}}}$ | Autoregressive coefficient for euro area inflation                                    | 0.3933                                               |                                                                    |
| $\beta^{\pi^{\text{EA}}}$  | Scalar constant for euro area inflation                                               | 0.0024                                               |                                                                    |
| $\sigma^{\pi^{\text{EA}}}$ | Standard deviation of euro area inflation                                             | 0.0023                                               |                                                                    |
| $\rho$                     | Adjustment coefficient of the policy rate                                             | 0.9287                                               |                                                                    |
| $r^*$                      | Real equilibrium interest rate                                                        | -0.0035                                              |                                                                    |
| $\zeta^{\pi}$              | Weight of the inflation target                                                        | 0.2854                                               |                                                                    |
| $\zeta^{\gamma}$           | Weight of economic growth                                                             | 1.3048                                               |                                                                    |
| $C$                        | Covariance matrix of euro area GDP and imports and exports                            |                                                      |                                                                    |

Note: Model parameters for the reference quarter 2014:Q1. Exogenous autoregressive coefficients and parameters of the Taylor rule are estimated over the sample 1997:Q1 to 2014:Q1. Note that the number of agents are not scaled here.

Table 6: Initial conditions (per-agent values)

| Initial condition     | Description                                                                   | Value     |
|-----------------------|-------------------------------------------------------------------------------|-----------|
| $P_{i,0}$             | Initial price of the $i^{\text{th}}$ firm                                     | 1.00      |
| $Y_{i,0}/Q_{i,0}^d$   | Initial production/demand of the $i^{\text{th}}$ firm (mln. Euro)             | 57,413.9  |
| $K_{i,0}$             | Initial capital of the $i^{\text{th}}$ firm (mln. Euro)                       | 6,849.5   |
| $M_{i,0}$             | Initial intermediate inputs of the $i^{\text{th}}$ firm (mln. Euro)           | 1,094.0   |
| $S_{i,0}$             | Initial finished goods inventory of the $i^{\text{th}}$ firm (mln. Euro)      | 0.00      |
| $N_{i,0}$             | Initial number of employees of the $i^{\text{th}}$ firm                       | 4.59      |
| $D_{i,0}$             | Initial deposits of the $i^{\text{th}}$ firm (mln. Euro)                      | 176,303.0 |
| $L_{i,0}$             | Initial debt of the $i^{\text{th}}$ firm (mln. Euro)                          | 782,191.0 |
| $\Pi_{i,0}$           | Initial profit of the $i^{\text{th}}$ firm (mln. Euro)                        | 67.21     |
| $b_{h,0}$             | Initial liquid assets (deposits) of the $h^{\text{th}}$ household (mln. Euro) | 367,679.0 |
| $k_{h,0}$             | Initial illiquid capital of the $h^{\text{th}}$ household (mln. Euro)         | 925,403.0 |
| $w_{h,0}$             | Initial wage of the $h^{\text{th}}$ household (mln. Euro)                     | 29.77     |
| $sb_0^{\text{inact}}$ | Initial pension/social benefits (mln. Euro)                                   | 12.69     |
| $sb_0^{\text{other}}$ | Initial other social benefits (mln. Euro)                                     | 3.57      |
| $L_0^G$               | Initial government debt (mln. Euro)                                           | 359,255.0 |
| $\Pi_0^{\text{bank}}$ | Initial bank profit (mln. Euro)                                               | 26,733.0  |
| $E_0^{\text{bank}}$   | Initial bank equity (mln. Euro)                                               | 105,485.0 |
| $E_0^{\text{CB}}$     | Initial central bank equity (mln. Euro)                                       | 491,979.0 |
| $D_0^{\text{RoW}}$    | Initial net position with rest of world (mln. Euro)                           | 0.00      |

Initial conditions for 2010Q1. At 1:100 scaling, each model agent represents 100 real-world counterparts; reported values are per model agent (i.e.,  $100 \times$  the real per-entity value). Per-firm values ( $K$ ,  $M$ ,  $S$ ,  $N$ ,  $\Pi$ ) represent mean values computed from model initialisation with random seed 123. Per-household  $k_{h,0}$  is the value before microdata-based resampling (Section 3.4.2); see Table 7 for the sector total  $K^H$ .

Table 7: Initial conditions for institutional sectors (2010Q1)

| Symbol   | Description                                | Value   |
|----------|--------------------------------------------|---------|
| $D^I$    | Firm sector deposits (mln. Euro)           | 176,303 |
| $L^I$    | Firm sector debt (mln. Euro)               | 782,191 |
| $\omega$ | Desired capacity utilisation rate          | 0.85    |
| $D^H$    | Household sector liquid assets (mln. Euro) | 367,679 |
| $K^H$    | Household sector capital (mln. Euro)       | 503,735 |

Initial conditions for 2010Q1.

where  $L^I$  is total firm debt from financial balance sheets (nasa\_10\_f\_bs, loans F.4, non-financial corporations S.11, non-consolidated liabilities). Initial deposits  $D^I$  are also from financial balance sheets (nasa\_10\_f\_bs, deposits F.2, S.11) and are distributed according to each firm's share of operating surplus:

$$D_{i,0} = D^I \frac{\max(\bar{\pi}_i Y_{i,0}, 0)}{\sum_{j=1}^I \max(\bar{\pi}_j Y_{j,0}, 0)}, \quad (37)$$

where the operating margin is  $\bar{\pi}_i = 1 - (1 + \tau^{\text{SIF}}) \bar{w}_i / \bar{\alpha}_i - \delta_i / \kappa_i - 1 / \beta_i^M - \tau_i^{\text{PROD}} - \tau_i^Y$ .<sup>11</sup> Initial profits are:

$$\Pi_{i,0} = \bar{\pi}_i Y_{i,0} - \bar{r}_0 L_{i,0} + r_0 D_{i,0}, \quad (38)$$

where  $r_0$  is the initial policy rate and  $\bar{r}_0 = r_0 + \mu$  is the initial lending rate including the risk

<sup>11</sup>Here  $\tau_i^{\text{PROD}}$  is the net tax rate on production for industry  $i$ , obtained from input-output tables, distinct from the capital tax rate  $\tau^K$  on illiquid asset returns in the household budget constraint (Section 2.2.2).

premium. Initial inventories of finished goods  $S_{i,0}$  are set to zero (no reliable data available) and initial prices  $P_{i,0}$  are normalised to one.

### 3.3.2 Households

We calibrate household initial conditions in two steps. First, following Poledna et al. (2023), we set aggregate positions from national accounts and disaggregate to the household level using income shares as a proxy for wealth. Second, we resample household wealth positions from the CBS microdata-based empirical distribution (Section 3.4.2).

Initial liquid assets  $b_{h,0}$  are sourced from financial balance sheets (nasa\_10\_f\_bs, deposits F.2, household and NPISH sector S14\_S15) and disaggregated by income share:

$$b_{h,0} = D^H \frac{Y_{h,0}}{\sum_{j=1}^H Y_{j,0}}, \quad (39)$$

where  $Y_{h,0}$  follows equation (4). Initial household capital  $k_{h,0}$  is set to the value of dwellings (N111N) from balance sheets for non-financial assets (nama\_10\_nfa\_st), disaggregated analogously:

$$k_{h,0} = K^H \frac{Y_{h,0}}{\sum_{j=1}^H Y_{j,0}}. \quad (40)$$

Initial wages are:

$$w_{h,0} = \begin{cases} \bar{w}_i & \text{if employed by firm } i, \\ w^{\text{UB}} / \theta^{\text{UB}} & \text{if unemployed,} \end{cases} \quad (41)$$

where  $\theta^{\text{UB}}$  is the unemployment benefit replacement rate (Table 5). Initial unemployment benefits  $w^{\text{UB}}$  are computed as total unemployment payments (GF.1005, gov\_10a\_exp) divided by the number of unemployed persons (cens\_11an\_r2).

Government transfers to households in the form of social benefits (D.62) are allocated to household types based on employment status using COFOG-classified government expenditure data (gov\_10a\_exp). All social benefits  $sb_0^{\text{inact}}$  and  $sb_0^{\text{other}}$  are distributed in equal proportion across household types, ensuring individual flows sum to total macroeconomic flows. Net transfers other than consumption, savings, taxes, and subsidies, comprising property and interest income (D.4), other current transfers (D.7), adjustments for changes in pension entitlements (D.8), and capital transfers other than capital taxes (D.9 – D.91), are treated as a net transfer from the government to the household sector.

### 3.3.3 Government, Bank, and Central Bank

Initial government debt  $L_0^G$  equals consolidated gross debt (GD) of the Dutch government (S.13) from the government deficit/surplus dataset (gov\_10dd\_edpt1).

Initial bank equity  $E_0^{\text{bank}}$  is sourced from financial balance sheets (nasa\_10\_f\_bs, equity F.5 and financial net worth BF.90, monetary financial institutions excl. the central bank, S122\_S123). Initial bank profit is:

$$\Pi_0^{\text{bank}} = \bar{r}_0 \sum_{j=1}^I L_{j,0} - r_0 \left( \sum_{j=1}^I D_{j,0} + \sum_{j=1}^H b_{j,0} \right), \quad (42)$$

where initial advances from the central bank  $D_0^{\text{bank}}$  are determined according to equation (97).

Initial central bank equity  $E_0^{\text{CB}}$  is the residual on the central bank’s balance sheet: assets (government debt  $L_0^G$ ) minus liabilities (bank reserves  $D_0^{\text{bank}}$  and the net position with the rest of the world  $D_0^{\text{RoW}}$ ), computed according to equation (105). The initial balance of trade with the rest of the world is assumed to be zero.

### 3.4 Household Sector Calibration

The national-accounts-based disaggregation described above assigns wealth in proportion to income, which cannot capture the rich joint distribution of income, liquid wealth, and illiquid wealth that Kaplan et al. (2014) and Fagereng et al. (2021) show is critical for realistic MPC heterogeneity. This section describes how we use CBS microdata to overcome this limitation.

#### 3.4.1 Liquid and Illiquid Asset Classification

We map the diverse household balance sheets in the CBS microdata to the two asset categories of the model. Following Statistics Netherlands conventions, we classify net liquid assets  $b_{h,t}$  as all deposits in checking and savings accounts (VEHW1111BANH). Illiquid assets  $k_{h,t}$  are defined as the difference between net wealth (VEHW1000VERH) and liquid assets, comprising real estate wealth net of mortgage debt, financial assets including government bonds, corporate and private equity, and remaining properties.

#### 3.4.2 Empirical Distribution

We estimate the empirical joint density of household income, liquid wealth, and illiquid wealth using CBS microdata. These three dimensions correspond directly to the state variables in the household optimisation problem: wages  $w_{h,t}$ , deposits  $b_{h,t}$ , and capital  $k_{h,t}$ . The empirical distribution serves two purposes: initialising the model so that the simulated agent distribution matches observed data, and providing the sampling ranges for neural network training (Section 2.3).

We combine three datasets by linking them using personal identification numbers: personal income (INPATAB), household income (INHATAB), and household wealth (VEHTAB), for the years 2011–2019; for 2010, we use the 2011 distribution as earlier data are unavailable. Household income  $Y_{h,t}$  is constructed from personal income data and aggregated to the household level. Due to data privacy regulations, we apply a cutoff to the upper tail to prevent inference about the highest incomes when extracting data from the CBS servers.

We define bin edges  $X_j$  for  $j \in \{1, \dots, 51\}$ , yielding 50 bins per dimension and  $50^3 = 125,000$  three-dimensional bins. With approximately 170,000 model households, many bins are sparsely populated or empty; we treat empty bins as having zero probability, so households are never sampled into unobserved wealth–income combinations. Increasing the number of bins would exacerbate sparsity without improving density estimation. We compute the three-dimensional histogram and normalise it to obtain the empirical joint density  $f(w, k, b)$ .

**Conditional Sampling Procedure.** The labour market module inherited from Poledna et al. (2023) assigns wages based on sectoral affiliation, which precludes substantial within-sector wage heterogeneity. To reconcile this with the empirical wealth distribution, we employ a two-step conditional sampling procedure:

1. **Wage assignment:** Household  $h$  receives wage  $w_{h,0}$  from the sectoral wage structure of Poledna et al. (2023): employed households receive the firm-specific wage  $\bar{w}_i$ ; unemployed households receive unemployment benefits.
2. **Conditional wealth draw:** Given  $w_{h,0}$ , we draw capital and deposit holdings from the conditional distribution:

$$(k_{h,0}, b_{h,0}) \sim f(k, b \mid w = w_{h,0}), \quad (43)$$

where the conditional density is computed from the empirical joint histogram by summing over the  $k$  and  $b$  bins for the wage bin containing  $w_{h,0}$ :

$$f(k, b \mid w) = \frac{f(w, k, b)}{\sum_{k'} \sum_{b'} f(w, k', b')}. \quad (44)$$

This preserves the empirical correlation between income and wealth: high-wage households are more likely to hold larger capital and deposit positions, consistent with the positive income–wealth correlation in the microdata. In practice, we identify the wage bin containing  $w_{h,0}$  and sample  $(k_{h,0}, b_{h,0})$  from the corresponding two-dimensional slice of the histogram. If a wage falls outside the observed range, we use the nearest boundary bin. The procedure is applied separately for each quarterly period using the corresponding year’s microdata.

Generalising this approach to allow within-sector wage heterogeneity, sampling from the full joint distribution  $f(w, k, b)$  rather than conditioning on pre-assigned wages, would require modifications to the labour market module, which is beyond the scope of this paper.

### 3.4.3 Calibration of the Household Policy Functions

The neural network solution method described in Section 2.3 produces household policy functions parameterised by the preference vector  $\Gamma$ . This section reports the calibration results: the estimated preference parameters and the model’s ability to replicate target moments.

Figure 5 presents the evolution of calibrated preference parameters across the 40 quarterly periods (2010Q1–2019Q4) for each of the four model variants described in Section 2.3 (baseline, bond liquidity, capital liquidity, and full liquidity). The core parameters, namely risk aversion  $\sigma$ , discount factor  $\beta$ , subsistence consumption  $c^*$ , and the Stone–Geary intercept  $c_{\text{int}}$ , exhibit reasonable stability across time, suggesting that household preferences are well-identified by the moment conditions.

The risk aversion parameter  $\sigma$  varies around 7.5 for most specifications, which is at the upper end of the bias-corrected range of 2–7 reported by the meta-analysis of Elminejad et al. (2025). The discount factor  $\beta$  fluctuates around 0.75, which may appear low relative to the standard DSGE calibration of  $\beta \approx 0.99$  quarterly. However, this value is consistent with the quasi-hyperbolic discounting literature, which distinguishes between short-run and long-run discount factors.<sup>12</sup>

<sup>12</sup>In the  $\beta$ – $\delta$  framework of Laibson (1997), agents discount the immediate future more heavily than distant pe-

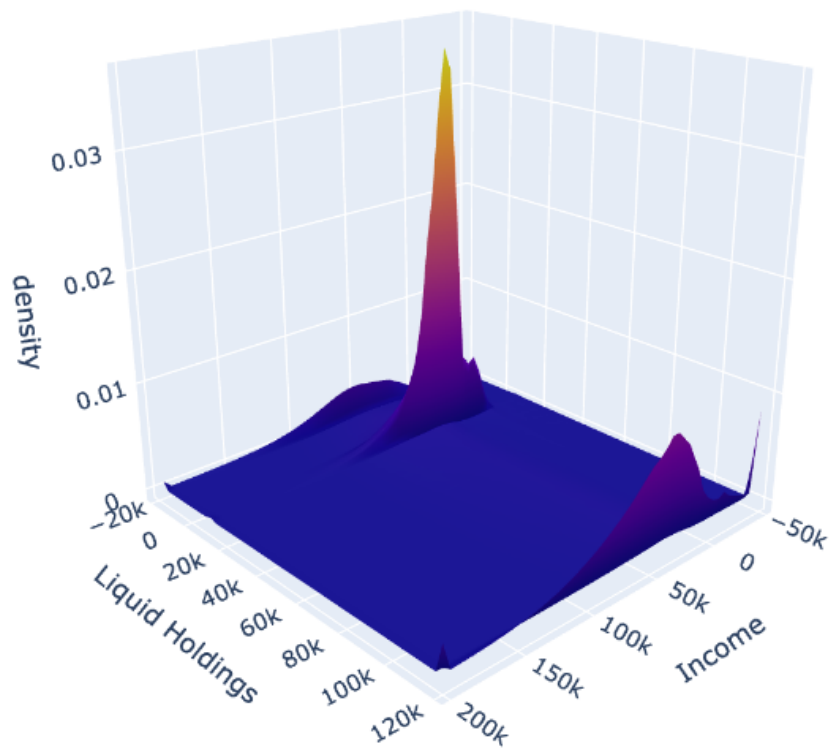


Figure 4: Empirical joint distribution of household income ( $w$ ), liquid wealth ( $b$ ), and illiquid wealth ( $k$ ), constructed from CBS microdata for 2011–2019. A cutoff is applied to the upper tail to comply with data privacy regulations. Conditional slices (fixing  $w$ ) are used to sample household wealth positions given assigned wages.



Figure 5: Evolution of calibrated preference parameters across quarters (2010Q1–2019Q4) for four model variants: baseline without liquidity preference (blue), bond liquidity preference (orange), capital liquidity preference (green), and full liquidity preference (purple). Parameters shown are risk aversion ( $\sigma$ ), discount factor ( $\beta$ ), subsistence consumption ( $c^*$ ), Stone-Geary intercept ( $c_{int}$ ), and liquidity preference parameters ( $\sigma_k, \sigma_b$ ).

The liquidity preference parameters  $\sigma_k$  and  $\sigma_b$  display more variation across specifications. In the capital liquidity variant,  $\sigma_k$  takes moderate values generating meaningful wealth-in-utility effects for illiquid capital; the bond liquidity variant yields similarly well-identified  $\sigma_b$  parameters. However, in the full liquidity specification, one of the two parameters typically hits its upper bound ( $\sigma = 10$ ), indicating that both liquidity channels cannot be separately identified when included simultaneously. This collinearity suggests that a single liquidity preference parameter suffices.

Figure 6 displays the model's ability to replicate the five target moments across specifications. The dashed lines indicate empirical targets:  $APC \approx 0.86$ ,  $MPC = 0.20$ ,  $MPC_k = 0.02$ ,  $MPC_b = 0.02$ , and the capital-to-deposits ratio from national accounts. The bond liquidity and capital liquidity specifications track these targets closely throughout the sample. The baseline and full liquidity specifications exhibit substantial deviations, particularly in the average propensity to consume.

The bottom-right panel of Figure 6 reports the sum of squared moment errors on a logarithmic scale. The capital liquidity and bond liquidity variants achieve errors two to four orders of magnitude smaller than the baseline and full liquidity specifications. This difference reflects two distinct failure modes. The baseline model lacks sufficient flexibility to simultaneously match the MPC hierarchy ( $MPC > MPC_b \approx MPC_k$ ) and the aggregate consumption rate. The full liquidity model, while flexible, suffers from identification problems: with two liquidity parameters competing to explain similar variation, the optimiser struggles to find a stable solution, producing the erratic APC behaviour visible in the top-left panel.

riods. Structural estimates from lifecycle models with liquid and illiquid assets find short-run discount factors between 0.50 and 0.85, combined with long-run factors near unity (Angeletos et al., 2001; Laibson et al., 2024). Since our Euler equation identifies only the composite one-period discount factor, our estimate of 0.75 is compatible with moderate present bias (e.g.,  $\beta \approx 0.76$ ,  $\delta \approx 0.99$ ). Meta-analyses of experimental studies report present-bias parameters averaging 0.68–0.94 depending on reward type (Frederick et al., 2002). The low estimated  $\beta$  is consistent with this literature but may also partly reflect model misspecification in other dimensions.

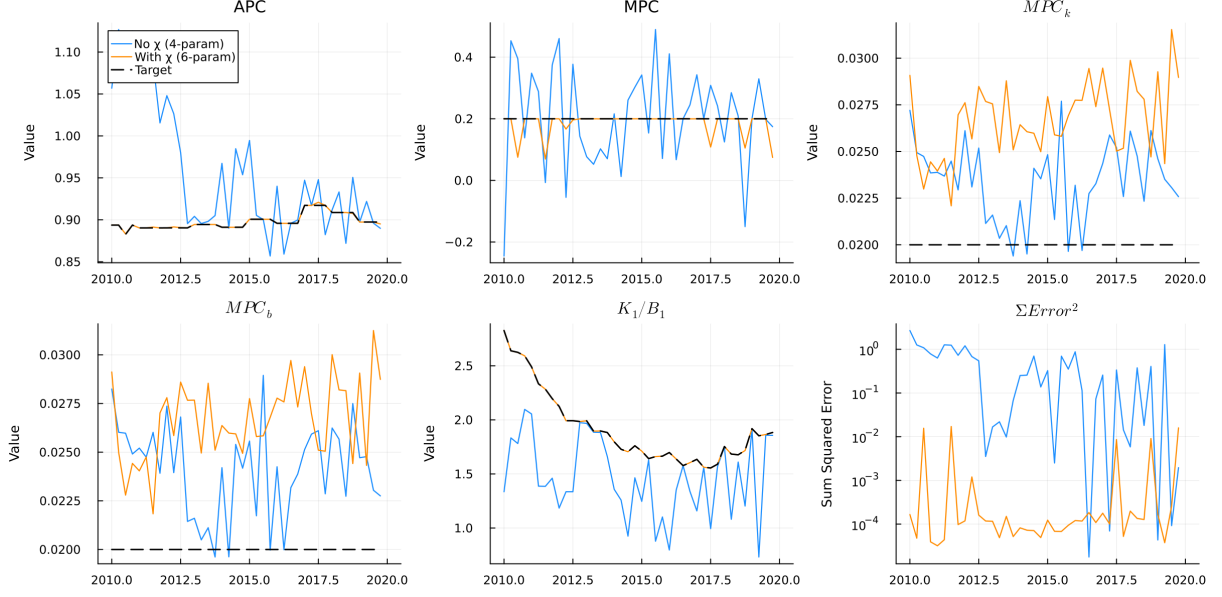


Figure 6: Moment achievement across quarters for four model variants. Dashed lines indicate empirical targets; coloured lines show model-implied values. Panels display average propensity to consume (APC), marginal propensity to consume out of income (MPC), marginal propensities with respect to capital ( $MPC_k$ ) and bonds ( $MPC_b$ ), the capital-to-bond ratio ( $K_1/B_1$ ), and the sum of squared moment errors on a logarithmic scale.

The calibration results support the inclusion of a single liquidity preference channel, either for capital or for bonds, rather than the baseline or the full two-parameter extension. Both intermediate specifications successfully replicate the empirical moment hierarchy while maintaining parameter stability. For the forecasting exercises in Section 5, we adopt the bond liquidity specification as our preferred model, as it achieves the lowest average calibration error across the sample period.

## 4 Heterogeneous MPCs and Intertemporal Responses

A central motivation for replacing the heuristic consumption block with optimising households is to generate empirically realistic heterogeneity in marginal propensities to consume. Section 3 documented that the OC-ABM matches the targeted cross-sectional means  $MPC_y = 0.20$  and  $MPC_w = 0.03$  from the DNB Household Survey (Christelis et al., 2019). Matching the mean is necessary but not sufficient: the calibrated mean may be achieved by a policy that lacks the buffer-stock shape across the state space, in which case the model would generate the right aggregate response to income shocks but the wrong dynamic path of consumption. We assess two dimensions of the policy beyond the mean MPC: the cross-sectional gradient across the empirical Dutch state distribution, and the five-year intertemporal MPC (iMPC) profile under a transitory income shock. Both are computed on the OC-ABM and on the heuristic ladder using identical methodology, allowing direct comparison.

### 4.1 Cross-sectional MPC heterogeneity

We measure  $MPC_y$ ,  $MPC_b$ , and  $MPC_k$  at each of the 15,976 ABM households using finite differences with  $\delta = 0.01$  (matching the calibration convention). For the OC-ABM, the policy is the calibrated 21-input belief network at  $\theta^* = (\beta, \gamma, \psi, c_{\text{int}}, \chi_b^0, \chi_b^1) = (0.981, 6.10, 1.50, 0.164, 0.136, 1.14)$ ,

with the values read from the `calibrated_params` block stored alongside the trained network weights. For each heuristic version we use the parameters calibrated to the same DNB MPC targets and aggregate APC at the same forecast origin. All measurements are taken at the empirical Dutch joint distribution  $f(y, k, b)$  from CBS microdata, not on a synthetic grid.

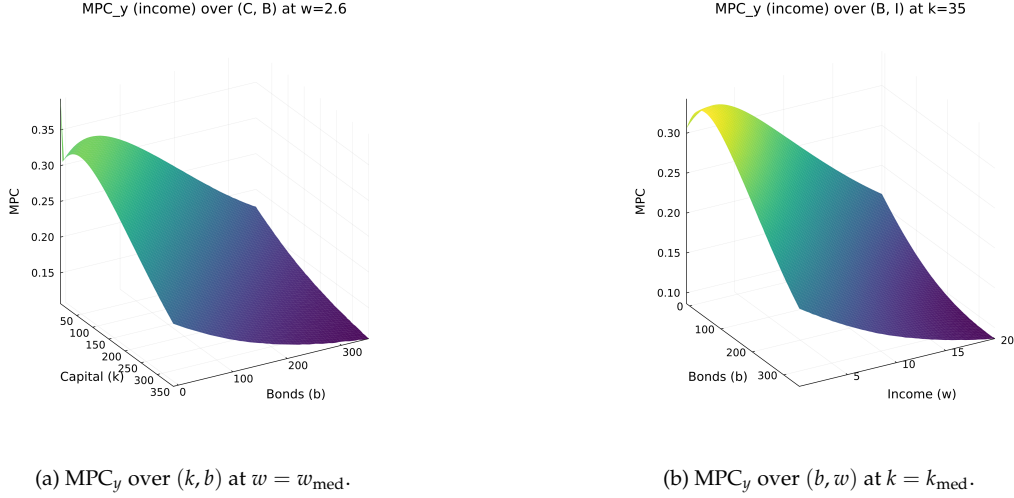


Figure 7: OC-ABM income MPC over the household state space. Surfaces are evaluated at the calibrated  $\theta^*$  on a  $60 \times 60$  grid spanning the empirical Dutch range. MPC<sub>y</sub> declines smoothly with each wealth dimension, peaking near 0.65 at the borrowing constraint and falling to  $\sim 0.15$  for wealth-rich households — the canonical buffer-stock signature (Carroll, 1997).

Figure 7 displays MPC<sub>y</sub> over two state slices. The surfaces are concave and monotone decreasing in both bond and capital holdings, with the steepest gradient at low liquid wealth. At the borrowing-constraint corner MPC<sub>y</sub>  $\approx 0.32$ ; at the upper tail of the wealth distribution MPC<sub>y</sub> falls to  $\approx 0.10$ . This is the policy shape predicted by buffer-stock theory: liquidity-constrained households consume nearly the full marginal unit of income, while wealth-rich households smooth across periods using their stock of assets. The cross-sectional mean over the empirical distribution is  $\overline{\text{MPC}}_y = 0.314$  with standard deviation 0.081. The mean exceeds the DNB target of 0.20 by approximately 50 per cent; the calibration loss recorded with the network weights is 0.020, indicating that the Stage 2 optimiser stopped short of fully matching the targeted moment. We treat the calibrated network as the policy of record and report the realised cross-section rather than rescaling to the target, since the policy is what propagates shocks through the forecasting exercise.<sup>13</sup>

Figures 8 and 9 display continuous MPC profiles for the full heuristic ladder against the OC-ABM, separated for legibility into static-and-habit rules (H1–H5) and dynamic and channel-equalised rules (H6–H10). Each panel reports a partial derivative of consumption with respect to one of the three household state variables (income  $y$ , capital  $k$ , bonds  $b$ ), holding the other two at their empirical sample medians. The OC-ABM exhibits the canonical buffer-stock signature: MPC<sub>y</sub> declines monotonically with income (top-left panels), with the sharpest gradient at the borrowing-constraint corner ( $b \approx -\underline{b}$ , top-right panels), and MPC<sub>k</sub> and MPC<sub>b</sub> remain small relative to MPC<sub>y</sub> as the calibration prescribes (middle and bottom rows).

<sup>13</sup>We report  $\delta = 0.01$  MPCs throughout this section for consistency with the cross-sectional MPC literature, which uses small perturbations to identify local derivatives (Fagereng et al., 2021; Kaplan et al., 2014).

Continuous MPC profiles (H1-H5 + OC-ABM 30in)

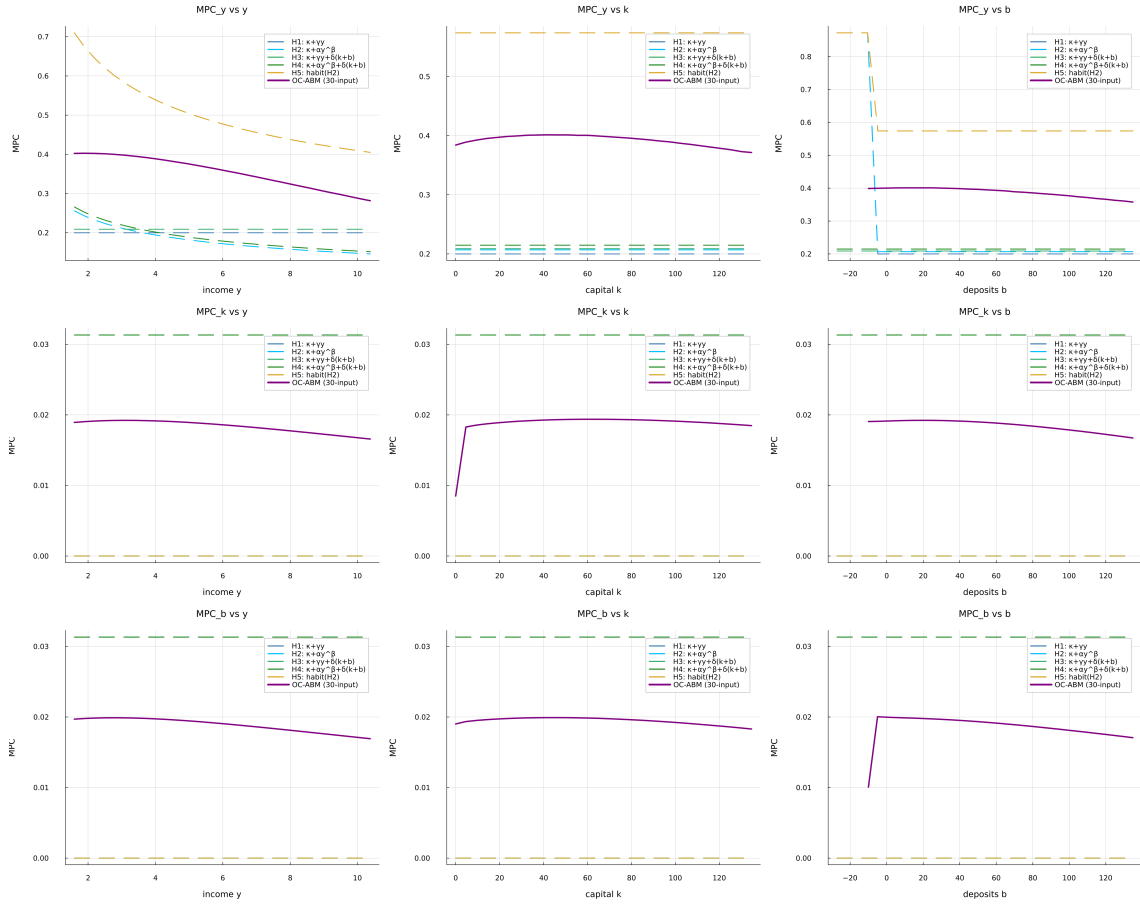


Figure 8: Continuous MPC profiles for the static and habit-extended heuristics (H1–H5) against the OC-ABM. Rows:  $MPC_y$ ,  $MPC_k$ ,  $MPC_b$ . Columns: dependence on income  $y$ , capital  $k$ , and bonds  $b$ , with the remaining two states held at empirical sample medians. The static linear rules (H1, H3) produce flat MPC profiles in every dimension by construction. The concave-income rule H2 generates a monotone  $MPC_y(y)$  gradient that overshoots the OC-ABM at low income. None of H1–H5 reproduces the borrowing-constraint kink in  $MPC_y(b)$  at  $b \approx -\underline{b}$ .

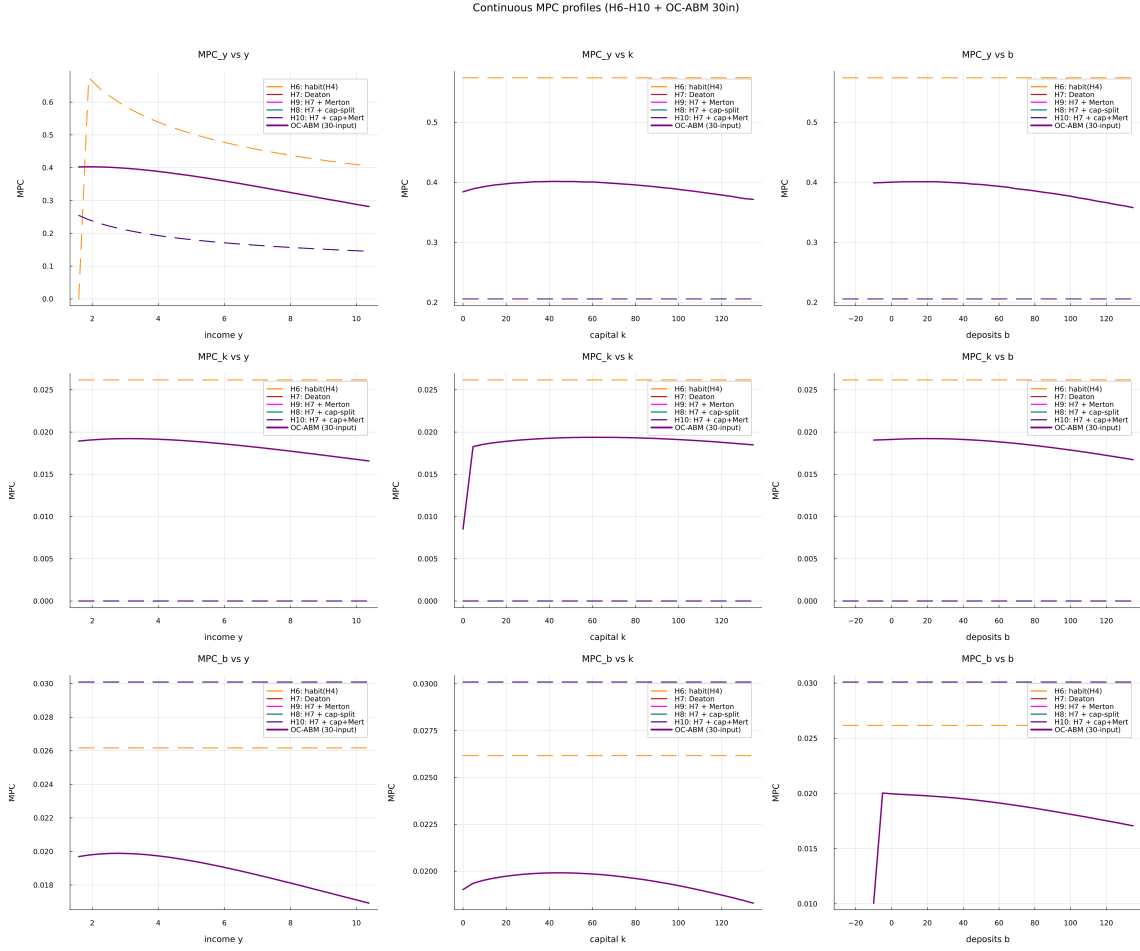


Figure 9: Continuous MPC profiles for the dynamic and channel-equalised heuristics (H6–H10) against the OC-ABM. Conventions as in Figure 8. The Deaton buffer-stock rule H7 and the Merton-portfolio variant H9 produce essentially flat  $MPC_y(y)$  profiles, whereas the capital-return cap-split variants H8 and H10 reproduce the OC-ABM’s concave income-MPC gradient. None of the heuristic variants reproduces the borrowing-constraint kink in  $MPC_y(b)$ .

The cross-sectional comparisons reveal which structural features reproduce which features of the OC-ABM policy. The static linear rules H1 and H3 are flat in income by construction. The concave-income rules H2 and H4 are concave in  $y$  but inherit a steeper-than-OC-ABM gradient at low income because the calibrated MPC moment is matched on the mean rather than on the gradient. The habit rules H5 and H6 attenuate the income response without reshaping it; the Deaton buffer-stock H7 and the Merton-portfolio variant H9 likewise leave the  $MPC_y(y)$  profile essentially flat. The two heuristics that partition the income aggregate into labour and capital components — H8 (Christelis cap-split alone) and H10 (cap-split combined with Merton portfolio) — generate a monotone declining  $MPC_y(y)$  gradient that closely tracks the OC-ABM over the empirical income range. This is a structural identification result: closing the capital-return channel through cap-split is the heuristic operation that reproduces the OC-ABM’s income MPC shape, while the Merton portfolio rule on its own does not. The remaining feature that no heuristic reproduces, however, is the borrowing-constraint kink in  $MPC_y(b)$  at the lower bound of the bond distribution: even the channel-equalised heuristic H10 is smooth across  $b = -\underline{b}$ , whereas the OC-ABM exhibits the sharp jump expected when the constraint binds. This is the policy-shape signature of Euler-equation discipline, and it cannot be replicated by a reduced-form rule.

## 4.2 Intertemporal MPC profile

The intertemporal MPC matrix  $M_{t,s} = \partial c_t / \partial y_s$  (Auclert et al., 2024) characterises the dynamic path of consumption following a one-time, transitory income shock at date  $s$ . We focus on the column  $M_{t,0}$  for  $t = 1, \dots, 20$  quarters, the partial-equilibrium response of a household to a unit shock at  $t = 1$  that returns to baseline at  $t \geq 2$ . For each model, we shock every household's income by  $\delta = 0.1$  at  $t = 1$ , propagate the state  $(k, b)$  forward using the model's own policy function, and record  $\Delta c_t / \delta$  averaged across the empirical distribution. The shocked path differs from the baseline path only via accumulated assets after the impact period.

The canonical empirical benchmark for the small-shock iMPC is the Italian Survey of Consumer Expectations (ISCE) hypothetical-lottery question analysed by Jappelli et al. (2024). Their 1,000 shock corresponds to roughly 2% of mean quarterly income — the same order of magnitude as our  $\delta = 0.1$  in normalised NN units. The ISCE iMPC declines from 0.265 at year one to  $\sim 0.012$  at year five; the Norwegian lottery study of Fagereng et al. (2021) reports 0.50, 0.18, 0.10, 0.05, 0.03 for substantially larger windfalls. Table 3 also reports HANK simulation iMPCs from Jappelli et al. (2024) for comparison: a one-asset HANK model with precautionary saving and a two-asset HANK model in the spirit of Kaplan et al. (2018).

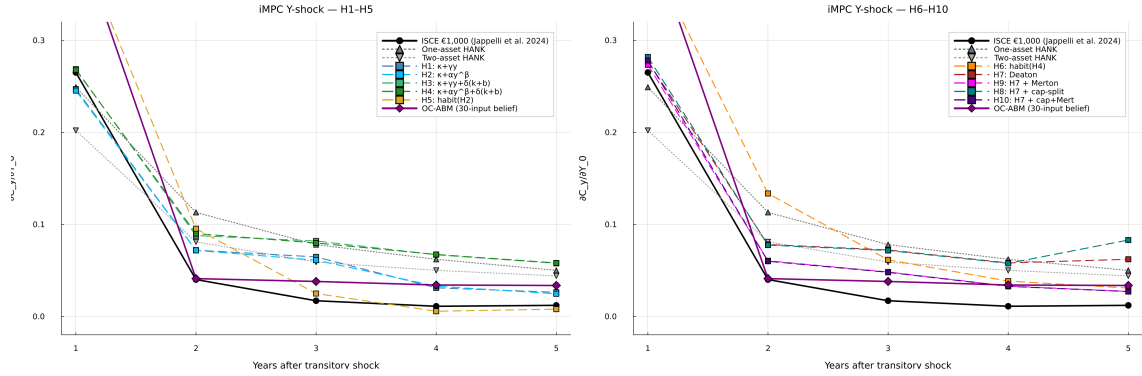


Figure 10: Five-year annual iMPC under transitory income shock  $\delta = 0.1$ . Left panel: OC-ABM (magenta, diamonds) against the static and habit heuristics H1–H5 (dashed), the canonical small-shock empirical benchmark ISCE 1,000 (black circles), and the one- and two-asset HANK reference profiles (grey, dotted). Right panel: OC-ABM against the dynamic and channel-equalised heuristics H6–H10, with the same empirical and HANK references. Years 2–5 are aggregated from quarterly iMPCs by summing within calendar years. The OC-ABM impact iMPC lies inside the HANK  $\{0.20, 0.25\}$  envelope and reproduces the level and slope of the empirical tail more closely than any heuristic variant.

Figure 10 shows the five-year iMPC profile for the OC-ABM, the full H1–H10 heuristic ladder, the ISCE empirical benchmark, and the HANK reference profiles. The static-and-habit panel (left, H1–H5) and the dynamic-and-channel-equalised panel (right, H6–H10) reveal qualitatively different failure modes. Among H1–H5, the linear-income rule H1 hits the impact MPC because the rule has the right slope at the cross-sectional mean but produces a flat tail because consumption is unresponsive to accumulated savings. H2 overshoots the impact through its steeper concave gradient. H3 and H4 generate tails that are too high and too persistent: each unit of saved windfall produces a quasi-permanent boost to consumption through the additive  $\delta(k + b)$  term. H5 attenuates the impact through the habit term but inherits the absent wealth dynamics of H2. Among H6–H10, the buffer-stock variants (H7, H8, H9, H10) all match the OC-ABM impact within a few percentage points but flatten too slowly in the tail, plateauing at iMPC  $\approx 0.05$ – $0.07$  rather than continuing to decay. H6 (habit on H4) overshoots the impact through the habit term and then crosses the empirical curve from above, decaying more

rapidly than the buffer-stock variants. Closing the capital-return channel (H8) or the portfolio channel (H9), individually or jointly (H10), shifts the tail level only marginally relative to H7 and does not bring the tail toward the ISCE benchmark.

The OC-ABM matches the impact iMPC closely and produces the lowest tail among the model class. The under-prediction of the deep ISCE tail beyond year three is a manifestation of the well-known missing intertemporal MPCs puzzle (Auclert et al., 2024; Kaplan and Violante, 2022), in which calibrated HANK models with realistic mean MPCs struggle to generate the sharp decay seen in small-shock lottery data. This puzzle is shared with the one-asset and two-asset HANK references in Figure 10 and remains an open question for the OC-ABM. The channel-equalised heuristic H10 does not resolve the puzzle either, which suggests that the missing-iMPC mechanism operates through a feature of household behaviour that is absent both from optimising and from reduced-form rules in our specification — candidate mechanisms include life-cycle effects, deeper-than- $b$  liquidity constraints, and discount-rate heterogeneity, none of which are present in any model variant we evaluate.

| Model                                         | Y1           | Y2           | Y3           | Y4           | Y5           | MSE vs ISCE    |
|-----------------------------------------------|--------------|--------------|--------------|--------------|--------------|----------------|
| ISCE 1,000 (Jappelli et al., 2024)            | 0.265        | 0.040        | 0.017        | 0.011        | 0.012        | —              |
| One-asset HANK (Jappelli et al., 2024)        | 0.249        | 0.113        | 0.078        | 0.062        | 0.050        | 0.00267        |
| Two-asset HANK (Jappelli et al., 2024)        | 0.202        | 0.081        | 0.059        | 0.050        | 0.044        | 0.00199        |
| <b>OC-ABM (this paper)</b>                    | <b>0.364</b> | <b>0.053</b> | <b>0.048</b> | <b>0.045</b> | <b>0.042</b> | <b>0.00251</b> |
| H1: $\kappa + \gamma y$                       | 0.247        | 0.072        | 0.065        | 0.031        | 0.026        | 0.00085        |
| H4: $\kappa + \alpha y^\beta + \delta(k + b)$ | 0.268        | 0.090        | 0.080        | 0.067        | 0.058        | 0.00255        |
| H6: habit(H4)                                 | 0.397        | 0.134        | 0.061        | 0.038        | 0.031        | 0.00583        |
| H7: Deaton buffer-stock                       | 0.277        | 0.078        | 0.072        | 0.058        | 0.062        | 0.00226        |

Table 8: Annual iMPC for OC-ABM and a representative subset of the heuristic ladder (H1, H4, H6, H7), evaluated under a transitory  $\delta = 0.1$  shock at  $t = 1$  with state propagation through each model’s own policy. MSE is computed against the ISCE 1,000 empirical benchmark over years 1–5. The OC-ABM achieves the lowest MSE of any structural model, including the HANK reference profiles, by matching both the impact MPC and the slow tail. The remaining heuristics H2, H3, H5, H8, H9, H10 follow qualitatively similar profiles to their nearest tabulated neighbour (H2/H3/H5 close to H4; H8/H9/H10 close to H7) and are shown in Figure 10.

Table 8 ranks the models by MSE against the ISCE benchmark. The OC-ABM achieves  $\text{MSE} = 2.5 \times 10^{-3}$ , on a par with the two-asset HANK reference ( $1.99 \times 10^{-3}$ ) and below the one-asset HANK reference ( $2.67 \times 10^{-3}$ ) but above the linear-income heuristic H1 ( $8.5 \times 10^{-4}$ ). H1 attains the lowest MSE in mechanical terms because the empirical ISCE tail collapses toward zero and the rule produces an artificially flat decay; this happens to track the data well over years 2–5 but at the cost of a tail driven by the additive consumption cap rather than by wealth-effect dynamics, leaving the rule unable to propagate shocks through the wealth distribution. The OC-ABM combination of a borrowing constraint that delivers the constrained-household impact response, capital adjustment costs that slow rebalancing toward illiquid assets, and Euler-equation discipline that smooths the tail produces a profile shape that no individual structural channel reproduces alone, even when the level overshoots at impact.

### 4.3 Discussion

The cross-sectional and intertemporal MPC diagnostics together substantiate the forecasting argument advanced in the Discussion and refine the channel decomposition reported in Section 5.3. The static heuristics H1–H4 hit the cross-sectional mean MPC by construction but produce flat or over-concave MPC gradients across the wealth distribution and tails with the wrong slope. They forecast the aggregate consumption mean well at short horizons but cannot reproduce the dynamic propagation of shocks because the policy lacks the heterogeneity that channels shocks toward constrained households. The Deaton buffer-stock H7 and the Merton-portfolio H9 generate partial cross-sectional heterogeneity but cannot match the iMPC tail because their wealth correction is static rather than tracking accumulated wealth deviations through the continuation value. The cap-split variants H8 and H10 reproduce the OC-ABM’s concave income MPC gradient over the empirical income range, but neither generates the borrowing-constraint kink in  $MPC_y(b)$ , and neither closes the iMPC tail gap relative to the OC-ABM. The OC-ABM matches the mean MPC, the cross-sectional gradient (including the borrowing-constraint kink), and the iMPC tail through the same set of preference parameters — a multi-moment match that no reduced-form rule reproduces. This is the reason the consumption forecast gain documented in Section 5 is concentrated at long horizons where shocks have had time to redistribute across the wealth distribution, and the reason the OC-ABM remains the only variant with statistically significant gains under every expectation mechanism even after the portfolio and capital-return channels are closed in H8–H10.

Three caveats apply. First, the Stage 2 calibration of the belief network used here ends with  $\overline{MPC}_y = 0.31$  rather than the targeted 0.20, with an objective loss of 0.020. The cross-sectional gradient is shaped correctly, but the level is offset upward. A tighter calibration — additional Stage 2 epochs or a moment-weight rebalance — would close the gap without affecting the qualitative results in this section. Second, the OC-ABM under-predicts the deep ISCE tail at years three through five, with iMPC plateauing near 0.05 rather than decaying to 0.01. This is the missing intertemporal MPCs puzzle and is shared with both HANK reference calibrations; it suggests that the present preference specification does not capture whatever feature of households generates the steep empirical tail. Candidate mechanisms include life-cycle effects, deeper liquidity constraints than the single  $\bar{b}$  floor, and discount-rate heterogeneity (Krusell and Smith, 1998; Kaplan and Violante, 2022), each of which is left to future work. Third, the iMPC comparison is partial-equilibrium: the shock does not propagate through general-equilibrium income or wealth feedbacks. This is consistent with the empirical iMPC definition (Auclert et al., 2024) but understates the cumulative aggregate consumption response that the OC-ABM generates inside the full ABM simulation, where income and prices respond endogenously.

## 5 Comparison of Forecast Results

Forecasting performance provides an external validity test for agent-based models: if a structurally richer model cannot outpredict parsimonious benchmarks, its additional complexity is not justified by the data. Conversely, forecast gains that concentrate in specific variables or horizons can reveal which structural features carry empirical content. We therefore evaluate whether the optimal-consumption household block introduced in Section 2.2.2 improves out-of-sample prediction, and at which horizons and for which variables the gains materialise.

We compare the OC-ABM, the base ABM of Poledna et al. (2023), and ten heuristic exten-

sions (H1-ABM to H10-ABM) against four time-series benchmarks: AR(1), ARX(1), VAR(1), and VARX(1). Each ABM variant is combined with all four aggregate expectation mechanisms (Section 2.6), yielding twelve model  $\times$  four mechanism combinations evaluated against the corresponding univariate and multivariate benchmarks. All ten heuristics share the OC-ABM's borrowing constraint (Section 2.4); H8–H10 additionally close the capital-return MPC channel (H8, H10) and the portfolio-choice channel (H9, H10), so cross-variant comparisons can be used to attribute the consumption forecast advantage to specific structural channels (Section 5.3).

We forecast real GDP, household consumption, investment, government consumption, exports, imports, Euro area real GDP and inflation, and the Euribor rate. GDP, household consumption, investment, government consumption, exports, imports, and Euro area GDP are entered as log levels. Inflation for the Netherlands and the Euro area is entered as the first difference of the log GDP deflator. The Euribor rate is entered as the quarterly rate. All time-series benchmarks are initially estimated over 1997Q1–2010Q1 and re-estimated each quarter through 2019Q4, so that the forecast origin advances with the data. Each ABM is calibrated to the corresponding reference quarter. In the ARX and VARX specifications, the 12-quarter-ahead paths of exogenous variables (imports, exports, government consumption, Euro area GDP growth, Euro area inflation, and the Euribor rate) are constructed as iterated forecasts from models estimated at the forecast origin, consistent with a genuine out-of-sample exercise; results are similar when treating these paths as known, though a systematic comparison is beyond the scope of this paper. We restrict the evaluation sample to the pre-COVID period 2010Q1–2019Q4 (40 reference quarters). We select the AR(1), ARX(1), VAR(1), and VARX(1) specifications because they minimise both the Akaike and Bayesian information criteria over the initial estimation sample.

Because ABM simulations are stochastic, we average over 100 Monte Carlo draws per reference quarter to obtain the point forecast. The 90% confidence interval displayed in the time-series figures spans the 5th and 95th percentiles of the simulation distribution. We measure accuracy by the root mean squared forecast error (RMSFE),

$$\text{RMSFE} = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{x}_t - x_t)^2}, \quad (45)$$

where  $\hat{x}_t$  is the forecast and  $x_t$  the realisation. We report percentage improvements in RMSFE relative to the benchmark,

$$\Delta\text{RMSFE} = \frac{\text{RMSFE}_{\text{benchmark}} - \text{RMSFE}_{\text{ABM}}}{\text{RMSFE}_{\text{benchmark}}} \times 100, \quad (46)$$

so that positive values indicate the ABM variant is more accurate. To test whether forecast accuracy differs significantly, we apply the modified Diebold–Mariano test of Harvey et al. (1997), which incorporates a small-sample correction. The test statistic is computed on the loss differential (difference between squared forecast errors) using Newey–West standard errors to account for serial correlation in overlapping multi-period forecasts (Diebold and Mariano, 1995). Because the ABM and time-series benchmarks belong to different model classes, they are non-nested and the standard Diebold–Mariano test is appropriate without the Clark–West (2006) size correction. Our primary family of interest comprises four tests of the same hypothesis (OC-ABM household consumption improvement at the 12-quarter horizon, one per expectation mechanism). Applying the Bonferroni correction for this family yields a corrected significance threshold of  $\alpha/4 = 0.0125$ ; since all four tests satisfy  $p < 0.01$ , the corrected  $p$ -value is at most 0.04, confirming significance at the 5% familywise level. Because the four tests

share the same model and sample, they are positively correlated, so Bonferroni is conservative and the true familywise error rate is lower.

Figure 11 summarises the OC-ABM’s 12-quarter RMSFE improvements across all four expectation mechanisms, measured relative to the AR(1) benchmark. Six patterns emerge. First, the OC-ABM delivers significant consumption gains under every mechanism, improving on the corresponding univariate benchmark by 9.9–21.8% ( $p < 0.01$  in each case), with the largest gains under mechanisms without exogenous predictors (AR: 21.8%, VAR: 18.6%). Second, GDP improvements of 11–15% are present under all mechanisms but are not uniformly significant; exact  $p$ -values are reported in Section 5.1 (AR) and Appendix F (ARX, VAR, VARX). Third, inflation forecasts show small negative differences from the benchmark, ranging from negligible under VARX ( $-0.1\%$ ) to a statistically significant  $-2.0\%$  under AR; this is expected because the OC-ABM modifies only the household block, while inflation is determined by the identical price-setting and labour market modules across all variants. Fourth, investment gains of 8–10% are robust and significant across all mechanisms, suggesting that the OC-ABM’s portfolio channel spills over to aggregate investment dynamics. Fifth, the static heuristics without dynamic structure (H1–H4) show large consumption losses (between  $-26\%$  and  $-308\%$  at 12 quarters) under AR expectations, while the dynamic specifications (H6–H10) deliver positive consumption point estimates ranging from  $+12.6\%$  (H8) to  $+31.6\%$  (H6); only the OC-ABM produces gains that are statistically significant under every mechanism. Sixth, the H8–H10 channel-decomposition heuristics isolate the source of the OC-ABM’s residual advantage over the buffer-stock baseline H7: closing the capital-return channel alone (H8) adds essentially zero consumption signal ( $+12.6$  vs  $+13.7\%$ ), whereas closing the portfolio-choice channel alone (H9) accounts for  $+11.7$  pp of additional consumption improvement, recovering the majority of the OC-ABM’s  $+21.8\%$  gain; the combined variant H10 ( $+21.2\%$ ) does not exceed H9, indicating that the two channels are substitutes rather than complements (Section 5.3). The remainder of this section presents the AR expectations case in detail (Section 5.1), examines robustness across mechanisms (Section 5.2), and decomposes the consumption gains by structural channel (Section 5.3); full horizon-by-horizon tables for ARX, VAR, and VARX expectations appear in Appendix F.

|             |        |           |            |            |             |
|-------------|--------|-----------|------------|------------|-------------|
| Euribor     | 0.20%  | -5.0%     | -5.0%      | -5.0%      | -5.0%       |
| Investment  | 13.27% | 8.8%      | 7.8%       | 8.2%       | 8.9%        |
| Consumption | 3.68%  | 21.7%     | 20.7%      | 18.5%      | 19.8%       |
| Inflation   | 0.60%  | -1.7%     | 0.0%       | 0.0%       | 0.0%        |
| GDP         | 3.47%  | 14.7%     | 10.4%      | 15.3%      | 12.4%       |
|             | AR(1)  | OCM<br>AR | OCM<br>ARX | OCM<br>VAR | OCM<br>VARX |

Figure 11: Twelve-quarter RMSFE of the OC-ABM relative to the AR(1) benchmark across four aggregate expectation mechanisms (AR, ARX, VAR, VARX). Positive values (green) indicate forecast improvement. Each cell averages over 40 reference quarters (2010Q1–2019Q4) and 100 Monte Carlo simulations per quarter.

### 5.1 AR expectations

Under AR expectations, aggregate variables follow univariate AR(1) processes updated each period. This is the simplest expectation mechanism and provides the cleanest test of whether the household optimisation block alone, without multivariate or exogenous information, improves forecasts.

Figure 12 displays the 12-quarter relative RMSFE for all nine ABM variants under AR expectations. The OC-ABM achieves a 21.8% consumption improvement ( $p < 0.01$ ) and a 14.6% GDP improvement (not significant,  $p = 0.21$ ). Among heuristic models, the habit-on-wealth-augmented rule H6-ABM produces the largest point estimate for consumption (+31.6%, marginally significant at  $p = 0.06$ ) and the largest GDP point estimate (+17.1%, not significant), and the buffer-stock rule H7-ABM yields a +13.7% consumption gain (not significant). The static income-only and income-plus-wealth heuristics (H1–H4) show large consumption losses ranging from  $-25.9\%$  for H3 to  $-307.9\%$  for H2, indicating that a static rule lacking either habit smoothing or buffer-stock target wealth cannot match the consumption dynamics of the data even after the borrowing-constraint asymmetry is removed. The OC-ABM is the only specification whose consumption gain is statistically significant at this horizon, reflecting both the structural shape constraints imposed by the Euler equation and the lower forecast variance that follows from optimisation-driven smoothing.

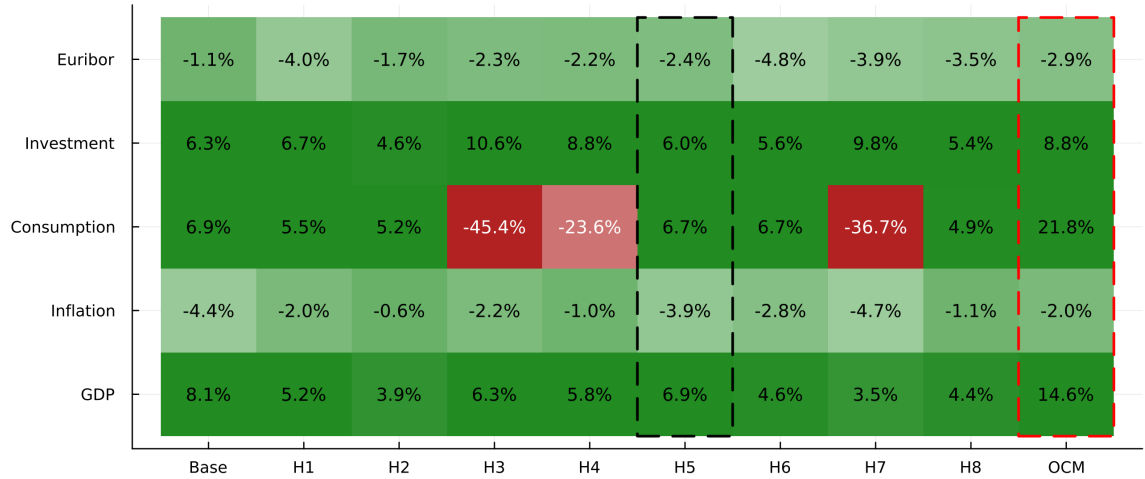


Figure 12: Twelve-quarter relative RMSFE of heuristic ABMs (H1–H7) and the OC-ABM under AR expectations, relative to the AR(1) benchmark. Evaluation period: 2010Q1–2019Q4, 100 Monte Carlo simulations.

Table 9 benchmarks the ABMs against the VAR(1). At short horizons (1–4 quarters), the VAR(1) dominates: its multivariate structure captures contemporaneous comovements that the ABM, operating under univariate AR expectations, cannot exploit. At 12 quarters, this advantage reverses. The OC-ABM achieves a 37.1% RMSFE improvement for GDP ( $p < 0.05$ ), 57.6% for inflation ( $p < 0.01$ ), and 26.9% for household consumption ( $p < 0.01$ ). Among the heuristics, H6-ABM produces a +32.9% consumption improvement at 12 quarters (marginally significant,  $p = 0.05$ ), while H7-ABM shows a +9.8% point estimate that is not statistically distinguishable from zero. The static heuristic H2-ABM, by contrast, shows a –281.3% consumption loss at 12 quarters, indicating that the income-only specification does not generate stable long-horizon dynamics once the borrowing-constraint asymmetry is removed. The short-run disadvantage and long-run advantage of the OC-ABM are consistent with the ABM’s structural content adding value only once the time-series model’s autoregressive momentum dissipates.

Table 9: Forecast performance under AR expectations, relative to VAR(1)

|        | GDP                                                                       | Inflation       | Household cons. | Gov. cons.       | Exports        | Imports         | GDP EA          | Infl. EA       | Euribor         |
|--------|---------------------------------------------------------------------------|-----------------|-----------------|------------------|----------------|-----------------|-----------------|----------------|-----------------|
| VAR(1) | <i>RMSFE for different forecast horizons</i>                              |                 |                 |                  |                |                 |                 |                |                 |
| 1q     | 0.39                                                                      | 0.75            | 0.6             | 0.57             | 1.26           | 1.38            | 0.36            | 0.15           | 0.04            |
| 2q     | 0.7                                                                       | 0.7             | 0.84            | 0.94             | 2.61           | 2.35            | 0.76            | 0.16           | 0.07            |
| 4q     | 1.52                                                                      | 0.85            | 1.4             | 1.66             | 5.42           | 4.85            | 1.69            | 0.18           | 0.16            |
| 8q     | 3.45                                                                      | 1.11            | 2.76            | 3.04             | 9.88           | 8.85            | 3.71            | 0.2            | 0.37            |
| 12q    | 4.71                                                                      | 1.44            | 3.94            | 4.38             | 13.08          | 11.45           | 4.95            | 0.28           | 0.5             |
| ABM    | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                 |                  |                |                 |                 |                |                 |
| 1q     | -7.3(0.61, )                                                              | 22.4(0.01, ***) | 10.1(0.26, )    | 7.1(0.46, )      | -17.9(0.35, )  | -74.1(0.0, ***) | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -4.4(0.09, *)                                                             | 18.5(0.0, ***)  | 1.5(0.66, )     | 3.2(0.0, ***)    | 12.4(0.0, ***) | -54.5(0.0, ***) | 11.7(0.0, ***)  | 2.3(0.18, )    | -20.5(0.0, ***) |
| 4q     | -10.8(0.04, **)                                                           | 31.8(0.0, ***)  | -6.5(0.0, ***)  | -1.6(0.33, )     | 32.6(0.0, ***) | -20.5(0.0, ***) | 31.1(0.0, ***)  | 11.0(0.0, ***) | 20.2(0.0, ***)  |
| 8q     | 28.8(0.01, ***)                                                           | 47.3(0.0, ***)  | 8.8(0.31, )     | -13.8(0.01, ***) | 39.0(0.0, ***) | -9.3(0.29, )    | 36.3(0.0, ***)  | 22.7(0.0, ***) | 53.8(0.0, ***)  |
| 12q    | 27.5(0.08, *)                                                             | 58.7(0.0, ***)  | 10.3(0.2, )     | -16.6(0.02, **)  | 32.6(0.0, ***) | -19.0(0.15, )   | 28.6(0.16, )    | 40.4(0.0, ***) | 58.8(0.0, ***)  |
| H6-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                 |                  |                |                 |                 |                |                 |
| 1q     | -7.2(0.62, )                                                              | 22.4(0.01, ***) | 2.8(0.82, )     | 3.3(0.73, )      | -18.6(0.33, )  | -69.1(0.0, ***) | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -0.9(0.72, )                                                              | 17.2(0.0, ***)  | 12.1(0.0, ***)  | 2.4(0.04, **)    | 12.6(0.0, ***) | -55.0(0.0, ***) | 9.5(0.0, ***)   | 3.6(0.04, **)  | -21.8(0.0, ***) |
| 4q     | -6.7(0.15, )                                                              | 32.4(0.0, ***)  | 6.3(0.0, ***)   | -1.8(0.28, )     | 33.3(0.0, ***) | -21.2(0.0, ***) | 28.9(0.0, ***)  | 13.7(0.0, ***) | 18.4(0.0, ***)  |
| 8q     | 35.8(0.0, ***)                                                            | 46.8(0.0, ***)  | 21.2(0.0, ***)  | -17.1(0.0, ***)  | 40.2(0.0, ***) | -10.6(0.2, )    | 36.7(0.0, ***)  | 22.3(0.0, ***) | 53.2(0.0, ***)  |
| 12q    | 38.9(0.04, **)                                                            | 58.7(0.0, ***)  | 36.1(0.0, ***)  | -18.4(0.01, **)  | 34.0(0.0, ***) | -20.7(0.1, )    | 29.2(0.15, )    | 38.6(0.0, ***) | 58.5(0.0, ***)  |
| OC-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                 |                  |                |                 |                 |                |                 |
| 1q     | -4.8(0.73, )                                                              | 22.4(0.01, ***) | -50.7(0.0, ***) | 6.3(0.49, )      | -15.7(0.41, )  | -80.4(0.0, ***) | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -14.5(0.0, ***)                                                           | 19.1(0.0, ***)  | -20.6(0.0, ***) | 3.7(0.01, ***)   | 16.2(0.0, ***) | -58.4(0.0, ***) | 8.3(0.0, ***)   | 1.4(0.49, )    | -23.1(0.0, ***) |
| 4q     | -19.7(0.0, ***)                                                           | 33.8(0.0, ***)  | -2.7(0.18, )    | 1.3(0.41, )      | 33.6(0.0, ***) | -16.5(0.0, ***) | 30.3(0.0, ***)  | 14.4(0.0, ***) | 18.9(0.0, ***)  |
| 8q     | 36.8(0.0, ***)                                                            | 46.8(0.0, ***)  | 15.1(0.13, )    | -16.7(0.0, ***)  | 40.0(0.0, ***) | -6.9(0.41, )    | 37.0(0.0, ***)  | 16.2(0.0, ***) | 53.0(0.0, ***)  |
| 12q    | 37.1(0.02, **)                                                            | 57.6(0.0, ***)  | 26.9(0.0, ***)  | -15.9(0.03, **)  | 34.0(0.0, ***) | -17.7(0.18, )   | 29.5(0.14, )    | 38.4(0.0, ***) | 57.9(0.0, ***)  |

ABMs under AR expectations vs. VAR(1). See Section 5 for estimation details. Diebold–Mariano  $p$ -values in parentheses; \*, \*\*, \*\*\*: 10%, 5%, 1% significance.

Table 10 compares the ABMs against the univariate AR(1). The horizon-dependent pattern persists: ABMs underperform at 1–4 quarters but improve at longer horizons. At 12 quarters, the OC-ABM achieves a 21.8% RMSFE reduction for household consumption ( $p < 0.01$ ) and 8.8% for investment ( $p < 0.01$ ). The GDP improvement of 14.6%, while economically meaningful, is not statistically significant ( $p = 0.21$ ), reflecting high cross-quarter variance in GDP forecast errors. Investment gains emerge earlier, reaching 6.6% at 8 quarters ( $p < 0.01$ ). These results confirm that the OC-ABM’s advantage is concentrated in household consumption, the variable directly affected by the optimisation block, with spillovers to GDP and investment that are directionally positive but noisier.

Table 10: Forecast performance under AR expectations, relative to AR(1)

|        | GDP                                                                      | Inflation      | Household consumption | Investment     | Euribor         |
|--------|--------------------------------------------------------------------------|----------------|-----------------------|----------------|-----------------|
| AR(1)  | <i>RMSFE for different forecast horizons</i>                             |                |                       |                |                 |
| 1q     | 0.41                                                                     | 0.57           | 0.56                  | 3.3            | 0.03            |
| 2q     | 0.67                                                                     | 0.57           | 0.82                  | 4.48           | 0.06            |
| 4q     | 1.32                                                                     | 0.56           | 1.47                  | 6.21           | 0.11            |
| 8q     | 2.59                                                                     | 0.58           | 2.72                  | 9.61           | 0.16            |
| 12q    | 3.47                                                                     | 0.6            | 3.68                  | 13.27          | 0.2             |
| ABM    | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                |                       |                |                 |
| 1q     | -2.1(0.81, )                                                             | -1.4(0.67, )   | 4.9(0.41, )           | 8.8(0.26, )    | -80.9(0.0, ***) |
| 2q     | -9.5(0.0, ***)                                                           | 0.5(0.11, )    | -0.7(0.6, )           | 0.9(0.21, )    | -37.7(0.0, ***) |
| 4q     | -27.0(0.0, ***)                                                          | -2.4(0.0, ***) | -1.3(0.79, )          | -4.9(0.0, ***) | -14.9(0.0, ***) |
| 8q     | 5.0(0.58, )                                                              | -0.4(0.32, )   | 7.4(0.25, )           | 1.4(0.03, **)  | -6.9(0.23, )    |
| 12q    | 1.7(0.88, )                                                              | 0.7(0.72, )    | 4.1(0.51, )           | 3.5(0.06, *)   | -0.7(0.93, )    |
| H6-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                |                       |                |                 |
| 1q     | -2.0(0.82, )                                                             | -1.4(0.67, )   | -2.9(0.71, )          | 8.8(0.26, )    | -80.9(0.0, ***) |
| 2q     | -5.8(0.04, **)                                                           | -1.0(0.0, ***) | 10.1(0.0, ***)        | 1.4(0.06, *)   | -39.1(0.0, ***) |
| 4q     | -22.4(0.0, ***)                                                          | -1.5(0.0, ***) | 10.9(0.01, **)        | -4.8(0.0, ***) | -17.4(0.0, ***) |
| 8q     | 14.3(0.15, )                                                             | -1.2(0.0, ***) | 20.0(0.04, **)        | 7.0(0.0, ***)  | -8.4(0.15, )    |
| 12q    | 17.1(0.28, )                                                             | 0.7(0.62, )    | 31.6(0.06, *)         | 8.2(0.0, ***)  | -1.5(0.85, )    |
| OC-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                |                       |                |                 |
| 1q     | 0.2(0.98, )                                                              | -1.4(0.67, )   | -59.5(0.0, ***)       | 8.8(0.26, )    | -80.9(0.0, ***) |
| 2q     | -20.0(0.0, ***)                                                          | 1.3(0.0, ***)  | -23.3(0.0, ***)       | 0.2(0.79, )    | -40.6(0.0, ***) |
| 4q     | -37.3(0.0, ***)                                                          | 0.7(0.02, **)  | 2.3(0.53, )           | -6.9(0.0, ***) | -16.7(0.0, ***) |
| 8q     | 15.7(0.05, *)                                                            | -1.4(0.0, ***) | 13.8(0.01, **)        | 6.6(0.01, ***) | -8.9(0.13, )    |
| 12q    | 14.6(0.21, )                                                             | -2.0(0.0, ***) | 21.8(0.01, ***)       | 8.8(0.0, ***)  | -2.9(0.69, )    |

ABMs under AR expectations vs. AR(1). See Section 5 for estimation details. Diebold–Mariano  $p$ -values in parentheses; \*, \*\*, \*\*\*: 10%, 5%, 1% significance.

Figures 13 and 14 illustrate the quarterly trajectories for two representative forecast origins: one launched during the Eurocrisis (origin 2012Q1, horizon through 2015Q4) and one during the subsequent recovery (origin 2015Q1, horizon through 2018Q4). These are separate 12-quarter forecast exercises from different origins, not a partition of the evaluation sample.

During the Eurocrisis (Figure 13), both models overpredict GDP and household consumption. Realised GDP remains roughly flat while both models project positive growth; household consumption falls and recovers only slowly, lying below the confidence intervals of both models for much of the period. Neither the OC-ABM nor the AR model captures the pronounced investment dip of 2013. The overprediction likely reflects the absence of a financial accelerator or balance-sheet deleveraging channel: without explicit debt-deflation dynamics, the ABM lacks a mechanism to propagate the sovereign-debt stress that depressed Dutch domestic demand during this period.

The recovery period (Figure 14) presents a markedly different picture. The OC-ABM tracks realised GDP closely, with observations remaining within the 90% confidence interval throughout. The AR model initially underpredicts and then overshoots. For household consumption, the OC-ABM's trajectory aligns well with the steady upward trend in the data.

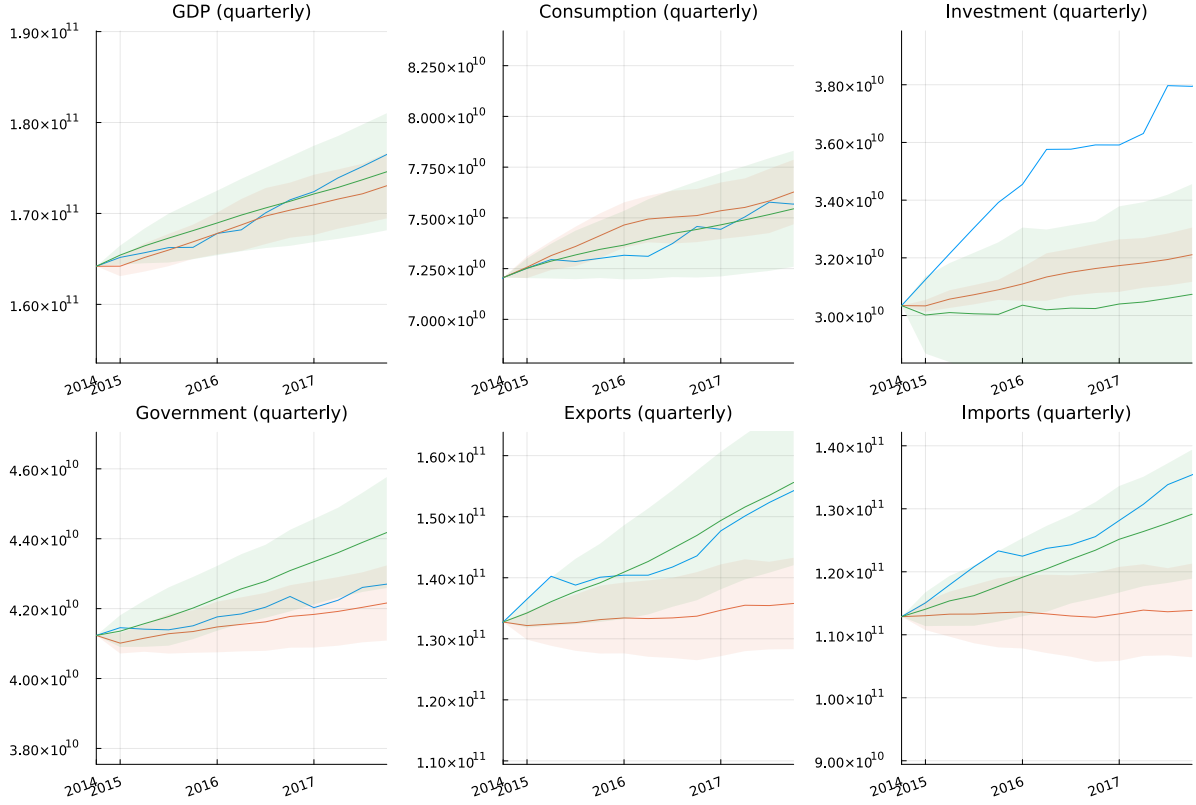


Figure 13: Quarterly forecasts from 2012Q1–2015Q4 (Eurocrisis period): OC-ABM under AR expectations (red) vs. AR(1) (green dashed) vs. realised Eurostat data (black). Shaded area: 90% confidence interval from 100 Monte Carlo simulations. Both models overpredict during the sovereign-debt stress period.

## 5.2 Robustness across expectation mechanisms

The household consumption gain holds under each of the three remaining expectation mechanisms. At the 12-quarter horizon the OC-ABM achieves consumption RMSFE reductions of 10.7% under ARX, 18.6% under VAR, and 9.9% under VARX, each relative to the matched univariate benchmark (Diebold–Mariano  $p < 0.01$  in all three cases). Against the multivariate benchmarks the reductions reach 28.2% (ARX vs. VARX(1)), 23.9% (VAR vs. VAR(1)), and 27.5% (VARX vs. VARX(1)). Investment gains lie between 8.7% and 9.9% across the three mechanisms and are statistically significant; GDP gains of 11–15% are directionally positive but not significant. Full horizon-by-horizon tables and the per-mechanism heatmaps appear in Appendix F.

The size of the consumption gain depends primarily on whether the benchmark conditions on exogenous predictors, not on whether it is multivariate. Mechanisms without exogenous predictors (AR, VAR) yield consumption gains of 18–22%, while ARX and VARX yield 10–11%. Euro area variables and the Euribor rate absorb part of the OC-ABM’s predictive content, consistent with the small-open-economy character of the Dutch data. Across the heuristic ladder, the same rule families dominate under every mechanism: H6 (habit on income with additive wealth) produces the highest point estimates for consumption (+15.4% under ARX, +31.6% under AR, +32.9% under VAR, +16.1% under VARX, with marginal significance under AR and VAR only), and H7 (Deaton buffer-stock) yields smaller and statistically insignificant gains. Static income-only and income-plus-wealth heuristics (H1–H5) produce consumption losses under every mechanism, ranging from –10% (H3 under VAR) to –338% (H2 under ARX). The OC-ABM is the only specification whose consumption gain is statistically significant under

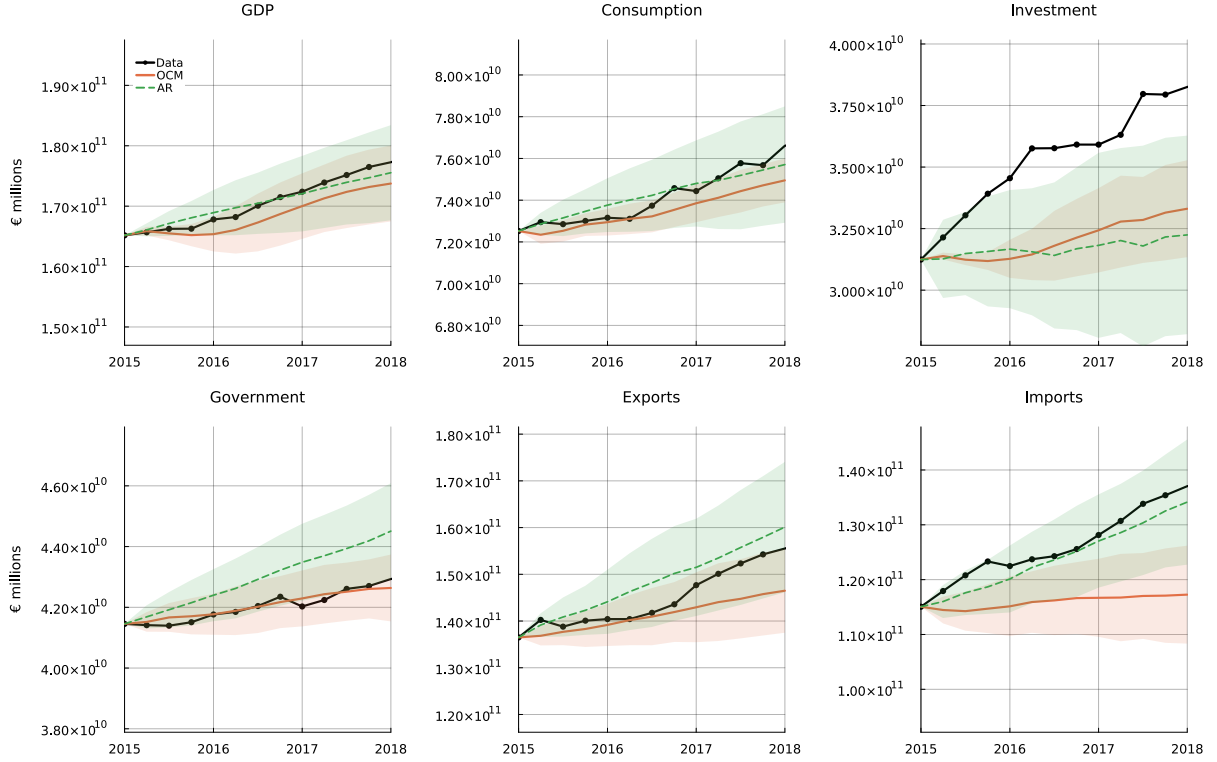


Figure 14: Quarterly forecasts from 2015Q1–2018Q4 (recovery period): OC-ABM under AR expectations (red) vs. AR(1) (green dashed) vs. realised Eurostat data (black). Shaded area: 90% confidence interval from 100 Monte Carlo simulations. The OC-ABM tracks realised GDP and consumption within the confidence interval throughout.

every mechanism ( $p < 0.01$  in all four cases), reflecting both the structural shape constraints imposed by the Euler-equation policy and the lower variance of its forecast errors relative to the noisier H6 estimates.

### 5.3 Channel decomposition

The H1–H7 ladder equalises the borrowing constraint but leaves three structural channels open between the heuristic class and the OC-ABM (Section 2.4, Table 2): the capital-return MPC channel, the portfolio-choice channel, and the optimisation/Euler-equation channel. Heuristics H8 and H9 close the first two individually, and H10 closes them jointly. The optimisation channel cannot be replicated by any reduced-form rule, so the residual gap between H10 and the OC-ABM attributes to Euler-equation discipline alone.

Table 11 reports the 12-quarter consumption RMSFE for the channel-decomposition heuristics, the H7 baseline, and the OC-ABM relative to the AR(1) benchmark. Three findings emerge. First, the capital-return MPC channel contributes essentially no consumption signal: H8 produces +12.6% versus H7's +13.7%, a 1 pp difference that is well within the dispersion of either point estimate. This is consistent with the empirical evidence that capital-return income is small relative to labour income for the median household and varies little within the 2010–2019 evaluation sample. Second, the portfolio-choice channel accounts for the majority of the residual gap between the buffer-stock baseline and the OC-ABM: H9 reaches +25.4%, an 11.7 pp gain over H7 that recovers approximately the full magnitude of the OC-ABM's +21.8% improvement. State-contingent portfolio rebalancing toward a Merton target, rather than holding the liquid/illiquid split fixed at the calibration ratio, is therefore the substantive mechanism

distinguishing the OC-ABM from the buffer-stock heuristic at the point-estimate level. Third, the combined variant H10 (+21.2%) lies between H8 and H9 rather than exceeding either, indicating that the two channels are substitutes in this specification: scaling capital-return contributions to income by  $\alpha_{\text{cap}}$  reduces the wealth signal that drives portfolio rebalancing, partially offsetting the Merton effect.

Table 11: Channel decomposition: 12-quarter consumption RMSFE under AR expectations vs. AR(1)

| Model                    | Channels closed beyond H7                       | $\Delta\text{RMSFE}$ (%) | DM $p$ -value |
|--------------------------|-------------------------------------------------|--------------------------|---------------|
| H7 (Deaton buffer-stock) | —                                               | +13.7                    | 0.23          |
| H8 (cap-split)           | Capital-return MPC asymmetry                    | +12.6                    | 0.27          |
| H9 (Merton)              | Portfolio choice                                | +25.4                    | 0.13          |
| H10 (combined)           | Capital-return + Portfolio choice               | +21.2                    | 0.23          |
| OC-ABM (bond-fixed)      | All three (Euler discipline implicit in policy) | +21.8***                 | < 0.01        |

Note: 12-quarter household consumption RMSFE relative to AR(1) under AR expectations, evaluated over 2010Q1–2019Q4 with 100 Monte Carlo simulations per reference quarter. All heuristics share the H7 borrowing constraint. H8 partitions the income aggregate by Equation (22) with  $\alpha_{\text{cap}} = 0.5$ . H9 implements the Merton portfolio rule of Equations (23)–(24). H10 combines both. Diebold–Mariano  $p$ -values use Newey–West HAC standard errors and the small-sample correction of Harvey et al. (1997); \*\*\* denotes  $p < 0.01$ .

A fourth observation concerns the form in which Euler-equation discipline contributes. The H9 point estimate (+25.4%) is statistically indistinguishable from the OC-ABM (+21.8%) at conventional levels, but the OC-ABM is the only variant whose gain is significant at  $p < 0.01$ . Closing the portfolio channel is what brings the heuristic point estimate up to the OC-ABM range; what the OC-ABM additionally delivers is a reduction in forecast variance. The Euler-equation discipline acts as a shape constraint over the entire state space, generating policy responses that are well-behaved at calibration moments and at states never observed in the calibration sample. The heuristics, by contrast, are pinned only at the calibrated moments and behave according to whatever functional form happens to extrapolate. The consequence in this exercise is a similar central tendency but wider sampling distribution of forecast errors, which lowers Diebold–Mariano significance even when the mean improvement is competitive.

Taken together, the channel decomposition closes the identification gap that the H1–H7 ladder alone could not. The borrowing constraint accounts for the move from  $-26\%$  to  $-308\%$  losses under H1–H4 to mild deviations under H7; dynamic structure (habit smoothing or buffer-stock target wealth) carries the gain from near-zero to  $+13.7\%$ ; portfolio choice contributes an additional  $+11.7$  pp; capital-return MPC asymmetry contributes negligibly; and Euler-equation discipline, while not changing the point estimate, removes the residual forecast variance that prevents the channel-equalised heuristic from achieving statistical significance.

#### 5.4 Sectoral forecasts

The ABM’s multi-sector structure enables forecasting at a level of granularity unavailable to aggregate time-series models. We exploit this to test whether ABM gains concentrate in specific sectors, which can shed light on the structural channels driving aggregate improvements. We decompose the model into ten sectors following the statistical classification of economic activities (NACE Rev. 2) and forecast the log differences of sectoral gross value added (GVA).

We benchmark against a sectoral VAR(1), selected by information criteria, estimated from 1997Q1 to 2010Q1 and re-estimated quarterly through 2019Q4. Under AR expectations, we compare the base ABM, H6-ABM (the best-performing heuristic for aggregate consumption), and OC-ABM.

Table 12: Sectoral GVA forecast performance under AR expectations, relative to VAR(1)

|        | A                                                                         | B, C, D and E    | F                | G, H and I     | J              | K                | L                | M and N        | O, P and Q     | R and S        |
|--------|---------------------------------------------------------------------------|------------------|------------------|----------------|----------------|------------------|------------------|----------------|----------------|----------------|
| VAR(1) | <i>RMSFE for different forecast horizons</i>                              |                  |                  |                |                |                  |                  |                |                |                |
| 1q     | 1.2                                                                       | 0.68             | 0.79             | 0.98           | 1.28           | 1.12             | 1.15             | 1.12           | 1.01           | 1.22           |
| 2q     | 0.97                                                                      | 1.14             | 1.52             | 1.91           | 2.64           | 2.17             | 1.88             | 2.26           | 2.08           | 2.14           |
| 4q     | 1.36                                                                      | 2.23             | 2.94             | 3.8            | 5.34           | 4.22             | 3.11             | 4.52           | 4.24           | 4.11           |
| 8q     | 2.49                                                                      | 4.26             | 5.54             | 7.59           | 10.75          | 8.15             | 5.72             | 9.01           | 8.5            | 7.89           |
| 12q    | 3.92                                                                      | 6.11             | 8.21             | 11.41          | 16.13          | 11.68            | 8.8              | 13.54          | 12.6           | 11.48          |
| ABM    | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                  |                  |                |                |                  |                  |                |                |                |
| 1q     | -358.7(0.0, ***)                                                          | -231.9(0.0, ***) | -286.2(0.0, ***) | 12.4(0.15, )   | -2.9(0.85, )   | -131.7(0.0, ***) | -131.6(0.0, ***) | 8.6(0.39, )    | 24.2(0.0, ***) | 33.3(0.0, ***) |
| 2q     | -720.0(0.0, ***)                                                          | -154.5(0.0, ***) | -182.9(0.0, ***) | 17.9(0.0, ***) | 37.4(0.0, ***) | -58.3(0.0, ***)  | -166.5(0.0, ***) | 32.2(0.0, ***) | 51.5(0.0, ***) | 22.5(0.0, ***) |
| 4q     | -565.4(0.0, ***)                                                          | -74.8(0.0, ***)  | -122.1(0.0, ***) | 33.3(0.0, ***) | 51.3(0.0, ***) | -9.4(0.0, ***)   | -175.4(0.0, ***) | 41.0(0.0, ***) | 60.5(0.0, ***) | 35.4(0.0, ***) |
| 8q     | -309.6(0.0, ***)                                                          | -51.9(0.0, ***)  | -120.1(0.0, ***) | 57.7(0.0, ***) | 53.9(0.0, ***) | -0.6(0.94, )     | -111.2(0.0, ***) | 46.7(0.0, ***) | 59.2(0.0, ***) | 52.4(0.0, ***) |
| 12q    | -99.1(0.0, ***)                                                           | -68.1(0.0, ***)  | -104.6(0.0, ***) | 65.7(0.0, ***) | 56.4(0.0, ***) | -16.0(0.23, )    | -53.6(0.0, ***)  | 52.7(0.0, ***) | 58.0(0.0, ***) | 51.2(0.0, ***) |
| H6-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                  |                  |                |                |                  |                  |                |                |                |
| 1q     | -358.7(0.0, ***)                                                          | -231.9(0.0, ***) | -286.2(0.0, ***) | 12.4(0.15, )   | -2.9(0.85, )   | -131.7(0.0, ***) | -131.6(0.0, ***) | 8.6(0.39, )    | 24.2(0.0, ***) | 33.3(0.0, ***) |
| 2q     | -716.5(0.0, ***)                                                          | -156.4(0.0, ***) | -182.1(0.0, ***) | 20.9(0.0, ***) | 38.9(0.0, ***) | -58.5(0.0, ***)  | -162.2(0.0, ***) | 33.2(0.0, ***) | 51.3(0.0, ***) | 27.4(0.0, ***) |
| 4q     | -555.7(0.0, ***)                                                          | -77.8(0.0, ***)  | -118.6(0.0, ***) | 36.7(0.0, ***) | 54.7(0.0, ***) | -10.5(0.0, ***)  | -171.6(0.0, ***) | 43.3(0.0, ***) | 61.8(0.0, ***) | 37.9(0.0, ***) |
| 8q     | -300.0(0.0, ***)                                                          | -57.7(0.0, ***)  | -111.5(0.0, ***) | 61.2(0.0, ***) | 59.6(0.0, ***) | -4.4(0.63, )     | -104.1(0.0, ***) | 51.5(0.0, ***) | 60.1(0.0, ***) | 59.1(0.0, ***) |
| 12q    | -90.4(0.0, ***)                                                           | -64.3(0.0, ***)  | -92.1(0.0, ***)  | 66.8(0.0, ***) | 63.4(0.0, ***) | -14.4(0.32, )    | -50.2(0.0, ***)  | 57.8(0.0, ***) | 59.7(0.0, ***) | 63.9(0.0, ***) |
| OC-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                  |                  |                |                |                  |                  |                |                |                |
| 1q     | -358.7(0.0, ***)                                                          | -231.9(0.0, ***) | -286.2(0.0, ***) | 12.4(0.15, )   | -2.9(0.85, )   | -131.7(0.0, ***) | -131.6(0.0, ***) | 8.6(0.39, )    | 24.2(0.0, ***) | 33.3(0.0, ***) |
| 2q     | -718.2(0.0, ***)                                                          | -151.6(0.0, ***) | -181.7(0.0, ***) | 15.8(0.0, ***) | 37.0(0.0, ***) | -54.9(0.0, ***)  | -177.4(0.0, ***) | 33.0(0.0, ***) | 51.2(0.0, ***) | 4.6(0.03, **)  |
| 4q     | -554.6(0.0, ***)                                                          | -74.6(0.0, ***)  | -122.1(0.0, ***) | 30.4(0.0, ***) | 49.7(0.0, ***) | -3.6(0.12, )     | -178.6(0.0, ***) | 38.3(0.0, ***) | 60.2(0.0, ***) | 30.6(0.0, ***) |
| 8q     | -302.2(0.0, ***)                                                          | -60.4(0.0, ***)  | -118.0(0.0, ***) | 65.2(0.0, ***) | 56.7(0.0, ***) | -7.0(0.44, )     | -104.1(0.0, ***) | 50.5(0.0, ***) | 58.4(0.0, ***) | 54.0(0.0, ***) |
| 12q    | -92.6(0.0, ***)                                                           | -68.7(0.0, ***)  | -97.3(0.0, ***)  | 69.8(0.0, ***) | 60.2(0.0, ***) | -18.1(0.2, )     | -48.3(0.0, ***)  | 56.1(0.0, ***) | 58.7(0.0, ***) | 55.4(0.0, ***) |

Sectors: Agriculture, forestry and fishing (A); Industry except construction (B, C, D and E); Construction (F); Wholesale and retail trade, transport, accommodation and food service activities (G, H and I); Information and communication (J); Financial and insurance activities (K); Real estate activities (L); Professional, scientific and technical activities, administrative and support service activities (M and N); Public administration, defence, education, human health and social work activities (O, P and Q); Arts, entertainment, recreation, other service activities (R and S). ABMs under AR expectations vs. sectoral VAR(1). See Section 5 for estimation details. Diebold–Mariano  $p$ -values in parentheses; \*, \*\*, \*\*\*: 10%, 5%, 1% significance.

Table 12 reveals a systematic pattern: ABMs improve most where the VAR(1) struggles. Sectors with high baseline RMSFE, such as wholesale, retail and transport (G, H, I), professional services (M, N), and public administration (O, P, Q), show ABM improvements of 50–70% at the 12-quarter horizon. These are sectors characterised by heterogeneous firm populations, complex demand linkages through the input–output network, and labour market dynamics that the ABM explicitly models. Conversely, sectors where the VAR(1) already forecasts well, such as agriculture (A), industry (B, C, D, E), and construction (F), see ABM losses, suggesting that the ABM’s stochastic firm-level dynamics introduce noise where aggregate regularities suffice. This pattern holds for both heuristic and optimal-consumption variants, suggesting that the sectoral gains derive primarily from the ABM’s production-side structure (firm heterogeneity, IO linkages, search-and-matching) rather than from the household optimisation block, though we cannot rule out indirect demand-side effects from the household block.

## 5.5 National accounting decompositions

A distinguishing feature of the ABM framework is its stock-flow consistency: every flow is an outflow for one agent and an inflow for another, following double-entry bookkeeping in accordance with ESA national accounting conventions. This property enables internally coherent decomposition of aggregate forecasts along multiple dimensions. Figure 15 illustrates this

by decomposing the same GDP forecast for 2014Q4–2017Q4 using the production approach (panel a: sectoral GVA contributions), the income approach (panel b: wages, social contributions, gross operating surplus, taxes), and the expenditure approach (panel c: household consumption, government consumption, capital formation, net exports). That all three sum to the same aggregate demonstrates an internal consistency property that purely statistical models cannot guarantee.<sup>14</sup>

This decomposition capability is not a forecasting result per se but an internal validity property of the modelling framework. It becomes practically useful when policymakers require not only an aggregate forecast but also a structural narrative, for instance whether a GDP revision is driven by a consumption shortfall or an investment downturn.

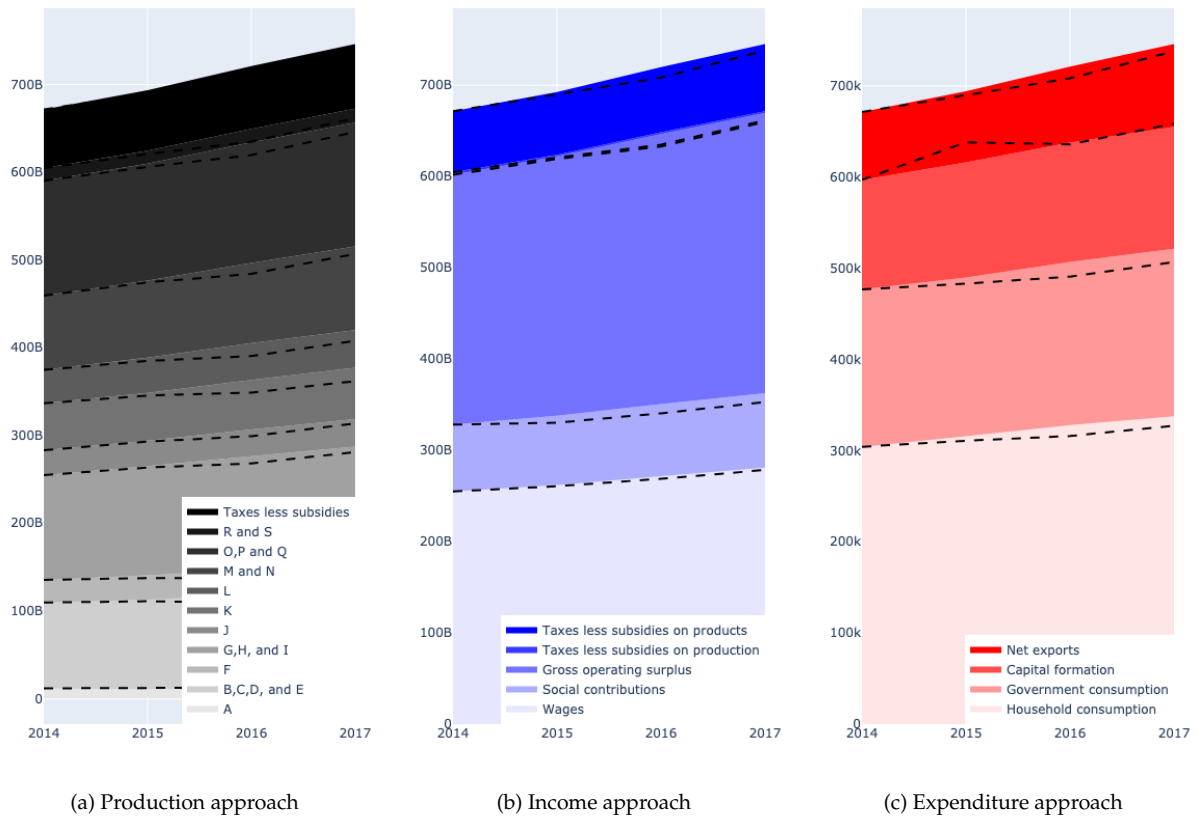


Figure 15: GDP forecast for 2014Q4–2017Q4 decomposed by production (a), income (b), and expenditure (c) approaches. Dashed lines: realised data. Shaded areas: average ABM forecast from 100 Monte Carlo simulations. All three panels sum to the same aggregate, demonstrating stock-flow consistency.

## 6 Internal Validity

Having presented the forecasting results in Section 5, we now assess whether the OC-ABM generates plausible macroeconomic dynamics. A model that forecasts well but produces implausible internal behaviour may be exploiting artefacts of the evaluation sample rather than capturing genuine economic mechanisms. We examine two diagnostic dimensions: persistence (auto-correlations) and comovement (cross-correlations), following the validation pro-

<sup>14</sup>We display a single representative forecast origin (2014Q4) to keep the figure legible; the stock-flow identity holds by construction at every reference quarter.

tocol of Caiani et al. (2016). Specifically, we adopt their approach of comparing HP-filtered cyclical components across the same set of variables (GDP, investment, consumption, wages, operating surplus, and the GDP deflator).

We focus on the OC-ABM under AR expectations, the simplest expectation mechanism, to isolate the contribution of the household optimisation block. All diagnostics are computed from a single calibration origin (2012Q1), which sits near the trough of the Eurocrisis. Results from this origin may therefore not be representative of the model’s typical behaviour; robustness across origins is left to future work. Appendix D presents the corresponding analysis under ARX, VAR, and VARX expectations, where the GDP deflator dynamics improve substantially.

## 6.1 Auto-correlations

Figure 16 displays the sample auto-correlation functions (ACFs) of the HP-filtered cyclical components for six variables: GDP, capital formation, household consumption, wages, operating surplus, and the GDP deflator.<sup>15</sup> ACFs are computed from 100 Monte Carlo simulations launched from the 2012Q1 calibration origin over a 20-quarter horizon.<sup>16</sup> The shaded band shows one standard deviation across simulations; the orange line shows the empirical ACF over 1997Q1–2019Q4.

The model matches the data well for GDP and operating surplus, reproducing the characteristic decay from strong positive auto-correlation at short lags to near-zero by lag 12–15. Capital formation shows a similar decay rate, though the model generates slightly more negative auto-correlations at longer lags. However, two discrepancies stand out. First, wages and household consumption decay more slowly in the model than in the data. For wages, this likely reflects insufficient labour-market frictions: wages adjust immediately to production changes via overtime payments, producing too much persistence, though the adaptive AR(1) expectation mechanism may also contribute. Because households’ expected future income tracks this persistent wage process, the excess persistence feeds through to consumption via the income channel. Introducing search frictions or time-to-fill for vacancies would directly address wage persistence; the consumption effect should diminish as a consequence. Second, the GDP deflator shows excessive persistence, indicating that the price-setting or expectation-formation mechanisms may be too sluggish.

<sup>15</sup>We apply the Hodrick–Prescott filter with the standard quarterly smoothing parameter  $\lambda = 1,600$ . Cyclical components are computed as percentage deviations from the HP trend in log levels for GDP, investment, and household consumption, and in levels for the GDP deflator growth rate.

<sup>16</sup>We use the biased (Box–Jenkins) auto-correlation estimator,  $r_k = \frac{\frac{1}{T} \sum_{t=1}^{T-k} (y_t - \bar{y})(y_{t+k} - \bar{y})}{\frac{1}{T} \sum_{t=1}^T (y_t - \bar{y})^2}$ , which divides both numerator and denominator by  $T$  rather than  $T - k$ . This is the standard convention in the time-series literature and ensures positive semi-definiteness of the implied autocovariance matrix. With a simulation horizon of  $T = 20$ , the bias correction difference between dividing by  $T$  and  $T - k$  is non-trivial at longer lags; this may partly account for differences between model ACFs (computed over  $T = 20$ ) and empirical ACFs (computed over  $T \approx 90$ ).

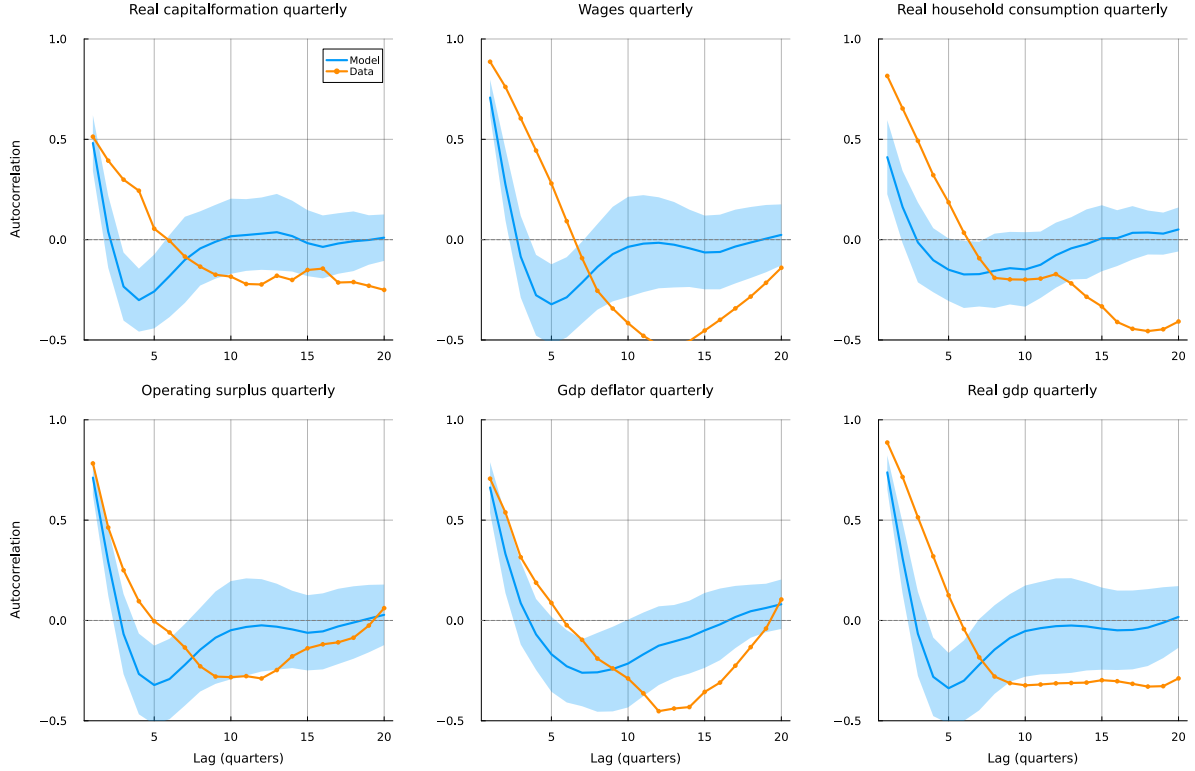


Figure 16: Auto-correlations of HP-filtered ( $\lambda = 1,600$ ) cyclical components. Blue line: mean across 100 Monte Carlo simulations from origin 2012Q1 (20-quarter horizon). Shaded area:  $\pm 1$  standard deviation. Orange line: empirical ACF (1997Q1–2019Q4). The model matches GDP, operating surplus, and capital formation well but shows slower decay for wages, household consumption, and the GDP deflator.

## 6.2 Cross-correlations

Figure 17 shows the cross-correlation between the HP-filtered cyclical component of real GDP at time  $t$  and other variables at time  $t - k$ , where negative lags indicate the variable leads GDP (i.e., the variable at  $t - k$  is correlated with GDP at  $t$ , so the variable precedes GDP by  $k$  quarters).<sup>17</sup> Simulation settings match Section 6.1.

Capital formation and household consumption match the data well: both peak at lag zero in the model and in the data, indicating coincident cyclical behaviour. Operating surplus shows the correct shape but with a flatter profile in the model.

Wages present a notable discrepancy. In the data, wages are pro-cyclical and leading, that is, positively correlated with GDP and peaking before it (around lag  $-5$ ). In the model, wages are coincident with GDP, peaking at lag zero. This arises because wages adjust immediately when production rises through overtime payments, eliminating the lead that hiring frictions generate in practice.

The GDP deflator cross-correlations provide the weakest match. The data show a negative cross-correlation at negative lags transitioning to positive at positive lags, indicating that deflator movements lag GDP. The model reproduces this sign pattern but generates a much flatter

<sup>17</sup>We compute the sample cross-correlation as  $\hat{\rho}_{xy}(k) = \hat{R}_{xy}(k) / [\hat{R}_{xx}(0) \hat{R}_{yy}(0)]^{1/2}$ , where  $\hat{R}_{xy}(k) = \frac{1}{T} \sum_{t=0}^{T-k-1} x_{t+k} y_t$  for  $k \geq 0$  and  $\hat{R}_{xy}(k) = \hat{R}_{yx}(-k)$  for  $k < 0$ . This is the standard Pearson-style normalisation ensuring  $\hat{\rho}_{xy}(k) \in [-1, 1]$ .

profile, with cross-correlations near zero across all lags rather than exhibiting the pronounced swing seen in the data. The GDP auto-correlation panel confirms that the model’s cyclical GDP dynamics are internally consistent, peaking at unity at lag zero as expected. Under ARX expectations (Appendix D), the deflator cross-correlations improve markedly: conditioning on exogenous variables disciplines the price-setting mechanism, aligning the model’s deflator dynamics more closely with the empirical pattern.

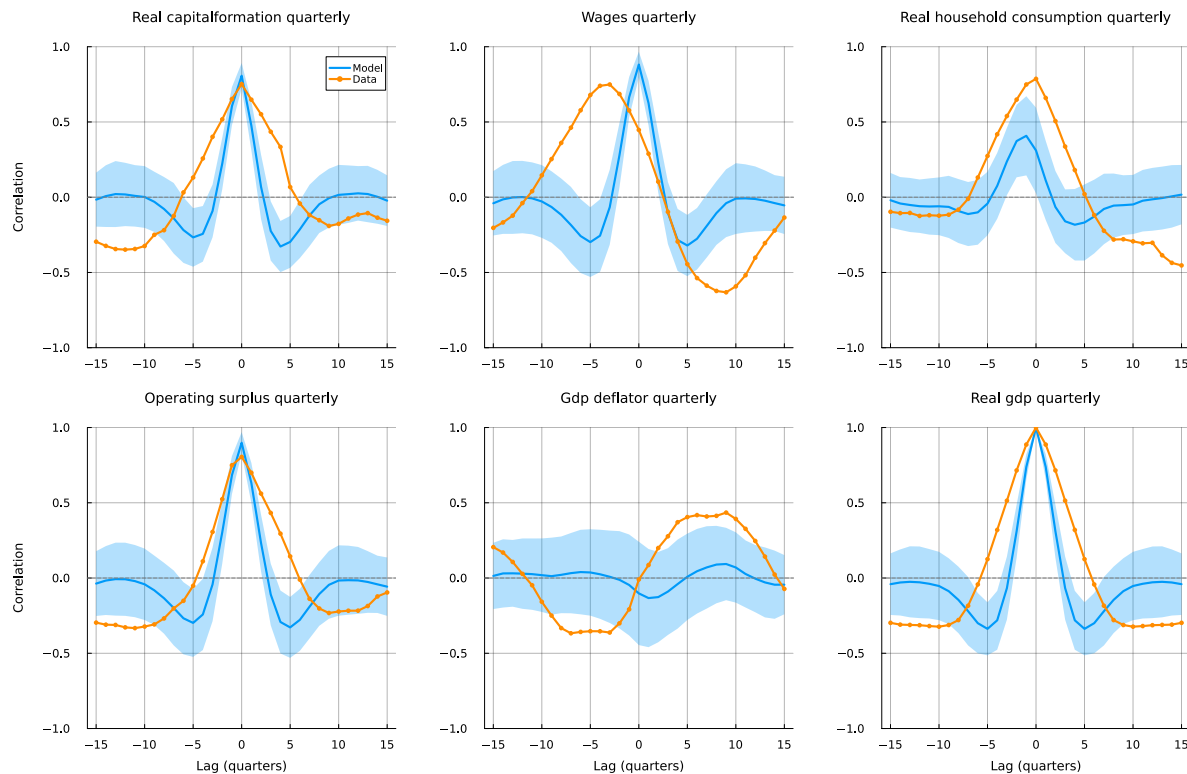


Figure 17: Cross-correlations between HP-filtered ( $\lambda = 1,600$ ) cyclical GDP and other variables. Positive lags: variable lags GDP. Negative lags: variable leads GDP. Blue line: mean across 100 Monte Carlo simulations from origin 2012Q1 (20-quarter horizon). Shaded area:  $\pm 1$  standard deviation. Orange line: empirical cross-correlation (1997Q1–2019Q4). Capital formation and consumption match well; wages peak at lag zero rather than leading as in the data.

## 7 Discussion

Our results indicate that intertemporal consumption-smoothing structure — whether through habit formation, buffer-stock dynamics, or Euler-equation optimisation — combined with heterogeneous MPCs and a state-contingent portfolio rule is the primary channel through which microfounded household behaviour improves macroeconomic forecasting. The OC-ABM is the only model variant that delivers *statistically significant* consumption forecast improvements under every expectation mechanism ( $p < 0.01$  in all four cases). Among the heuristic alternatives, the habit-formation rule H6 produces consumption point estimates that are higher than the OC-ABM’s under every mechanism (+15.4 to +32.9% relative to the matched univariate benchmark) but with substantially wider confidence intervals, achieving only marginal significance under AR and VAR. The buffer-stock rule H7 yields smaller point estimates that are not statistically distinguishable from zero. The Merton-portfolio variant H9 raises the H7 consumption gain by 11.7 pp to +25.4%, recovering most of the OC-ABM’s point estimate but again without reaching conventional significance. Static income-only and income-plus-

wealth heuristics (H1–H5) produce consumption losses ranging from  $-10\%$  to  $-338\%$  under all four mechanisms once the borrowing-constraint asymmetry is removed, confirming that MPC heterogeneity per se does not suffice. What matters is the joint presence of dynamic structure (habit, target-wealth, or Euler), through which households absorb transitory shocks via liquid-asset adjustment rather than passing them through to spending one-for-one, and a state-contingent portfolio rule, through which the liquid–illiquid composition adjusts to changes in expected returns and wealth.

Two robust patterns characterise the forecasting results. First, the OC-ABM delivers statistically significant RMSFE reductions for household consumption at the 12-quarter horizon across all four expectation mechanisms, ranging from 9.9% under VARX to 21.8% under AR expectations, each measured relative to its corresponding univariate benchmark (Figure 11; Diebold–Mariano  $p < 0.01$  in all cases). GDP improvements of 11–15% are directionally positive throughout but not statistically significant under any mechanism, reflecting high cross-quarter variance in GDP forecast errors (Tables 10, 23, 25, 27). Second, in our setting, ABMs underperform at short horizons (1–4 quarters) and outperform at longer horizons (8–12 quarters). This horizon dependence is consistent with the ABM’s structural content, including distributional dynamics, portfolio adjustment, and input–output propagation, compounding over time, while simple autoregressive structures capture the serial correlation that dominates short-run dynamics more efficiently.

The channel decomposition reported in Section 5.3 closes most of the identification problem within the paper. The OC-ABM differs from the heuristic ladder along four structural dimensions, enumerated in Table 2: the borrowing constraint, the treatment of capital returns, the choice of portfolio composition, and the optimisation framework. We equalise the first across all ten heuristics through the consumption cap  $c_{h,t} \leq a_{h,t}/(1 + \tau^{\text{VAT}})$ . H8 closes the capital-return channel by partitioning the income aggregate and scaling the capital-return contribution by  $\alpha_{\text{cap}} = 0.5$  (Christelis et al., 2019). H9 closes the portfolio-choice channel by replacing the fixed liquid–illiquid split with a Merton-targeted rebalancing rule (Merton, 1969). H10 closes both jointly. Only the optimisation/Euler channel remains uniquely closed by the OC-ABM, since reduced-form rules cannot impose intertemporal optimality at every state in the policy domain.

The decomposition attributes the OC-ABM’s residual advantage over the buffer-stock baseline as follows. The capital-return channel contributes negligibly: H8 (+12.6%) and H7 (+13.7%) are statistically indistinguishable, consistent with the small share of capital-return income for the median household and the low volatility of capital returns over the 2010–2019 evaluation sample. The portfolio-choice channel accounts for the majority of the residual gap: H9 (+25.4%) gains 11.7 pp over H7 and reaches the OC-ABM’s +21.8% point estimate. The two channels are substitutes in the combined variant H10 (+21.2%, between H8 and H9), indicating that scaling the capital-return contribution to income reduces the wealth signal that drives the Merton rebalancing. The Euler-equation channel, finally, cannot be replicated by any reduced-form rule. Its empirical signature in this exercise is not a higher point estimate — H6 and H9 match the OC-ABM at the level of the central tendency — but a lower forecast variance, which delivers the OC-ABM’s unique statistical significance under every mechanism while the channel-equalised heuristic H9 plateaus at  $p \approx 0.13$ .

Our methodological contribution, the amortised neural-network calibration procedure, separates solving the household optimisation problem from calibrating it. The approach relates to the extended policy functions of Kase et al. (2025), who treat preference parameters as pseudo-state variables (additional inputs to the policy network) when estimating nonlinear HANK

models. Our method differs in two respects: we target moment matching rather than likelihood estimation, and we employ online learning during simulation. While Kase et al. (2025) solve the policy function once and hold the network fixed during estimation, our network receives incremental updates each period as the perceived law of motion evolves. This online adaptation is important for embedding HANK-style households in an ABM where expectations change endogenously, though we cannot rule out that the two-asset structure or stock-flow accounting also contribute to the forecasting gains.

The OC-ABM and the heuristic ladder are equally parsimonious in their economic parameters: the OC-ABM calibrates four preference parameters ( $\beta, \sigma, c^*, c_{\text{int}}$ ) plus the bond-liquidity coefficients ( $\chi_b^0, \chi_b^1$ ), the same order of magnitude as the four to five parameters in H4 or H7. The neural-network policy is a numerical solver — a function approximator for the household’s optimal policy — not a vector of free parameters fit to moments. Its weights are pinned by Euler-equation residuals at every collocation point, a structural restriction with no in-sample analogue. The heuristics, by contrast, are calibrated to match cross-sectional MPC moments and aggregate APC; they impose no intertemporal optimality. This is precisely the dimension on which the comparison is informative.

The forecast results illustrate how this discipline matters. Within the static heuristic ladder, the more flexible four-parameter rule H4 (concave income plus linear wealth) is uniformly dominated by the three-parameter rule H3 (linear income plus linear wealth) at the 12-quarter horizon under every mechanism — a pattern characteristic of overfitting when functional flexibility is added without a corresponding structural restriction. The buffer-stock rule H7, although it has the same number of free parameters as H4, imposes a global shape restriction — consumption rises smoothly toward target wealth and falls away from it — that H4 does not, and the habit rule H6 imposes an analogous restriction through smoothing across periods. Both deliver positive consumption point estimates under every mechanism. The OC-ABM imposes the same kind of shape restriction at every state through the Euler equation, but uniquely produces gains that are statistically significant under every mechanism, reflecting the lower forecast variance that follows from optimisation-driven smoothing. The comparison is therefore not between a flexible OC-ABM and parsimonious heuristics, but between three equally parsimonious classes of intertemporal model (habit, buffer-stock, Euler) that differ in the kind of structural discipline they impose: smoothing across periods (H6), shape constraints around a target wealth (H7), or shape constraints from intertemporal optimality (OC-ABM).

The OC-ABM generates plausible auto-correlation decay for GDP, operating surplus, and capital formation, and the cross-correlation profiles of both household consumption and capital formation are well-matched at lag zero. Wages and household consumption show slower ACF decay than the data, the former driven by immediate wage adjustment via overtime payments and the latter by the resulting persistence in expected income. In the cross-correlations, two systematic weaknesses emerge: wages are coincident with GDP rather than leading by approximately 5 quarters, and the GDP deflator profile is substantially flatter than in the data. Both point to the supply side (labour and goods markets) rather than the household block. The deflator dynamics improve substantially under ARX expectations, suggesting that conditioning on exogenous variables disciplines the price-setting mechanism.

Several limitations qualify the interpretation of our results. The OC-ABM consistently underperforms time-series benchmarks at 1–4 quarter horizons, meaning it is not a replacement for time-series models in nowcasting or short-term forecasting applications. The model is calibrated to the Dutch economy using CBS microdata, Dutch input-output tables, and Eurostat national accounts, and generalisability to other countries remains untested. The evaluation

sample (2010Q1–2019Q4) excludes the COVID-19 pandemic, the subsequent inflation surge, and the energy crisis. Finally, the Eurocrisis episode (Section 5.1) reveals that the model overpredicts during sovereign-debt stress, likely because it lacks a financial accelerator or balance-sheet deleveraging channel.

The OC-ABM’s long-horizon consumption gains may have practical relevance for institutions that produce medium-term forecasts, such as central banks, fiscal councils, and statistical agencies. Our results suggest that microfounded heterogeneous consumption behaviour can improve medium-term forecasts, particularly for household consumption, the largest component of GDP in most advanced economies. The model’s stock-flow consistency and sectoral granularity offer an additional advantage: forecasts can be decomposed along production, income, and expenditure dimensions (Section 5.5), enabling policymakers to trace the sources of a GDP revision. Moreover, because the OC-ABM generates MPC heterogeneity endogenously from the joint distribution of income and wealth, it can in principle evaluate the aggregate impact of distributional policies — such as targeted transfers to liquidity-constrained households versus universal transfers — that reduced-form time-series models cannot distinguish by construction; we leave such policy counterfactuals to future work, as the present paper focuses on forecasting performance.

Several extensions follow from this analysis. Enriching the labour market with search frictions or vacancy-posting delays would address the wage-timing discrepancy identified in Section 6, and replacing homogeneous AR(1) expectations with heterogeneous expectation rules (Brock and Hommes, 1997) would test robustness when expectation heterogeneity itself becomes a source of aggregate dynamics. A formal variance-ratio test of the OC-ABM versus H9 squared forecast errors would corroborate the interpretation that Euler-equation discipline operates through reduced forecast dispersion; the present evidence is the differential significance level at indistinguishable point estimates. Adding a financial accelerator or balance-sheet deleveraging channel would address the Eurocrisis overprediction documented in Section 5.1. The framework is also extensible to 1:1 agent-to-household calibration using register data, life-cycle effects, other countries with comparable microdata, and endogenous credit constraints for crisis episodes.

In sum, the gains are robust across four expectation mechanisms, statistically significant at the 12-quarter horizon, and decomposable into three additive contributions: dynamic consumption-smoothing structure (the move from static rules H1–H4 to buffer-stock H7, worth roughly the full distance from negative to positive point estimates), state-contingent portfolio choice (the move from H7 to H9, worth +11.7 pp), and Euler-equation discipline (the residual that delivers the OC-ABM’s unique statistical significance at indistinguishable point estimates). Empirically realistic MPC heterogeneity arises as a by-product of these structural elements rather than being imposed directly. Short-horizon forecasts remain a comparative weakness, and GDP improvements, while consistently positive, are not statistically significant. These results suggest that, at least for the Dutch economy over our evaluation period, microfounded household behaviour and agent-based heterogeneity are complementary rather than competing modelling strategies, and that their combination may offer a productive framework for medium-term macroeconomic forecasting in other settings where distributional dynamics matter.

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## A Model details

This appendix provides the full specification of the model components not described in the main text. The household sector, including the optimisation problem, capital market, and expectation formation, is presented in Section 2. Here we detail the firm sector (production, pricing, investment, and finance), the government sector (fiscal rules, taxation, and debt management), the banking sector (loan provision and capital requirements), the central bank (monetary policy and balance sheet), and the trade sector (imports and exports). All equations follow Poledna et al. (2023) unless otherwise noted. Parameter values and their sources are reported in Table 5 and the sector parameter tables (Table 13).

### A.1 Firms

We follow Poledna et al. (2023) for the firm sector. The sector consists of heterogeneous firms across  $s \in \{1, \dots, S\}$  industries, each containing  $I_s$  firms for a total of  $I = \sum_{s=1}^S I_s$  firms. Each firm  $i \in \{1, \dots, I\}$  produces a principal product  $g \in \{1, \dots, G\}$  using labour, capital, and intermediate inputs from other firms. Demand for firm  $i$ 's products is the sum of final consumption, capital goods, material inputs, and intermediate goods demanded.

#### A.1.1 Sales

Realised demand  $Q_{i,t}^d$  is determined after the firm sets its price and production  $Y_{i,t}$ . Firms face demand from households, government entities, and other firms seeking capital or intermediate goods.

$$Q_{i,t}^d \begin{cases} < Y_{i,t} + S_{i,t-1} & \text{if demand from consumers is smaller than supply from firm } i, \text{ and} \\ = Y_{i,t} + S_{i,t-1} & \text{if demand from consumers exactly matches supply from firm } i, \text{ and} \\ > Y_{i,t} + S_{i,t-1} & \text{if demand from consumers is larger than supply from firm } i, \end{cases} \quad (47)$$

where  $S_{i,t-1}$  is the firm's inventory of finished goods. Which firm receives demand is determined by a search-and-matching mechanism that shapes the consumption and supply networks each period. Each period, a consumer visits a number of randomly chosen foreign or domestic firms selling good  $g$ . The probability that a consumer selects firm  $i$  depends on both the firm's price and its size. Two probabilities, one based on price competitiveness and one on firm size, are computed from exponential distributions and averaged to obtain the selection probability:

$$\begin{aligned} pr_{i,t}^{\text{price}} &= \frac{e^{-2P_{i,t}}}{\sum_{i \in I_{s=g}} e^{-2P_{i,t}}} \\ pr_{i,t}^{\text{size}} &= \frac{Y_{i,t}}{\sum_{i \in I_{s=g}} Y_{i,t}} \\ pr_{i,t}^{\text{cum}} &= \frac{pr_{i,t}^{\text{price}} + pr_{i,t}^{\text{size}}}{2} \end{aligned} \quad (48)$$

Here  $pr_{i,t}^{\text{price}}$  is the probability firm  $i$  is selected by a consumer due to its offering price.  $pr_{i,t}^{\text{size}}$  is the probability firm  $i$  is selected by a consumer due to its size.  $pr_{i,t}^{\text{cum}}$  is the average probability firm  $i$  is selected by a consumer due to its size and its offering price. If the most preferred

firm runs out of stock before the consumer has satisfied its demands, the consumer goes to the other firms. If a consumer cannot satisfy its demands for a product  $g$  it saves involuntarily. Consequently, actual sales is both determined on demand and supply available and is the minimum of the two quantities:

$$Q_{i,t} = \min\{Y_{i,t} + S_{i,t-1}, Q_{i,t}^d\} \quad (49)$$

Excess supply is the difference between production and sales:

$$\Delta S_{i,t} = Y_{i,t} - Q_{i,t}. \quad (50)$$

This reflects the firm's expectation error concerning demand. Excess supply is stored as inventory:

$$S_{i,t} = S_{i,t-1} + \Delta S_{i,t}. \quad (51)$$

In the next period these goods are available again for sale at the same firm together with the newly produced goods. We assume there is no depreciation of goods.

### A.1.2 Supply

The supply choice of the firm is determined by multiplying the expected growth rate by the previous period's demand for its product:

$$Q_{i,t}^s = Q_{i,t-1}^d (1 + \gamma_t^e). \quad (52)$$

Prices reflect the previous price, expected inflation, and cost-push inflation arising from the firm's cost structure:

$$P_{i,t} = P_{i,t-1} \cdot \underbrace{(1 + \pi_{i,t}^c)}_{\text{Cost-Push Inflation}} \cdot \underbrace{(1 + \pi_t^e)}_{\text{Built-In Inflation}}, \quad (53)$$

where

$$\begin{aligned} \pi_{i,t}^c = & \underbrace{\frac{(1 + \tau^{\text{SIF}}) \bar{w}_i}{\bar{\alpha}_i} \left( \frac{\bar{P}_t^{\text{HH}}}{P_{i,t}} - 1 \right)}_{\text{Increase of unit labour costs}} + \underbrace{\frac{1}{\beta_i^M} \left( \frac{\sum_g a_{sg} \bar{P}_{g,t-1}}{P_{i,t}} - 1 \right)}_{\text{Increase of unit material costs}} \\ & + \underbrace{\frac{\delta_i}{\kappa_i} \left( \frac{\bar{P}_t^{\text{CF}}}{P_{i,t}} - 1 \right)}_{\text{Increase of unit capital costs}} \quad \forall i \in I_S. \end{aligned} \quad (54)$$

$\bar{\alpha}_i$  denotes the average productivity of labour.  $\bar{w}_i$  are average gross wages indexed by the consumer price index  $\bar{P}_t^{\text{HH}}$ , and including employers' contribution to social insurance charged with a rate  $\tau^{\text{SIF}}$ .  $\frac{1}{\beta_i^M} \sum_g a_{sg}$  are unit expenditures (in real terms) on intermediate input by industry  $s$  on good  $g$  weighted by the average product price index for good  $g$  ( $\bar{P}_{g,t}$ ).  $\delta_i/\kappa_i$  are unit capital costs due to depreciation.  $\delta_i$  is the firm's capital depreciation rate and  $\kappa_i$  the productivity coefficient for capital.  $\bar{P}_t^{\text{CF}}$  is the average price of capital goods. Inflation rate is the log difference of the consumer price index (CPI):

$$\pi_t = \log \left( \frac{\bar{P}_t}{\bar{P}_{t-1}} \right) \quad (55)$$

where the producer price index is defined as

$$\bar{p}_t = \frac{\sum_i P_{i,t} Q_{i,t} + \sum_m P_{m,t} Q_{m,t}}{\sum_i Q_{i,t} + \sum_m Q_{m,t}}. \quad (56)$$

$P_{m,t}$  and  $Q_{m,t}$  are price and sales from foreign producers  $m$  respectively. By incorporating expectations about future costs, inflation, and profits alongside current costs, firms reduce the probability of bankruptcy due to irrational pricing behaviour. Under adaptive learning, the expectation gap gradually narrows toward the actual state of the economy.

### A.1.3 Production

In every period  $t$  each firm  $i$  produces output  $Y_{i,t}$  in the form of principal product  $g$  using labour  $N_{i,t}$ , intermediate goods and raw materials  $M_{i,t}$ , and capital  $K_{i,t}$ . The firms produce by means of a Leontief production function which is given by:

$$Y_{i,t} = \min \left( Q_{i,t}^S, \beta_i^M M_{i,t}, \alpha_{i,t} N_{i,t}, \kappa_i K_{i,t-1} \right), \quad (57)$$

where  $\alpha_{i,t}$  is the firm  $i$ 's labour productivity,  $\beta_i^M$  is the firm's intermediate input productivity coefficient and  $\kappa_i$  is the firm's capital productivity. Production may be lower than desired production if it cannot obtain enough resources for production.

### A.1.4 Investment

Every firm  $i$  makes an investment decision each period which adjusts next period real capital stock  $K_{i,t}$ . As today's investment is available to use for production tomorrow capital is durable and sticky. The desired investment in capital stock is given by:

$$I_{i,t}^d = \frac{\delta_i}{\kappa_i} \min(Q_{i,t}^S, \kappa_i K_{i,t}). \quad (58)$$

$\delta_i$  is the firm's capital depreciation rate. Capital stock is homogeneous for all firms. Each firm demands  $b_g^{CF} I_{i,t}^d$  from firms producing good  $g$ . Hence, firm  $i$ 's demand for capital investment in good  $g$  is

$$I_{ig,t}^d = b_g^{CF} I_{i,t}^d. \quad (59)$$

It is possible that firm  $i$  cannot realize its capital demand. This is dependent on the search-and-matching process. We have that realized capital investment is given by:

$$I_{i,t} \begin{cases} = \sum_g I_{ig,t}^d & \text{if the firm successfully realized the investment plan, and} \\ < \sum_g I_{ig,t}^d & \text{if all firms visited could not satisfy its demand} \end{cases} \quad (60)$$

Next period capital stock is determined by realized investment, current capital stock and depreciation due to capital use:

$$K_{i,t} = K_{i,t-1} - \frac{\delta_i}{\kappa_i} Y_{i,t} + I_{i,t}. \quad (61)$$

### A.1.5 Intermediate Inputs

Intermediate inputs are used by firms for production. Each firm  $i$  holds a stock of input goods  $M_{i,t}$  in real terms for each good  $g$ . The firm ensures a positive supply to prevent impeding production. Each period firm  $i$  demands intermediate goods such that it can produce its production plan. The desired amount of intermediate goods and raw materials  $\Delta M_{i,t}^d$  is given by:

$$\Delta M_{i,t}^d = \frac{\min\{Q_{i,t}^s, \kappa_i K_{i,t-1}\}}{\beta_i^M} \quad (62)$$

Moreover, firm  $i$  belongs to industry  $s$  and consumes intermediate input  $g$  according to sector-specific technology coefficients  $a_{sg}$  which gives

$$\Delta M_{ig,t}^d = a_{sg} \Delta M_{i,t}^d, \forall i \in I_s. \quad (63)$$

The intermediate inputs also depend on a search-and-matching process which gives:

$$\Delta M_{i,t} \begin{cases} = \sum_g \Delta M_{ig,t}^d & \text{if the firm successfully realized its plan, and} \\ < \sum_g \Delta M_{ig,t}^d & \text{if all firms visited could not satisfy its demand} \end{cases} \quad (64)$$

Failure in realizing the firm's production plan results in its production being limited. The evolution of the stock of good  $g$  of firm  $i$  depends on the usage of the good, the purchasing of the good and the stock of the good of the previous period. The law of motion is:

$$M_{i,t} = M_{i,t-1} - \frac{Y_{i,t}}{\beta_i^M} + \Delta M_{i,t}. \quad (65)$$

We assume no depreciation of raw materials, as inventories are continuously replenished.

### A.1.6 Employment

Labour  $N_{i,t}$  is used as a production input by firm  $i$  measured in number of persons employed. The firm demands labour based on its according to desired scale of activity  $Q_{i,t}^s$  and average labour productivity  $\bar{\alpha}_i$ :

$$N_{i,t}^d = \max \left\{ 1, \text{round} \left( \frac{\min\{Q_{i,t}^s, \kappa_i K_{i,t-1}\}}{\bar{\alpha}_i} \right) \right\}. \quad (66)$$

If a firm demands less than a half-time position, then its demand is rounded down. If it demands more than a half-time position, then demand is rounded up and a new employee is hired. If the operating workforce is higher than is needed, the firm fires  $N_{i,t} - N_{i,t}^d$  random employees. If the demand for labour is larger than the current work force, the firm posts labour vacancies:

$$V_{i,t} = N_{i,t}^d - N_{i,t-1}. \quad (67)$$

Vacancies are filled by means of the labour market search-and-matching mechanism.

$$N_{i,t} \begin{cases} = N_{i,t}^d & \text{if the firm successfully fills all vacancies, and} \\ < N_{i,t}^d & \text{if there are unfilled vacancies.} \end{cases} \quad (68)$$

Employees can work part-time, full-time, or overtime. This is reflected in the actual productivity of labour  $\alpha_{i,t}$ . Productivity of firm  $i$  is

$$\alpha_{i,t} = \bar{\alpha}_i \min \left( 1.5, \frac{\min \left( Q_{i,t}^S, \beta_i^M M_{i,t-1}, \kappa_i K_{i,t-1} \right)}{N_{i,t} \bar{\alpha}_i} \right). \quad (69)$$

The maximum work effort is set to 150 percent of full time employment. Wages are increased or decreased to pay for work efforts relative to normal working hours, giving

$$w_{i,t} = \bar{w}_i \frac{\min \left( Q_{i,t}^S, \beta_i^M M_{i,t-1}, \kappa_i K_{i,t-1} \right)}{N_{i,t} \bar{\alpha}_i}. \quad (70)$$

Here,  $w_{i,t}$  is the real wage paid by firm  $i$ . Nominal wages are increased according to inflation.

#### A.1.7 External Finance

Firms may need to finance current or future expenditures by borrowing. Each firm forecasts its future cash flow  $\Delta D_{i,t}^e$ , equal to the expected change in deposits  $D_{i,t}$ :

$$\Delta D_{i,t}^e = \underbrace{\Pi_{i,t}^e}_{\text{Exp. profit}} - \underbrace{\theta L_{i,t-1}}_{\text{Debt instalment}} - \underbrace{\tau^{\text{FIRM}} \max(0, \Pi_{i,t}^e)}_{\text{Corporate taxes}} - \underbrace{\theta^{\text{DIV}} (1 - \tau^{\text{FIRM}}) \max(0, \Pi_{i,t}^e)}_{\text{Dividend payout}}, \quad (71)$$

where

$$\Pi_{i,t}^e = \Pi_{i,t-1} (1 + \gamma_t^e) (1 + \pi_t^e), \quad (72)$$

is expected profit of firm  $i$  which is forecasted using previous period profits.  $\theta$  is the rate of debt instalments on firm  $i$ 's outstanding loans  $L_{i,t-1}$ .  $\tau^{\text{FIRM}}$  is the corporate tax rate.  $\theta^{\text{DIV}}$  is the dividend payout ratio. We assume that the dividends are paid out proportionally to the illiquid asset holdings of the households. If a firm cannot use internal finance to pay its expenditure, it will turn to a bank for a loan to cover remaining expected future expenditures:

$$\Delta L_{i,t}^d = \max(0, -\Delta D_{i,t}^e - D_{i,t-1}) \quad (73)$$

Whether a firm is granted a loan depends on the bank's capitalisation and the number of firms requesting loans. If a firm cannot obtain a loan, it may become insolvent.

#### A.1.8 Accounting

Firm profit  $\Pi_{i,t}$  is an accounting measure, defined as revenues from sales plus changes in inventories minus expenditures on labour, material, depreciation, interest payments and taxes

$$\begin{aligned} \Pi_{i,t} = & \underbrace{P_{i,t} Q_{i,t}}_{\text{Sales}} + \underbrace{P_{i,t} \Delta S_{i,t}}_{\text{Inventory change}} - \underbrace{(1 + \tau^{\text{SIF}}) w_{i,t} N_{i,t} \bar{P}_t^{\text{HH}}}_{\text{Labour costs}} \\ & - \underbrace{\frac{1}{\beta_i^M} \bar{P}_{i,t} Y_{i,t}}_{\text{Material costs}} - \underbrace{\frac{\delta_i}{\kappa_i} P_{i,t}^{\text{CF}} Y_{i,t}}_{\text{Depreciation}} - \underbrace{\tau_i^Y P_{i,t} Y_{i,t} - \tau_i^{\text{PROD}} P_{i,t} Y_{i,t}}_{\text{Net taxes/subsidies on products/production}} \\ & - \underbrace{r_t (L_{i,t-1} + \max(0, -D_{i,t-1}))}_{\text{Interest payments}} + \underbrace{\bar{r}_t \max(0, D_{i,t-1})}_{\text{Interest received}}, \end{aligned} \quad (74)$$

where  $r_t$  is the interest rate paid on outstanding loans. The actual prices paid by firm  $i$  for intermediate goods and investment in capital goods are  $\bar{P}_{i,t}$  and  $P_{i,t}^{\text{CF}}$ . These prices are an outcome of the search-and-matching process. Finally,  $\bar{P}_t^{\text{HH}}$  is the consumer price index (CPI) :

$$\bar{P}_t^{\text{HH}} = \sum_g b_g^{\text{HH}} \bar{P}_{g,t}. \quad (75)$$

The parameter  $b_g^{\text{HH}}$  is the household consumption coefficient for product  $g$ , and  $\bar{P}_{g,t}$  is the producer price index for good  $g$ :

$$\bar{P}_{g,t} = \frac{\sum_{i \in I_{s=g}} P_{i,t} Q_{i,t} + P_{m=g,t} Q_{m=g,t}}{\sum_{i \in I_{s=g}} Q_{i,t} + Q_{m=g,t}}. \quad (76)$$

Firm net cash flow is a measure how much liquidity is available for firms to use and is equal to changes in the deposit account:

$$\begin{aligned} \Delta D_{i,t} = & \underbrace{P_{i,t} Q_{i,t}}_{\text{Sales}} - \underbrace{\left(1 + \tau^{\text{SIF}}\right) w_{i,t} N_{i,t} \bar{P}_t^{\text{HH}}}_{\text{Labor costs}} - \underbrace{\bar{P}_{i,t} \Delta M_{i,t}}_{\text{Material costs}} \\ & - \underbrace{\tau_i^Y P_{i,t} Y_{i,t} - \tau_i^{\text{PROD}} P_{i,t} Y_{i,t}}_{\text{Net taxes/subsidies on products and production}} - \underbrace{\tau^{\text{FIRM}} \max(0, \Pi_{i,t})}_{\text{Corporate tax payments}} \\ & - \underbrace{\theta^{\text{DIV}} \left(1 - \tau^{\text{FIRM}}\right) \max(0, \Pi_{i,t})}_{\text{Dividend payments}} - \underbrace{r (L_{i,t-1} + \max(0, -D_{i,t-1}))}_{\text{Interest payments}} \\ & + \underbrace{\bar{r}_t \max(0, D_{i,t-1})}_{\text{Interest received}} - \underbrace{P_{i,t}^{\text{CF}} I_{i,t}}_{\text{Investment costs}} + \underbrace{\Delta L_{i,t}}_{\text{New credit}} - \underbrace{\theta L_{i,t-1}}_{\text{Debt instalment}}. \end{aligned} \quad (77)$$

Firms have to pay interest on loans and overdrafts (in case  $D_{i,t} < 0$ ) at rate  $r_t$ , which includes the bank's markup rate. When firms have a positive balance on the bank account, i.e.  $D_{i,t} > 0$ , the interest rate received is lower and is equal to the rate set by the central bank. Firm deposits are equal to deposits plus net cash flow:

$$D_{i,t} = D_{i,t-1} + \Delta D_{i,t} \quad (78)$$

Similarly, debt follows the following law of motion:

$$L_{i,t} = (1 - \theta) L_{i,t-1} + \Delta L_{i,t} \quad (79)$$

Firm equity  $E_{i,t}$  is computed as a balancing item on the firm's balance sheet taking into account deposits debts, capital and input goods. All goods are accounted for market-to-market:

$$E_{i,t} = D_{i,t} + \sum_g a_{sg} \bar{P}_{g,t} M_{i,t} + P_{i,t} S_{i,t} + \bar{P}_t^{\text{CF}} K_{i,t} - L_{i,t} \quad \forall i \in I_s \quad (80)$$

$\bar{P}_t^{\text{CF}}$  is defined as the capital goods price index (CGPI) which is given by:

$$\bar{P}_t^{\text{CF}} = \sum_g b_g^{\text{CF}} \bar{P}_{g,t}, \quad (81)$$

where  $b_g^{\text{CF}}$  is the capital formation coefficient for product  $g$ .

### A.1.9 Insolvency

When a firm has a negative deposit balance ( $D_{i,t} < 0$ ) and negative equity ( $E_{i,t} < 0$ ) simultaneously, it declares bankruptcy. The firm is then replaced by a new firm which enters the market. The real capital stock is taken over from the bankrupt firm at zero costs however the new firm has also to take on part of the liabilities of the old firm. The remaining liabilities of firm  $i$  amount to a fraction  $\zeta^b$  of its real capital stock for the new firm as a part of the loans is written off. The remaining liabilities are transferred to the balance sheet of the new firm. In the next period ( $t + 1$ ) liabilities of the new firm  $i$  are given by

$$L_{i,t+1} = \zeta^b \bar{P}_{i,t}^{\text{CF}} K_{i,t}. \quad (82)$$

The new firm starts with zero deposits:

$$D_{i,t+1} = 0. \quad (83)$$

The new firm's equity is initialised as

$$E_{i,t+1} = E_{i,t} + (L_{i,t} - D_{i,t} - \zeta^b \bar{P}_{i,t}^{\text{CF}} K_{i,t}) \quad (84)$$

## A.2 Government

The Dutch government is modelled to have two functions: to act as a consumer on the retail market to purchase government consumption and as a redistributive entity that redistributes levied taxes to provide social services to its citizens. Government consumption is assumed to be exogenous.

### A.2.1 Government Consumption

The government consists of governmental entities  $j \in \{1, \dots, J\}$  which participate in the markets as consumers. These entities represent the state, the provincial government, the municipalities and social security funds. Governmental demand follows the following rule:

$$\log(C_t^G) = \alpha^G \log(C_{t-1}^G) + \beta^G + \epsilon_{t-1}^G \quad (85)$$

where  $\epsilon_{t-1}^G$  is a random shock with zero mean and variance  $(\sigma^G)^2$ . The total nominal government consumption demand is then uniformly distributed to the  $J$  government entities and attributed to goods  $g$ :

$$C_{j,t}^d = \frac{C_t^G \sum_g c_g^G \bar{P}_{g,t-1} (1 + \pi_t^e)}{J} \quad (86)$$

and the consumption budget of the  $j$ -th government entity to purchase the  $g$ -th good is given by

$$C_{jg,t}^d = \frac{c_g^G \bar{P}_{g,t-1}}{\sum_g c_g^G \bar{P}_{g,t-1}} C_{j,t}^d \quad (87)$$

where  $c_g^G$  is the fraction of goods of type  $g$  demanded by the government. Realized government consumption is then determined again by the search-and-matching process on the consumption goods market:

$$C_{j,t} \begin{cases} = \sum_g C_{jg,t}^d & \text{if the government successfully realized the consumption plan, and} \\ < \sum_g C_{jg,t}^d & \text{if all firms visited could not satisfy its demand.} \end{cases}$$

The government also makes other expenditures such as interest payments, social benefits other than social transfers in kind, and subsidies. The government pays interest at rate  $r^G$  on outstanding government debt  $L_{t-1}^G$ . The government also pays social benefits for inactive households ( $\sum_{h \in H^{\text{inact}}} sb_t^{\text{inact}}$ ), social benefits for any household  $h$  ( $\sum_h sb_t^{\text{other}}$ ), and unemployment benefits for unemployed households ( $\sum_{h \in H_t^U} w_{h,t}$ ). The government also pays subsidies. These are paid to firms via subsidy rates (uniform for each industry, but different across industries) on products and production. These subsidies are incorporated in the net tax rates on products ( $\tau_i^Y$ ) and production ( $\tau_i^{\text{PROD}}$ ).

### A.2.2 Government Revenue

Taxes, social contributions and other transfers from all sectors form the revenue of the government:

$$\begin{aligned}
Y_t^G = & \underbrace{\left( \tau^{\text{SIF}} + \tau^{\text{SIW}} \right) \bar{P}_t^{\text{HH}} \sum_{h \in H_t^E} w_{h,t}}_{\text{Social security contributions}} + \underbrace{\tau^{\text{INC}} \left( 1 - \tau^{\text{SIW}} \right) \bar{P}_t^{\text{HH}} \sum_{h \in H_t^E} w_{h,t}}_{\text{Labor income taxes}} \\
& + \underbrace{\tau_h^{\text{VAT}} \sum_h C_{h,t}}_{\text{Value added taxes}} + \underbrace{\tau^{\text{INC}} \left( 1 - \tau^{\text{FIRM}} \right) \theta^{\text{DIV}} \left( \sum_i \max(0, \Pi_{i,t}) + \max(0, \Pi_{k,t}) \right)}_{\text{Capital income taxes}} \\
& + \underbrace{\tau^{\text{FIRM}} \left( \sum_i \max(0, \Pi_{i,t}) + \max(0, \Pi_{k,t}) \right)}_{\text{Corporate income taxes}} + \underbrace{\tau^{\text{CF}} \sum_h I_{h,t}}_{\text{Taxes on capital formation}} \\
& + \underbrace{\sum_i \tau_i^Y P_{i,t} Y_{i,t}}_{\text{Net taxes/subsidies on products}} + \underbrace{\sum_i \tau_i^{\text{PROD}} P_{i,t} Y_{i,t}}_{\text{Net taxes/subsidies on production}} + \underbrace{\tau_l^{\text{EXPORT}} \sum_l C_{l,t}}_{\text{Export taxes}}.
\end{aligned} \tag{88}$$

### A.2.3 Government Deficit

The government deficit (or surplus) is an accounting identity consisting of expenditures minus revenues:

$$\begin{aligned}
\Pi_t^G = & \underbrace{\sum_{h \in H^{\text{inact}}} \bar{P}_t^{\text{HH}} sb_t^{\text{inact}} + \sum_{h \in H_t^U} \bar{P}_t^{\text{HH}} \theta^{\text{UB}} w_{h,t} + \sum_h \bar{P}_t^{\text{HH}} sb_t^{\text{other}}}_{\text{Social benefits and transfers}} \\
& + \underbrace{\sum_j C_{p,j,t}}_{\text{Government consumption}} + \underbrace{r^G L_{t-1}^G}_{\text{Interest payments}} - \underbrace{Y_t^G}_{\text{Government revenues}}.
\end{aligned} \tag{89}$$

### A.2.4 Government Debt

The government debt is previous period government debt plus the current deficit:

$$L_t^G = L_{t-1}^G + \Pi_t^G. \tag{90}$$

We assume that the central bank buys all government debt.

### A.3 Bank

For simplicity, we assume that there is a single bank that represent the entire private financial system. The representative bank takes deposits from households and firms, extends loans to firms, and receives advances from (or deposits reserves at) the central bank.

#### A.3.1 Provision of Loans

The bank extends loans to firms according to a risk assessment of potential borrowers and is subject to minimum capital requirements imposed by the regulator. The bank uses fractional reserve banking, thus the bank is allowed to loan up to a multiple of its equity. The bank is however uncertain about the realized values of its equity due to fundamental uncertainty and thus the bank must forecast future equity value  $E_{k,t}^e$  and future outstanding loan value  $\sum_{i=1}^I (L_{i,t}^e + \Delta L_{i,t})$ . The bank has to adhere to a minimum capital requirement coefficient  $0 < \zeta < 1$ :

$$\frac{E_{k,t}^e}{\sum_{i=1}^I (L_{i,t}^e + \Delta L_{i,t})} = \frac{E_{k,t-1}}{\sum_{i=1}^I ((1 - \theta)L_{i,t-1} + \Delta L_{i,t})} \geq \zeta, \quad (91)$$

Therefore,  $1/\zeta$  is the maximum allowed leverage for the bank.  $\Delta L_{i,t}$  is the realized new loans to firm  $i$  in period  $t$ , which is equivalent to the new credit demanded by firms  $\Delta L_{i,t}^d$  if the bank capital requirements are met and the firm borrows within the maximum LTV. When bank capital and leverage requirements are not met, no lending can take place. Furthermore, the bank forms a risk assessment of a potential default on the part of firm  $i$  before extending a loan to it. This risk assessment is based on the borrower's leverage as measured by its loan-to-value ratio, that is the amount of loans over the market value of its capital stock. Thus, the bank will grant a loan to firm  $i$  only up to the point where the borrower's leverage (or loan-to-value) ratio after the loan (including overdrafts on deposit accounts), is below a regulated level which is  $\zeta^{\text{LTV}}$ . Due to fundamental uncertainty, the bank has to form expectations on the value of firm  $i$ 's capital stock  $(K_{i,t}^e)$ :

$$\frac{L_{i,t}^e + \Delta L_{i,t}}{K_{i,t}^e} = \frac{\overbrace{(1 - \theta)L_{i,t-1} + \Delta L_{i,t}}^{=L_{i,t}^e}}{\underbrace{\bar{P}_{t-1}^{\text{CF}} (1 + \pi_t^e) K_{i,t-1}}_{=K_{i,t}^e}} \leq \zeta^{\text{LTV}}. \quad (92)$$

Here,  $K_{i,t}^e$  denotes the amount of collateral the bank expects a certain firm  $i$  to hold at the moment when the firm asks for credit, while  $L_{i,t}^e$  is the amount of debt already in the books of the firm. Altogether, therefore, the amount of new credit extended to firm  $i$  by the bank ( $\Delta L_{i,t}$ ) depends on the credit demanded by the firm, the bank's risk assessment regarding the default of its potential borrower, and the minimum capital requirements imposed by the regulator:

$$\Delta L_{i,t} \begin{cases} = \Delta L_{i,t}^d & \text{if the borrower's loan-to-value ratio and capital requirements are satisfied} \\ < \Delta L_{i,t}^d & \text{otherwise.} \end{cases} \quad (93)$$

The order of arrival of firms at the bank is random and is important for loan provision. If financially robust (low leverage) firms come late to the bank, then it may be denied credit if it arrived "too late" (i.e. after other less robust firms).

### A.3.2 Accounting for Profits and Losses

The difference between interest revenues from loan provisions (including overdrafts on deposit accounts incurred by firms and households ( $D_{i,h,t-1} < 0$ )) and costs due to interest payments on deposits held with the bank by firms and households ( $D_{i,h,t-1} > 0$ ) are the bank's profits. They are computed as follows:

$$\begin{aligned} \Pi_{k,t} = & \underbrace{r_t \left( \sum_{i=1}^I L_{i,t-1} + \max(0, -D_{i,t-1}) + r_t \sum_{h=1}^H \max(0, -D_{h,t-1}) + \bar{r}_t \max(0, D_{k,t-1}) \right)}_{\text{Interest received}} \\ & - \underbrace{\bar{r}_t \sum_{i=1}^I \max(0, D_{i,t-1}) - \bar{r}_t \sum_{h=1}^H \max(0, D_{h,t-1}) - \bar{r}_t \max(0, -D_{k,t-1})}_{\text{Interest payments}} \end{aligned} \quad (94)$$

The bank pays policy rate  $\bar{r}_t$  on deposits. This rate is set exogenously by the central bank. The effective interest rate  $r_t$  faced for bank credit or overdrafting is determined by the policy rate  $\bar{r}_t$  and a fixed markup  $\mu$ :

$$r_t = \bar{r}_t + \mu \quad (95)$$

Bank equity is an accounting identity which is determined by bank profits or losses and is given by

$$\begin{aligned} E_{k,t} = & E_{k,t-1} + \Pi_{k,t} - \underbrace{\theta^{\text{DIV}} (1 - \tau^{\text{FIRM}}) \max(0, \Pi_{k,t})}_{\text{Dividend payments}} \\ & - \underbrace{\tau^{\text{FIRM}} \max(0, \Pi_{k,t})}_{\text{Corporate taxes}} - \underbrace{\sum_{i \in I'} (L_{i,t} - D_{i,t} - \zeta^b \bar{P}_{i,t}^{\text{CF}} K_{i,t})}_{\text{Write-off of bad debt}}, \end{aligned} \quad (96)$$

where  $I'$  is the set of insolvent borrowers. Moreover, the outstanding overdraft of firm  $i$ 's deposit account as well as a fraction  $(1 - \zeta^b) \bar{P}_{i,t}^{\text{CF}} K_{i,t}$  of loans extended to firm  $i$  have to be written off from the bank's balance sheet when firm  $i$  is declared bankrupt. The bank's balance sheet consists of (1) loans to firms (on the asset side); (2) deposits received (on the liability side), (3) equity capital; and (4) (net) central bank reserves held,  $D_{k,t}$ , or advances obtained by the bank from the central bank:

$$D_{k,t} = \sum_{i=1}^I D_{i,t} + \sum_{h=1}^H D_{h,t} + E_{k,t} - \sum_{i=1}^I L_{i,t} \quad (97)$$

### A.4 Central Bank

The central bank (CB) determines monetary policy by setting the policy rate  $\bar{r}_t$ . The central bank does so taking in consideration implicit inflation and growth targets. Its goal is to provide liquidity to the banking system, and to take deposits from the bank in the form of reserves deposited at the central bank. Finally, the central bank also purchases external assets (government bonds).

#### A.4.1 Determination of Interest Rates

A generalized Taylor rule (Taylor, 1993) determines the policy rate. We use the same rule as Blattner and Margaritov (2010), which includes a "growth" rule specification:

$$\bar{r}_t = \rho \bar{r}_{t-1} + (1 - \rho) \left( r^* + \pi^* + \xi^\pi \left( \pi_t^{\text{EA}} - \pi^* \right) + \xi^\gamma \gamma_t^{\text{EA}} \right) \quad (98)$$

where  $\rho$  is a measure for gradual adjustment of the policy rate,  $r^*$  is the real equilibrium interest rate,  $\pi^*$  is the inflation target of the central bank,  $\xi^\pi$  is the weight the central bank puts on inflation targeting and  $\xi^\gamma$  the weight placed on economic growth, respectively. Note that the output gap does not enter the equation. Inflation ( $\pi_t^{\text{EA}}$ ) and economic growth ( $\gamma_t^{\text{EA}}$ ) of the monetary union follow an AR(1) rule given by:

$$\log \left( 1 + \pi_t^{\text{EA}} \right) = \alpha^{\pi^{\text{EA}}} \log \left( 1 + \pi_{t-1}^{\text{EA}} \right) + \beta^{\pi^{\text{EA}}} + \epsilon_{t-1}^{\pi^{\text{EA}}} \quad (99)$$

and

$$\log \left( \gamma_t^{\text{EA}} \right) = \alpha^{\gamma^{\text{EA}}} \log \left( \gamma_{t-1}^{\text{EA}} \right) + \beta^{\gamma^{\text{EA}}} + \epsilon_{t-1}^{\gamma^{\text{EA}}} \quad (100)$$

where

$$\gamma_t^{\text{EA}} = \frac{\gamma_t^{\text{EA}}}{\gamma_{t-1}^{\text{EA}}} - 1 \quad (101)$$

and  $\epsilon_{t-1}^{\pi^{\text{EA}}}$  and  $\epsilon_{t-1}^{\gamma^{\text{EA}}}$  are random shocks with zero mean.  $\epsilon_{t-1}^{\pi^{\text{EA}}}$  has variance  $(\sigma^{\pi^{\text{EA}}})^2$ . We assume that the small open economy is part of a monetary union and has no influence on the policy rate.

#### A.4.2 Accounting for Profits and Losses

The difference between revenues from interest payments on government debt, as well as revenues ( $D_{k,t} < 0$ ) or costs ( $D_{k,t} > 0$ ) as a result of the net position in advances or reserves of the banking system is defined as the central bank's profits  $\Pi_t^{\text{CB}}$ . Profits are computed by:

$$\Pi_t^{\text{CB}} = r^G L_{t-1}^G - \bar{r}_t D_{k,t-1}. \quad (102)$$

The central bank's equity  $E_t^{\text{CB}}$  is an accounting identity dependent on its profits or losses and its past equity. It is computed by:

$$E_t^{\text{CB}} = E_{t-1}^{\text{CB}} + \Pi_t^{\text{CB}}. \quad (103)$$

The following law of motion determines the net creditor/debtor position of the national economy to the rest of the world ( $D_t^{\text{RoW}}$ ):

$$D_t^{\text{Row}} = D_{t-1}^{\text{Row}} - \underbrace{\left( 1 + \tau^{\text{EXPORT}} \right) \sum_l C_{l,t}}_{\text{Exports}} + \underbrace{\sum_m P_{m,t} Q_{m,t}}_{\text{Imports}}. \quad (104)$$

For instance, a balance of trade surplus ( $(1 + \tau^{\text{EXPORT}}) \sum_l C_{l,t} > \sum_m P_{m,t} Q_{m,t}$ ) enters with a negative sign. This is such as  $D_t^{\text{RoW}}$  is on the liability side of the central banks balance sheet. Thus a trade surplus would reduce national liabilities versus the RoW as it is an inflow of money into the national economy. Our financial system is closed via the accounting identity

due to inherent stock-flow consistency. This connects the change in the amount of deposits in the banking system to the government deficit (surplus) and to the balance of trade:

$$\begin{aligned} E_t^{\text{CB}} + D_t^{\text{RoW}} &= L_t^G - D_{k,t} \\ &= L_t^G - \sum_{i=1}^I D_{i,t} - \sum_{h=1}^H D_{h,t} - E_{k,t} + \sum_{i=1}^I L_{i,t}. \end{aligned} \quad (105)$$

## A.5 Imports and Exports

Trade with the rest of the world (RoW) is modelled by a set of foreign agents that trade with the domestic economy. Each sector is represented by a single foreign agent which supplies goods on the domestic market for intermediate, capital, and consumption goods. Foreign consumption is also modelled by a single agent for each market. We assume a small open economy setting and hence that demand for export and supply for import is exogenous. Hence, imports are constrained by the exogenously set foreign supply. Demand for imports and supply for export is however endogenous.

### A.5.1 Imports

Total supply of Imports  $Y_t^I$  in real terms follows an AR(1) rule:

$$\log(Y_t^I) = \alpha^I \log(Y_{t-1}^I) + \beta^I + \epsilon_{t-1}^I, \quad (106)$$

where  $\epsilon_{t-1}^I$  is a random shock with zero mean. A representative foreign firm for each sector imports goods from the rest of the world and supplies them to domestic markets. Thus the  $m$ -th, ( $m \in \{1, \dots, S\}$ ), foreign firm representing an industry  $s$  imports the principal product  $g$

$$Y_{m,t} = c_{g=s}^I Y_t^I \quad (107)$$

where  $c_{g=s}^I$  is the fraction of imported goods of type  $g$  as part of total imports. Import price is assumed to develop similarly as the average sectoral domestic adjusted for expected inflation.

$$P_{m,t} = \bar{P}_{g,t-1}(1 + \pi_t^e), \quad (108)$$

where  $m$  produces the principal product  $g$ . Sales of imports are equal to the realized demand which is an outcome of the search-and-matching process on the goods markets:

$$Q_{m,t} = \min(Y_{m,t}, Q_{m,t}^d) \quad (109)$$

where  $Q_{m,t}^d$  is subject to the search-and-matching mechanism specifying the demand by consumers from foreign firm  $m$ :

$$Q_{m,t}^d \begin{cases} < Y_{m,t} & \text{if demand from consumers is smaller than supply from foreign firm } m, \\ = Y_{m,t} & \text{if demand from consumers exactly matches supply from foreign firm } m, \text{ and} \\ > Y_{m,t} & \text{if demand from consumers is larger than supply from foreign firm } m. \end{cases} \quad (110)$$

### A.5.2 Exports

The export market consists of foreign consumers  $l \in \{1, \dots, L\}$  which act in the domestic market as consumer. Total demand for exports  $C_t^E$  is assumed to be exogenous and follows an AR(1) rule:

$$\log(C_t^E) = \alpha^E \log(C_{t-1}^E) + \beta^E + \epsilon_{t-1}^E, \quad (111)$$

where  $\epsilon_{t-1}^E$  is a random shock with zero mean. The total demand for exports is then uniformly distributed to the  $L$  foreign consumers and attributed to goods  $g$ :

$$C_{l,t}^d = \frac{C_t^E \sum_g c_{g,t}^E \bar{P}_{g,t-1} (1 + \pi_t^e)}{L}, \quad (112)$$

and the demand for exported goods by the  $l$ -th foreign consumer to purchase the  $g$ -th good is given by

$$C_{lg,t}^d = \frac{c_g^E \bar{P}_{g,t-1}}{\sum_g c_g^E \bar{P}_{g,t-1}} C_{l,t}^d, \quad (113)$$

where  $c_g^E$  is the fraction of exports of goods of type  $g$ . Realized consumption by foreign consumers is again determined by the search-and-matching process on goods markets:

$$C_{l,t} \begin{cases} = \sum_g C_{lg,t}^d & \text{if the foreign consumer successfully realized the consumption plan, and} \\ < \sum_g C_{lg,t}^d & \text{if all firms visited could not satisfy its demand.} \end{cases} \quad (114)$$

### A.6 Macroeconomic Aggregates

Macroeconomic aggregates follow Poledna et al. (2023). GDP in the model is calculated by adding the value of all final goods and services produced and purchased by agents in the model in a given period. The nominal GDP and its components of each period  $t$  can be measured by three different approaches: (1) production; (2) expenditure based, and (3) income

based:

$$\begin{aligned}
\text{GDP}_t &= \underbrace{\sum_i \tau_i^Y P_{i,t} Y_{i,t} + \sum_h \tau^{\text{VAT}} C_{h,t} + \sum_h \tau^{\text{CF}} I_{h,t} + \sum_j \tau^G C_{j,t} + \sum_l \tau^{\text{EXPORT}} C_{l,t}}_{\text{Taxes on products}} \\
&+ \underbrace{\sum_i \left(1 - \tau_i^Y\right) P_{i,t} Y_{i,t}}_{\text{Total sales of goods and services}} - \underbrace{\sum_i \frac{1}{\beta_i^M} \bar{P}_{i,t} Y_{i,t}}_{\text{Intermediate inputs}} \quad (\text{Production Approach}) \\
&= \underbrace{\sum_h \left(1 + \tau^{\text{VAT}}\right) C_{h,t}}_{\text{Household consumption}} + \underbrace{\sum_j \left(1 + \tau^G\right) C_{j,t}}_{\text{Government consumption}} + \underbrace{\sum_h \left(1 + \tau^{\text{CF}}\right) I_{h,t} + \sum_i P_{i,t}^{\text{CF}} I_{i,t}}_{\text{Gross fixed capital formation}} \\
&+ \underbrace{\sum_i P_{i,t} (Y_{i,t} - Q_{i,t}) + \bar{P}_{i,t} \left(\Delta M_{i,t} - \frac{1}{\beta_i^M} Y_{i,t}\right)}_{\text{Changes in inventories}} \quad (\text{Expenditure Approach}) \\
&+ \underbrace{\sum_l \left(1 + \tau^{\text{EXPORT}}\right) C_{l,t}}_{\text{Exports}} - \underbrace{\sum_m P_{m,t} Q_{m,t}}_{\text{Imports}} \\
&= \underbrace{\sum_i \tau_i^Y P_{i,t} Y_{i,t} + \sum_h \tau^{\text{VAT}} C_{h,t} + \sum_h \tau^{\text{CF}} I_{h,t} + \sum_j \tau^G C_{j,t} + \sum_l \tau^{\text{EXPORT}} C_{l,t}}_{\text{Taxes on products}} \\
&+ \underbrace{\sum_i P_{i,t} Y_{i,t} - \left(1 + \tau^{\text{SIF}}\right) \bar{P}_t^{\text{HH}} N_{i,t} w_{i,t} - \frac{1}{\beta_i^M} \bar{P}_{i,t} Y_{i,t} - \tau_i^Y P_{i,t} Y_{i,t} - \tau_i^{\text{PROD}} P_{i,t} Y_{i,t}}_{\text{Gross operating surplus and mixed income}} \\
&+ \underbrace{\sum_i \left(1 + \tau^{\text{SIF}}\right) \bar{P}_t^{\text{HH}} N_{i,t} w_{i,t}}_{\text{Compensation of employees}} + \underbrace{\sum_i \tau_i^{\text{PROD}} P_{i,t} Y_{i,t}}_{\text{Net taxes on production}} \quad (\text{Income Approach})
\end{aligned} \tag{115}$$

The GDP deflator is a measure of inflation and is an economy-wide average price of all goods and services produced and sold. All individual prices and sales are determined on the agent level by the search and matching mechanism. In our model, the GDP deflator is defined as nominal GDP divided by real GDP:

$$\text{GDP deflator}_t = \frac{\text{GDP}_t}{\text{real GDP}_t}, \tag{116}$$

where

$$\begin{aligned}
\text{real GDP}_t &= \sum_i \tau_i^Y Y_{i,t} + \sum_h \tau^{\text{VAT}} \frac{C_{h,t}}{\bar{P}_{h,t}} + \sum_h \tau^{\text{CF}} \frac{I_{h,t}}{\bar{P}_{h,t}^{\text{CF}}} + \sum_j \tau^G \frac{C_{j,t}}{\bar{P}_{j,t}} \\
&+ \sum_l \tau^{\text{EXPORT}} \frac{C_{l,t}}{\bar{P}_{l,t}} + \sum_i \left(1 - \tau_i^Y\right) Y_{i,t} - \sum_i \frac{1}{\beta_i^M} Y_{i,t}.
\end{aligned} \tag{117}$$

## B Sector Parameters

In this section we report the sector parameters. Parameters are shown for 2010Q1. Column headers correspond to the number of firms ( $I_s$ ), employment ( $N_s$ ), labour productivity ( $\alpha_s$ ), intermediate input productivity ( $\beta_s$ ), capital productivity ( $\kappa_s$ ), depreciation rate ( $\delta_s$ ), average wage ( $w_s$ ), net product tax rate ( $\tau_s^Y$ ), net production tax rate ( $\tau_s^{\text{PROD}}$ ), and the capital formation ( $b_g^{CF}$ ), household capital formation ( $b_g^{CFH}$ ), household consumption ( $b_g^{HH}$ ), government consumption ( $c_g^G$ ), export ( $c_g^E$ ), and import ( $c_g^I$ ) coefficients.

Table 13: Sectoral parameters

|        | $I_s$ | $N_s$ | $\alpha_s$ | $\beta_s$ | $\kappa_s$ | $\delta_s$ | $w_s$   | $\tau_s^Y$ | $\tau_s^{\text{PROD}}$ | $b_s^{CF}$ | $b_s^{CFH}$ | $b_s^{HH}$ | $c_s^G$ | $c_s^E$ | $c_s^I$ |
|--------|-------|-------|------------|-----------|------------|------------|---------|------------|------------------------|------------|-------------|------------|---------|---------|---------|
| A01    | 47    | 50    | 142.2422   | 1.6442    | 0.0697     | 0.0092     | 10.228  | 0.0081     | -0.0198                | 0.0038     | 0           | 0.0116     | 0.0003  | 0.0371  | 0.0341  |
| A02    | 1     | 1     | 62.7955    | 1.8045    | 0.325      | 0.0135     | 17.0328 | 0.0125     | 0                      | 0          | 0           | 0          | 0.0006  | 0.0006  | 0.0005  |
| A03    | 1     | 1     | 138.935    | 2.2311    | 0.0531     | 0.0083     | 12.2553 | 0          | 0.0038                 | 0          | 0           | 0.0002     | 0       | 0.0015  | 0.0009  |
| B      | 1     | 4     | 1363.5774  | 5.8705    | 0.3027     | 0.0238     | 42.0109 | 0.0006     | 0.0008                 | 0.0051     | 0           | 0.0122     | 0.0001  | 0.0456  | 0.1006  |
| C10-12 | 3     | 61    | 226.8615   | 1.2876    | 0.7901     | 0.029      | 18.8172 | 0.0058     | 0.004                  | 0.0025     | 0           | 0.0692     | 0       | 0.0805  | 0.0602  |
| C13-15 | 2     | 7     | 118.0405   | 1.5052    | 0.5931     | 0.0346     | 18.1308 | 0.0085     | 0.0003                 | 0.002      | 0           | 0.0206     | 0       | 0.019   | 0.0327  |
| C16    | 1     | 7     | 96.4733    | 1.5227    | 0.6919     | 0.0287     | 18.6946 | 0.0031     | -0.0046                | 0.0019     | 0           | 0.001      | 0       | 0.0017  | 0.0062  |
| C17    | 1     | 9     | 167.7744   | 1.3859    | 0.6517     | 0.0443     | 22.5258 | 0.0047     | 0.0005                 | 0.0001     | 0           | 0.0032     | 0       | 0.0088  | 0.0125  |
| C18    | 1     | 14    | 88.4743    | 1.7152    | 0.5085     | 0.0472     | 18.6204 | 0.0034     | 0                      | 0.0006     | 0           | 0.0007     | 0       | 0.001   | 0       |
| C19    | 1     | 3     | 2298.3141  | 1.0264    | 1.3098     | 0.0306     | 40.1587 | 0.0003     | 0.0003                 | 0.0006     | 0           | 0.0124     | 0       | 0.07    | 0.0525  |
| C20    | 1     | 22    | 469.3842   | 1.269     | 1.2584     | 0.0668     | 29.883  | 0.0018     | 0.0002                 | 0.004      | 0           | 0.0056     | 0       | 0.0979  | 0.0797  |
| C21    | 1     | 8     | 187.765    | 1.9297    | 0.3421     | 0.0467     | 27.5745 | 0.0014     | 0.0014                 | 0.0014     | 0           | 0.0044     | 0.0145  | 0.0408  | 0.049   |
| C22    | 1     | 15    | 119.6428   | 1.5382    | 0.684      | 0.0385     | 19.6085 | 0.0055     | -0.0022                | 0.0039     | 0           | 0.002      | 0       | 0.0157  | 0.0191  |
| C23    | 1     | 13    | 109.248    | 1.5451    | 0.4147     | 0.0228     | 20.1007 | 0.0088     | 0.0004                 | 0.0014     | 0           | 0.0014     | 0       | 0.0037  | 0.0076  |
| C24    | 2     | 11    | 190.4082   | 1.3076    | 0.8387     | 0.0399     | 22.3391 | -0.0006    | -0.0046                | 0.0007     | 0           | 0.0002     | 0       | 0.0266  | 0.0318  |
| C25    | 4     | 42    | 105.1637   | 1.5537    | 0.6349     | 0.0255     | 19.0012 | 0.003      | -0.0021                | 0.0146     | 0           | 0.0025     | 0       | 0.0117  | 0.0144  |
| C26    | 1     | 14    | 464.8173   | 1.2193    | 0.3015     | 0.0141     | 26.1131 | 0.0008     | -0.002                 | 0.0502     | 0           | 0.0135     | 0.0045  | 0.1258  | 0.1399  |
| C27    | 1     | 10    | 129.0447   | 1.6943    | 0.5589     | 0.0932     | 22.1635 | 0.0041     | -0.0041                | 0.006      | 0           | 0.0041     | 0.0001  | 0.021   | 0.0271  |
| C28    | 2     | 36    | 152.8241   | 1.6006    | 0.8951     | 0.0506     | 23.3451 | 0.0025     | -0.0034                | 0.0587     | 0           | 0.0007     | 0       | 0.0579  | 0.0512  |
| C29    | 1     | 10    | 155.0001   | 1.4389    | 0.6434     | 0.0463     | 20.0447 | 0.0039     | -0.0037                | 0.0523     | 0           | 0.0134     | 0       | 0.0147  | 0.0323  |
| C30    | 1     | 9     | 189.84     | 1.4251    | 1.5109     | 0.0514     | 21.3257 | 0.0018     | -0.0031                | 0.0293     | 0           | 0.0039     | 0       | 0.0108  | 0.0102  |
| C31_32 | 6     | 71    | 40.5554    | 2.0417    | 0.76       | 0.0399     | 13.212  | 0.0093     | -0.0067                | 0.022      | 0           | 0.0104     | 0.0151  | 0.0208  | 0.0274  |
| C33    | 2     | 18    | 114.3401   | 1.5387    | 3.119      | 0.1063     | 24.5799 | 0.0018     | -0.0008                | 0.0036     | 0           | 0.0003     | 0       | 0.0025  | 0.0017  |
| D      | 1     | 13    | 357.6926   | 1.6992    | 0.3929     | 0.0507     | 25.709  | 0.0007     | 0.0011                 | 0.0147     | 0           | 0.018      | 0       | 0.0022  | 0.0019  |
| E36    | 1     | 3     | 148.0054   | 3.1081    | 0.0634     | 0.0173     | 21.8796 | 0.0843     | 0.0189                 | 0.0007     | 0           | 0.0039     | 0       | 0       | 0       |
| E37-39 | 2     | 14    | 133.0105   | 1.5777    | 0.1107     | 0.0094     | 20.7124 | 0.0103     | -0.0049                | 0.0003     | 0           | 0.0012     | 0.0005  | 0.0078  | 0.0088  |
| F      | 74    | 184   | 124.4548   | 1.5484    | 0.6466     | 0.0163     | 21.1838 | 0.0053     | -0.0002                | 0.3977     | 0           | 0.0019     | 0.0031  | 0.006   | 0       |
| G45    | 16    | 57    | 75.3683    | 1.9114    | 0.6931     | 0.0391     | 16.4425 | 0.0112     | -0.0022                | 0.0099     | 0           | 0.0251     | 0       | 0.0027  | 0       |
| G46    | 43    | 219   | 96.3222    | 2.3218    | 0.7276     | 0.0364     | 21.374  | 0.0035     | -0.0044                | 0.0015     | 0           | 0.0506     | 0.0055  | 0.0884  | 0.0086  |
| G47    | 58    | 334   | 27.2404    | 2.6443    | 0.5602     | 0.042      | 7.856   | 0.0064     | -0.0085                | 0.0028     | 0           | 0.1184     | 0.0099  | 0.0031  | 0       |
| H49    | 9     | 91    | 74.3942    | 1.8315    | 0.2525     | 0.0153     | 17.3844 | 0.0303     | 0.0015                 | 0.0058     | 0           | 0.0213     | 0.0038  | 0.0193  | 0.0099  |
| H50    | 3     | 8     | 232.2778   | 1.4592    | 0.2618     | 0.0274     | 21.8882 | 0.0042     | -0.008                 | 0.0004     | 0           | 0.0013     | 0       | 0.0131  | 0.0041  |
| H51    | 1     | 13    | 190.6406   | 1.4254    | 0.673      | 0.0623     | 29.3681 | -0.0014    | 0.0056                 | 0.0008     | 0           | 0.0061     | 0       | 0.0131  | 0.0042  |
| H52    | 3     | 37    | 123.618    | 1.9797    | 0.0742     | 0.0096     | 23.051  | 0.0118     | 0.0034                 | 0.0036     | 0           | 0.0027     | 0.0097  | 0.0135  | 0.0064  |
| H53    | 3     | 32    | 40.8171    | 2.2236    | 1.616      | 0.0634     | 11.3271 | 0.0419     | -0.0026                | 0.0004     | 0           | 0.0014     | 0       | 0.0032  | 0.0022  |
| I      | 25    | 144   | 34.9064    | 1.929     | 0.4168     | 0.0188     | 7.0638  | 0.0226     | 0.0017                 | 0.0004     | 0           | 0.0661     | 0.0003  | 0.0003  | 0.0098  |
| J58    | 2     | 16    | 103.6125   | 2.0038    | 2.5062     | 0.1048     | 21.8103 | -0.0002    | 0.0005                 | 0.0041     | 0           | 0.0084     | 0       | 0.0023  | 0.0039  |
| J59_60 | 6     | 10    | 111.2788   | 1.8703    | 0.868      | 0.0512     | 20.7718 | 0.0169     | -0.0016                | 0.0013     | 0           | 0.0066     | 0       | 0.0008  | 0.0013  |
| J61    | 1     | 16    | 281.9909   | 2.1403    | 0.2185     | 0.0376     | 27.6394 | 0.0015     | -0.0006                | 0.0038     | 0           | 0.0317     | 0       | 0.0033  | 0.0041  |
| J62_63 | 26    | 71    | 86.4875    | 2.3887    | 1.7509     | 0.0953     | 26.2104 | 0.0019     | -0.0061                | 0.063      | 0           | 0.0031     | 0.0007  | 0.0141  | 0.0107  |
| K64    | 6     | 61    | 214.2981   | 3.3865    | 0.4488     | 0.0176     | 36.3846 | 0.0202     | 0.0015                 | 0.0025     | 0           | 0.0171     | 0.0001  | 0.02    | 0.008   |
| K65    | 1     | 41    | 129.5476   | 1.9359    | 0.8417     | 0.074      | 25.8076 | 0.0189     | 0.0073                 | 0.0032     | 0           | 0.0519     | 0.0001  | 0.0032  | 0.0015  |
| K66    | 8     | 24    | 65.63      | 3.4499    | 3.4776     | 0.1987     | 27.7649 | 0.0299     | 0.0015                 | 0.001      | 0           | 0.0007     | 0       | 0.0005  | 0.0003  |
| L68A   | 14    | 34    | 528.3746   | 1.9306    | 0.1355     | 0.0515     | 22.8184 | 0.0218     | 0.0297                 | 0.0168     | 0           | 0.1842     | 0.0141  | 0.0004  | 0       |
| M69_70 | 69    | 151   | 85.6611    | 2.2496    | 1.5542     | 0.0585     | 26.3086 | 0.0007     | -0.0009                | 0.0218     | 0           | 0.0076     | 0.0022  | 0.0195  | 0.0439  |
| M71    | 19    | 51    | 77.2271    | 2.3867    | 0.6943     | 0.0417     | 24.3274 | 0.0019     | -0.0041                | 0.0717     | 0           | 0.0007     | 0.0001  | 0.0038  | 0.0001  |
| M72    | 2     | 16    | 75.1256    | 2.2247    | 2.3388     | 0.839      | 25.2507 | 0.0007     | -0.2664                | 0.0269     | 0           | 0.0002     | 0.0038  | 0.0021  | 0.0025  |
| M73    | 18    | 21    | 100.7593   | 1.6633    | 2.3769     | 0.1005     | 18.566  | 0.002      | -0.0002                | 0.0009     | 0           | 0.0001     | 0       | 0.0014  | 0.0031  |
| M74_75 | 22    | 22    | 68.635     | 1.9237    | 1.2497     | 0.0643     | 11.6511 | 0.0047     | -0.0085                | 0.0008     | 0           | 0.003      | 0       | 0.0011  | 0.0014  |
| N77    | 5     | 13    | 210.1032   | 2.6242    | 1.1406     | 0.0429     | 19.0141 | 0.0302     | 0.0442                 | 0.0055     | 0           | 0.0182     | 0       | 0.0314  | 0.0674  |
| N78    | 7     | 249   | 18.2019    | 4.9294    | 9.3894     | 0.1897     | 10.0747 | 0.0009     | -0.01                  | 0.0029     | 0           | 0.0002     | 0.0023  | 0.0003  | 0.0009  |
| N79    | 2     | 10    | 174.4929   | 1.2992    | 1.5567     | 0.0317     | 13.9171 | -0.0133    | -0.0027                | 0.0005     | 0           | 0.0155     | 0       | 0.0023  | 0       |
| N80-82 | 16    | 101   | 35.9623    | 2.9084    | 0.9643     | 0.0381     | 11.081  | 0.0045     | -0.0087                | 0.0022     | 0           | 0.0107     | 0       | 0.0014  | 0.0013  |
| O      | 5     | 254   | 77.5868    | 2.6845    | 0.1595     | 0.0273     | 24.7495 | 0.0512     | 0.0031                 | 0.0253     | 0           | 0.0143     | 0.3768  | 0.0019  | 0.0003  |
| P      | 22    | 245   | 40.3055    | 5.2785    | 0.2315     | 0.0348     | 19.4    | 0.027      | 0.0012                 | 0.0245     | 0           | 0.0101     | 0.1724  | 0.0006  | 0       |
| Q86    | 36    | 233   | 46.0174    | 3.4401    | 0.3195     | 0.0182     | 17.2718 | 0.0299     | -0.0407                | 0.0114     | 0           | 0.024      | 0.1831  | 0.0003  | 0       |
| Q87_88 | 12    | 368   | 23.1359    | 3.7579    | 0.3653     | 0.0235     | 11.5328 | 0.0292     | 0.0009                 | 0.0014     | 0           | 0.0137     | 0.1695  | 0       | 0       |
| R90-92 | 28    | 28    | 76.7376    | 2.5182    | 0.3728     | 0.0309     | 14.8073 | 0.0068     | 0.0024                 | 0.0031     | 0           | 0.0183     | 0.0054  | 0.0005  | 0.0005  |
| R93    | 9     | 28    | 51.9745    | 1.9427    | 0.1051     | 0.0057     | 12.055  | 0.0451     | 0.0025                 | 0.0005     | 0           | 0.0167     | 0.0001  | 0       | 0       |
| S94    | 7     | 36    | 43.1646    | 2.6983    | 0.2396     | 0.0147     | 16.9867 | 0.0502     | -0.002                 | 0.0005     | 0           | 0.0147     | 0.0011  | 0.0001  | 0       |
| S95    | 4     | 4     | 52.1333    | 1.9679    | 3.6944     | 0.2735     | 9.0876  | 0.0151     | 0.0013                 | 0.0005     | 0           | 0.0013     | 0       | 0.0002  | 0.0003  |
| S96    | 32    | 32    | 41.2095    | 2.6526    | 0.2908     | 0.0233     | 8.1205  | 0.0188     | -0.0093                | 0.0004     | 0           | 0.0158     | 0       | 0       | 0.0005  |

Note: Sectoral parameters are shown for 2010Q1.

## C Neural Network Architecture and Training

This appendix provides the implementation details for the amortised deep-learning solution method described in Section 2.3.

### C.1 Network Architecture

The policy function is approximated by a multilayer perceptron (MLP) with two hidden layers of 256 units each, using the Swish activation function. The input layer receives the current state vector  $\mathbf{s} = (k_{h,t}, b_{h,t}, w_{h,t}, r_t^k, r_t^b, \pi_t)$  concatenated with the preference parameter vector  $\Gamma$ , giving 10 inputs for the baseline specification and 12 for the liquidity variants (which add  $\sigma_k$  and/or  $\sigma_b$ ). All state inputs are normalised to  $[-1, 1]$  via min-max scaling; returns and inflation are z-scored. The output layer produces five values with no activation: three control shares  $(\zeta_c, \zeta_k, \zeta_b)$  passed through sigmoid functions to enforce  $[0, 1]$  bounds, and two Lagrange multipliers  $(\lambda^b, \lambda^k)$  passed through softplus functions to enforce non-negativity. Consumption, capital, and bond holdings are then recovered from the control shares via the budget constraint. Weights are initialised with Glorot uniform; biases are initialised to zero. The total number of parameters is approximately 70,000.

### C.2 Three-Stage Training Procedure

Training proceeds in three stages, totalling 40,000 epochs per quarterly calibration period.

**Stage 1: Joint training (25,000 epochs).** Both network weights and the preference parameter vector  $\Gamma$  are updated simultaneously. The loss function combines KKT residuals with moment-matching targets:

$$\mathcal{L} = \mathcal{L}_{\text{KKT}} + w_t \cdot \mathcal{L}_{\text{moments}}, \quad (118)$$

where the moment weight  $w_t$  follows a schedule that increases the frequency of moment evaluations from every 50 epochs to every epoch. The network is optimised with Adam ( $\eta_{\text{net}} = 10^{-3}$ ) and the parameters with Adam ( $\eta_{\Gamma} = 10^{-2}$ ), both subject to gradient clipping at unit norm. The learning rate follows a cosine-annealing schedule with exponential decay (period 6,000,  $\gamma = 0.99995$ ), cycling between  $10^{-3}$  and  $10^{-4}$ .

**Stage 2: KKT-focused refinement (5,000 epochs).** The preference parameters are frozen at their Stage 1 values and the network alone is refined on KKT residuals. To concentrate accuracy near the calibrated  $\Gamma^*$ , 70% of parameter samples are drawn from a truncated normal centred at  $\Gamma^*$  (with standard deviation equal to 10% of the parameter range), while the remaining 30% are drawn uniformly across the full range. This stage is skipped entirely if the Stage 1 KKT residuals are already below  $10^{-2}$ .

**Stage 3: Moment calibration (10,000 epochs).** The network is frozen and only  $\Gamma$  is optimised via gradient descent through the fixed network to minimise weighted moment errors. The moment loss assigns weights  $w_{\text{MPC}} = w_{\text{MPC}_k} = w_{\text{MPC}_b} = w_{K_1/B_1} = 1$  and  $w_{\text{APC}} = 10$ , reflecting

the importance of matching the aggregate consumption rate. The optimiser is Adam with  $\eta = 10^{-2}$ .

**Warm-starting.** Networks are trained independently for each quarterly period, initialised from a common starting network pre-trained on the first calibration quarter.

### *C.3 Batch Construction*

Each training epoch samples a batch of 256 state–parameter pairs. State variables (capital  $k$ , bonds  $b$ , wage income  $w$ ) are drawn uniformly from their empirical bounds. Returns and inflation are sampled from log-normal processes calibrated to the data. Preference parameters are drawn uniformly from their optimisation bounds (or from the focused distribution in Stage 2). For the cross-product loss, two independent sets of transition shocks are drawn per batch, each containing innovations to capital returns, bond returns, wages, and inflation.

## D Internal Validity under Alternative Expectations

This appendix presents the auto- and cross-correlation diagnostics for the OC-ABM under ARX, VAR, and VARX expectations, complementing the AR-expectations analysis in Section 6. The simulation setup is identical: 100 Monte Carlo simulations from the 2012Q1 calibration origin over a 20-quarter horizon, with HP-filtered ( $\lambda = 1,600$ ) cyclical components.

### D.1 ARX Expectations

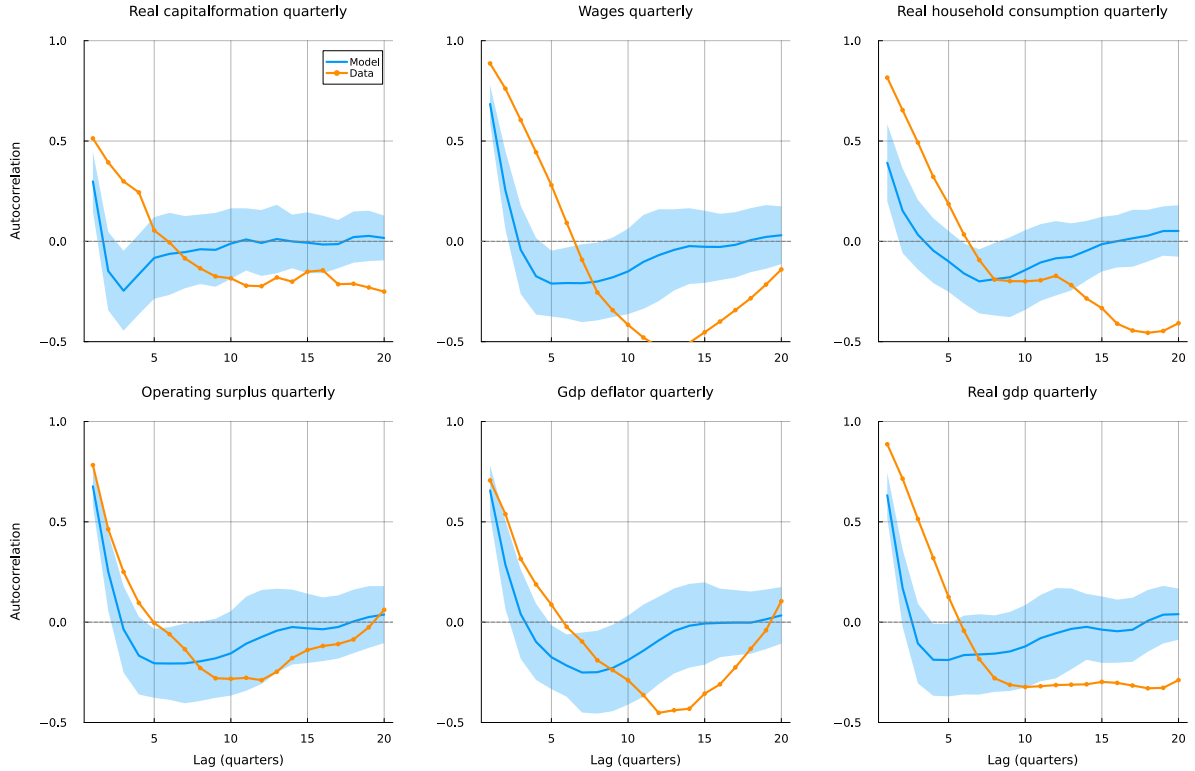


Figure 18: Auto-correlations of HP-filtered ( $\lambda = 1,600$ ) cyclical components under ARX expectations. Blue line: mean across 100 Monte Carlo simulations from origin 2012Q1 (20-quarter horizon). Shaded area:  $\pm 1$  standard deviation. Orange line: empirical ACF (1997Q1–2019Q4). Compared to Figure 16 (AR expectations), the GDP deflator persistence is reduced under ARX expectations.

### D.2 VAR Expectations

### D.3 VARX Expectations

### D.4 Interpretation

Comparing the alternative-expectations diagnostics with the AR-expectations results in Section 6, two improvements are evident across all three mechanisms. First, the GDP deflator auto-correlations decay more rapidly under ARX, VAR, and VARX expectations, reducing the excess persistence identified under AR expectations. This suggests that richer expectation processes, whether conditioning on exogenous variables (ARX, VARX) or exploiting cross-variable

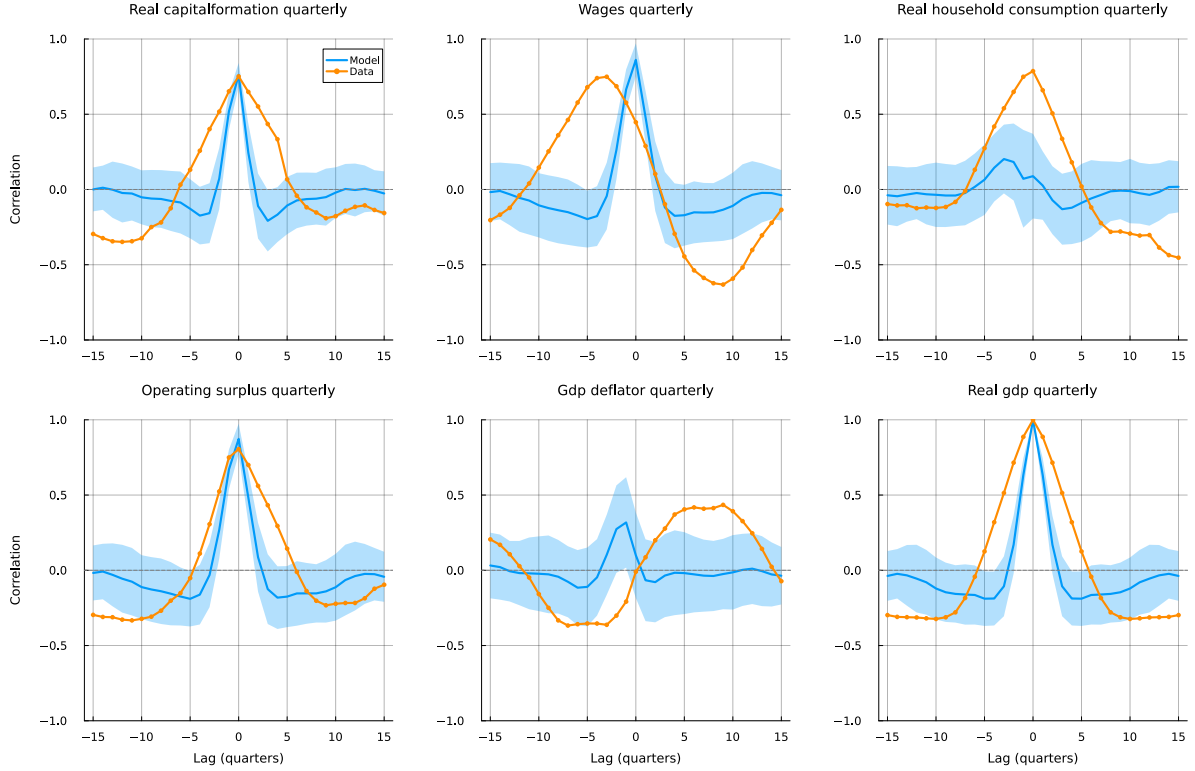


Figure 19: Cross-correlations between HP-filtered ( $\lambda = 1,600$ ) cyclical GDP and other variables under ARX expectations. Positive lags: variable lags GDP. Negative lags: variable leads GDP. Blue line: mean across 100 Monte Carlo simulations from origin 2012Q1 (20-quarter horizon). Shaded area:  $\pm 1$  standard deviation. Orange line: empirical cross-correlation (1997Q1–2019Q4). The GDP deflator cross-correlations improve markedly relative to AR expectations (Figure 17), aligning more closely with the empirical pattern.

dynamics (VAR, VARX), discipline the model’s price dynamics, anchoring inflation expectations closer to their empirical counterparts. Second, the GDP deflator’s cross-correlations with GDP at positive lags improve, with the model generating a more realistic decay pattern that aligns more closely with the empirical profile.

The wage timing discrepancy persists under all alternative expectations: wages remain coincident with GDP rather than leading. This confirms that the wage timing discrepancy originates in the labour market module’s immediate wage adjustment mechanism rather than in the expectation formation process. The capital formation and household consumption cross-correlations remain well-matched at lag zero across all specifications, providing consistent evidence that the demand-side dynamics of the OC-ABM are robust to the choice of aggregate expectations.

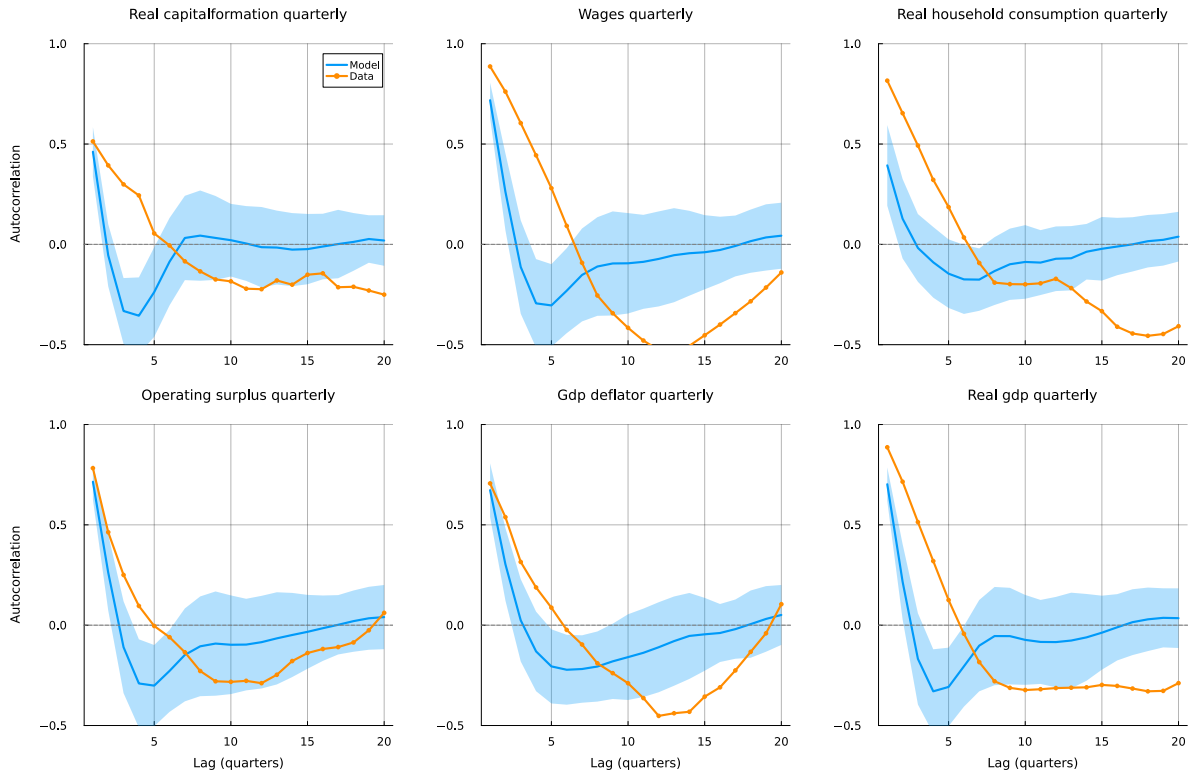


Figure 20: Auto-correlations of HP-filtered ( $\lambda = 1,600$ ) cyclical components under VAR expectations. Blue line: mean across 100 Monte Carlo simulations from origin 2012Q1 (20-quarter horizon). Shaded area:  $\pm 1$  standard deviation. Orange line: empirical ACF (1997Q1–2019Q4).

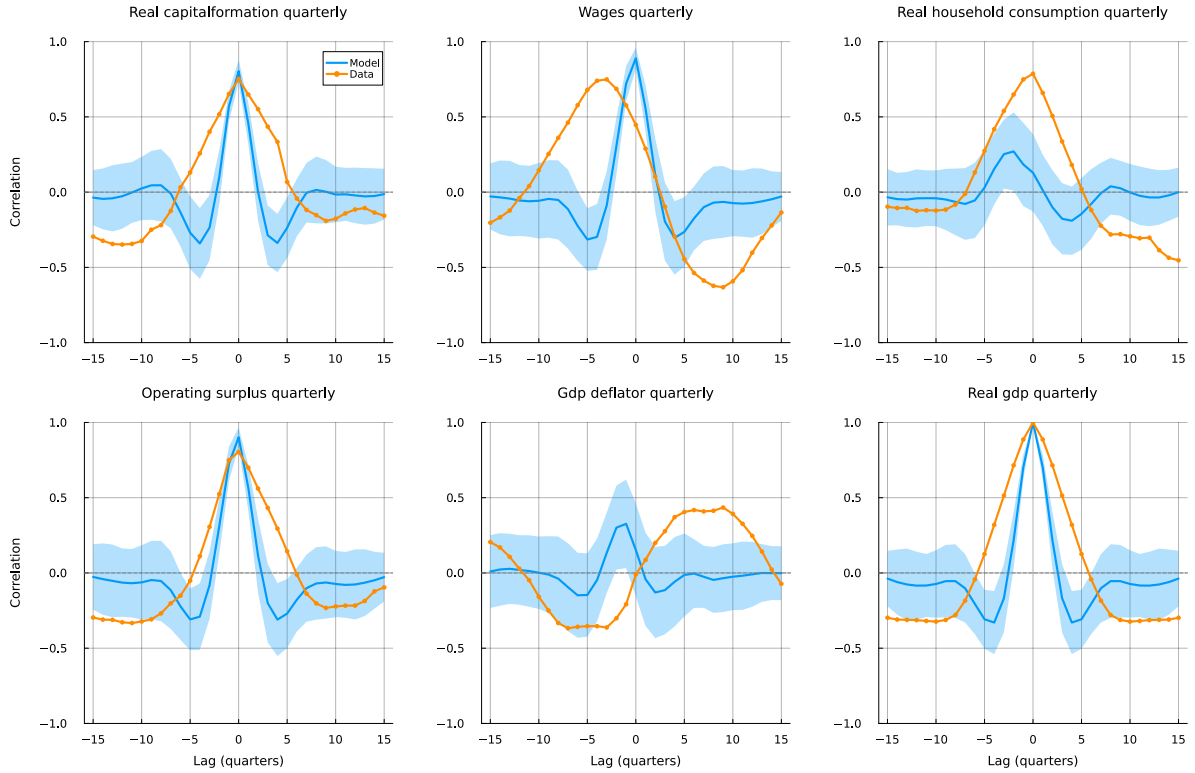


Figure 21: Cross-correlations between HP-filtered ( $\lambda = 1,600$ ) cyclical GDP and other variables under VAR expectations. Positive lags: variable lags GDP. Negative lags: variable leads GDP. Blue line: mean across 100 Monte Carlo simulations from origin 2012Q1 (20-quarter horizon). Shaded area:  $\pm 1$  standard deviation. Orange line: empirical cross-correlation (1997Q1–2019Q4).

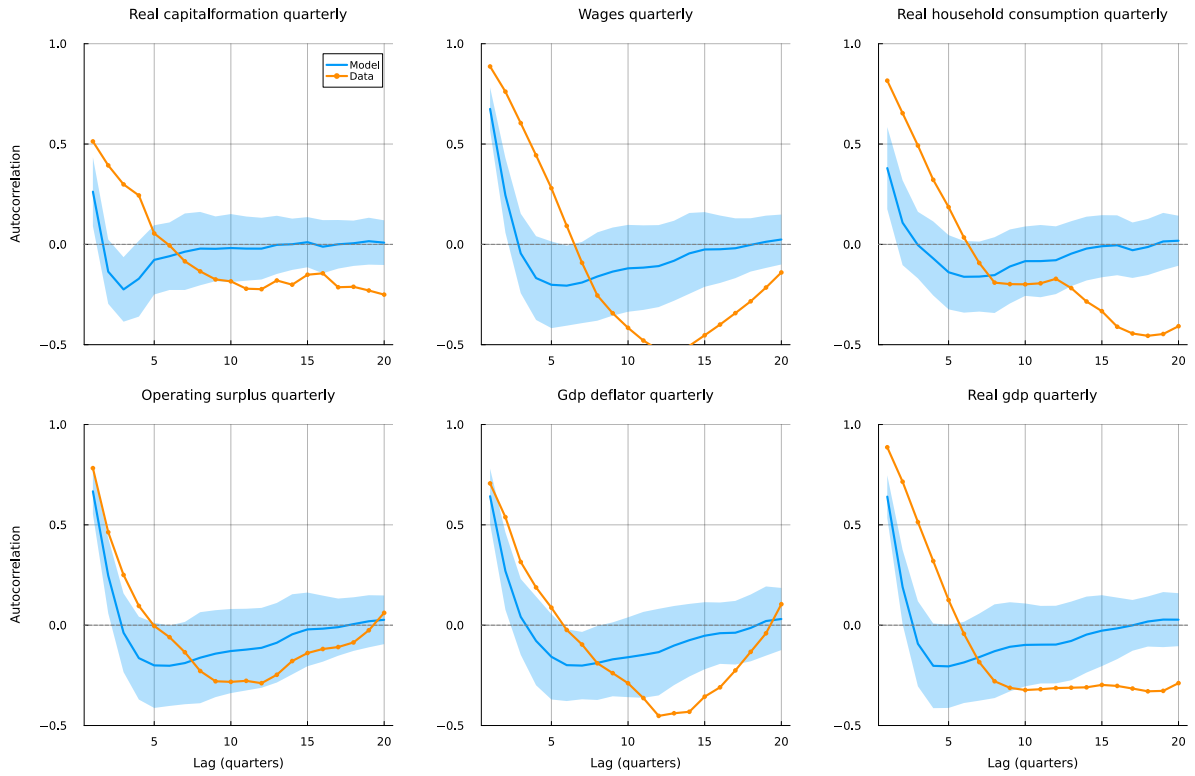


Figure 22: Auto-correlations of HP-filtered ( $\lambda = 1,600$ ) cyclical components under VARX expectations. Blue line: mean across 100 Monte Carlo simulations from origin 2012Q1 (20-quarter horizon). Shaded area:  $\pm 1$  standard deviation. Orange line: empirical ACF (1997Q1–2019Q4).

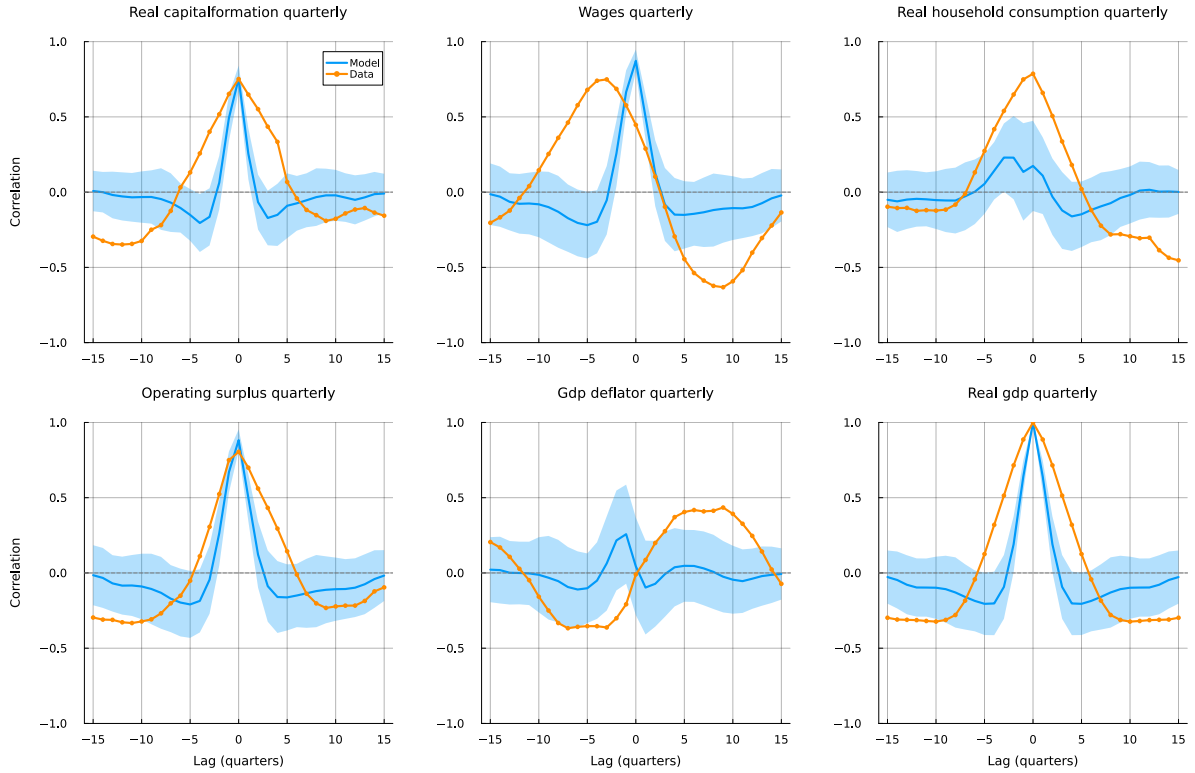


Figure 23: Cross-correlations between HP-filtered ( $\lambda = 1,600$ ) cyclical GDP and other variables under VARX expectations. Positive lags: variable lags GDP. Negative lags: variable leads GDP. Blue line: mean across 100 Monte Carlo simulations from origin 2012Q1 (20-quarter horizon). Shaded area:  $\pm 1$  standard deviation. Orange line: empirical cross-correlation (1997Q1–2019Q4).

## E Forecasting Results of Heuristic ABMs

This appendix reports the full forecast results for all seven heuristic ABM variants (H1–H7, defined in Section 2.4) under AR and ARX expectations. Section 5 presents selected comparisons; here we provide the complete set for reference. All seven heuristics share the same borrowing constraint as the OC-ABM (Section 2.4).

### *E.1 ARX expectations*

Table 14: Heuristic ABM forecast performance under AR expectations, relative to AR(1)

|        | GDP                                                                      | Inflation       | Household consumption | Investment       | Euribor         |
|--------|--------------------------------------------------------------------------|-----------------|-----------------------|------------------|-----------------|
| AR(1)  | <i>RMSFE for different forecast horizons</i>                             |                 |                       |                  |                 |
| 1q     | 0.41                                                                     | 0.57            | 0.56                  | 3.3              | 0.03            |
| 2q     | 0.67                                                                     | 0.57            | 0.82                  | 4.48             | 0.06            |
| 4q     | 1.32                                                                     | 0.56            | 1.47                  | 6.21             | 0.11            |
| 8q     | 2.59                                                                     | 0.58            | 2.72                  | 9.61             | 0.16            |
| 12q    | 3.47                                                                     | 0.6             | 3.68                  | 13.27            | 0.2             |
| H1-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -0.6(0.94, )                                                             | -1.4(0.67, )    | -35.3(0.01, **)       | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -17.9(0.0, ***)                                                          | -2.5(0.0, ***)  | -61.0(0.0, ***)       | 0.7(0.38, )      | -38.9(0.0, ***) |
| 4q     | -60.8(0.0, ***)                                                          | -2.9(0.0, ***)  | -103.5(0.0, ***)      | -8.5(0.0, ***)   | -17.7(0.0, ***) |
| 8q     | -29.6(0.03, **)                                                          | -2.6(0.0, ***)  | -195.5(0.0, ***)      | -2.3(0.32, )     | -8.9(0.13, )    |
| 12q    | -91.9(0.0, ***)                                                          | -2.4(0.0, ***)  | -292.2(0.0, ***)      | -14.0(0.01, **)  | -5.5(0.43, )    |
| H2-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -0.6(0.95, )                                                             | -1.4(0.67, )    | -38.2(0.01, ***)      | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -18.8(0.0, ***)                                                          | -1.3(0.0, ***)  | -65.0(0.0, ***)       | 0.4(0.65, )      | -39.2(0.0, ***) |
| 4q     | -61.0(0.0, ***)                                                          | -2.2(0.0, ***)  | -107.1(0.0, ***)      | -9.1(0.0, ***)   | -16.9(0.0, ***) |
| 8q     | -31.4(0.02, **)                                                          | 0.5(0.08, *)    | -214.7(0.0, ***)      | -2.3(0.32, )     | -6.3(0.3, )     |
| 12q    | -96.9(0.0, ***)                                                          | 0.8(0.02, **)   | -307.9(0.0, ***)      | -14.8(0.01, ***) | -2.7(0.72, )    |
| H3-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -1.3(0.88, )                                                             | -1.4(0.67, )    | 2.6(0.77, )           | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -7.5(0.01, **)                                                           | 0.5(0.14, )     | 10.1(0.0, ***)        | 1.2(0.1, *)      | -39.4(0.0, ***) |
| 4q     | -31.1(0.0, ***)                                                          | -2.6(0.0, ***)  | -13.6(0.01, **)       | -5.1(0.0, ***)   | -15.7(0.0, ***) |
| 8q     | 9.4(0.38, )                                                              | 0.4(0.8, )      | -30.7(0.0, ***)       | 7.3(0.0, ***)    | -8.8(0.14, )    |
| 12q    | 6.2(0.73, )                                                              | -3.4(0.01, ***) | -25.9(0.05, *)        | 6.5(0.0, ***)    | -3.4(0.64, )    |
| H4-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -1.3(0.89, )                                                             | -1.4(0.67, )    | 3.0(0.73, )           | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -4.6(0.13, )                                                             | 3.4(0.0, ***)   | 8.8(0.0, ***)         | 1.7(0.03, **)    | -40.5(0.0, ***) |
| 4q     | -29.5(0.0, ***)                                                          | -1.9(0.0, ***)  | -15.1(0.01, ***)      | -4.2(0.0, ***)   | -17.6(0.0, ***) |
| 8q     | 3.4(0.76, )                                                              | 0.0(0.93, )     | -37.2(0.0, ***)       | 5.3(0.0, ***)    | -7.3(0.22, )    |
| 12q    | -0.4(0.98, )                                                             | -0.7(0.0, ***)  | -37.0(0.0, ***)       | 5.6(0.0, ***)    | -2.8(0.7, )     |
| H5-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -1.3(0.88, )                                                             | -1.4(0.67, )    | -3.5(0.69, )          | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -12.8(0.0, ***)                                                          | 0.1(0.73, )     | -8.8(0.0, ***)        | 0.6(0.41, )      | -39.4(0.0, ***) |
| 4q     | -43.1(0.0, ***)                                                          | -0.5(0.75, )    | -42.4(0.0, ***)       | -7.3(0.0, ***)   | -16.1(0.0, ***) |
| 8q     | -1.8(0.87, )                                                             | -1.9(0.0, ***)  | -26.8(0.04, **)       | 2.7(0.04, **)    | -8.0(0.2, )     |
| 12q    | -4.0(0.82, )                                                             | 1.4(0.18, )     | -27.0(0.22, )         | 4.2(0.0, ***)    | -2.4(0.77, )    |
| H6-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -2.0(0.82, )                                                             | -1.4(0.67, )    | -2.9(0.71, )          | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -5.8(0.04, **)                                                           | -1.0(0.0, ***)  | 10.1(0.0, ***)        | 1.4(0.06, *)     | -39.1(0.0, ***) |
| 4q     | -22.4(0.0, ***)                                                          | -1.5(0.0, ***)  | 10.9(0.01, **)        | -4.8(0.0, ***)   | -17.4(0.0, ***) |
| 8q     | 14.3(0.15, )                                                             | -1.2(0.0, ***)  | 20.0(0.04, **)        | 7.0(0.0, ***)    | -8.4(0.15, )    |
| 12q    | 17.1(0.28, )                                                             | 0.7(0.62, )     | 31.6(0.06, *)         | 8.2(0.0, ***)    | -1.5(0.85, )    |
| H7-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -2.7(0.76, )                                                             | -1.4(0.67, )    | -40.7(0.01, ***)      | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -1.6(0.52, )                                                             | 2.3(0.0, ***)   | -15.7(0.0, ***)       | 1.6(0.03, **)    | -39.7(0.0, ***) |
| 4q     | -14.7(0.01, **)                                                          | -1.2(0.0, ***)  | 7.5(0.0, ***)         | -3.7(0.0, ***)   | -17.1(0.0, ***) |
| 8q     | 13.0(0.1, )                                                              | 2.3(0.0, ***)   | 9.7(0.08, *)          | 5.2(0.0, ***)    | -8.8(0.12, )    |
| 12q    | 11.2(0.37, )                                                             | -1.4(0.01, **)  | 13.7(0.23, )          | 6.9(0.0, ***)    | -3.2(0.68, )    |

Note: The table depicts the RMSFE of the forecast period 2010Q1 to 2019Q4. The AR(1) is estimated starting in 1997Q1 and is re-estimated each quarter. The ABM is calibrated to 40 different reference quarters from 2010Q1 to 2019Q4 and the ABM results are obtained as an average of 100 Monte Carlo simulations. We report the p-values of the Diebold-Mariano tests (Harvey et al., 1997). We test the null hypothesis whether the tested models have the same accuracy as the AR(1) model. We denote 10 percent, 5 percent and 1 percent significance by \*, \*\*, and \*\*\* respectively.

Table 15: Heuristic ABM forecast performance under AR expectations, relative to VAR(1)

|        | GDP                                                                       | Inflation       | Household cons.  | Gov. cons.       | Exports        | Imports          | GDP EA          | Infl. EA       | Euribor         |
|--------|---------------------------------------------------------------------------|-----------------|------------------|------------------|----------------|------------------|-----------------|----------------|-----------------|
| VAR(1) | <i>RMSFE for different forecast horizons</i>                              |                 |                  |                  |                |                  |                 |                |                 |
| 1q     | 0.39                                                                      | 0.75            | 0.6              | 0.57             | 1.26           | 1.38             | 0.36            | 0.15           | 0.04            |
| 2q     | 0.7                                                                       | 0.7             | 0.84             | 0.94             | 2.61           | 2.35             | 0.76            | 0.16           | 0.07            |
| 4q     | 1.52                                                                      | 0.85            | 1.4              | 1.66             | 5.42           | 4.85             | 1.69            | 0.18           | 0.16            |
| 8q     | 3.45                                                                      | 1.11            | 2.76             | 3.04             | 9.88           | 8.85             | 3.71            | 0.2            | 0.37            |
| 12q    | 4.71                                                                      | 1.44            | 3.94             | 4.38             | 13.08          | 11.45            | 4.95            | 0.28           | 0.5             |
| H1-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                  |                |                  |                 |                |                 |
| 1q     | -5.7(0.69, )                                                              | 22.4(0.01, ***) | -27.8(0.02, **)  | 5.6(0.47, )      | -16.2(0.39, )  | -75.2(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -12.5(0.0, ***)                                                           | 16.0(0.0, ***)  | -57.5(0.0, ***)  | -0.6(0.67, )     | 16.0(0.0, ***) | -58.7(0.0, ***)  | 8.0(0.0, ***)   | 1.1(0.55, )    | -21.5(0.0, ***) |
| 4q     | -40.3(0.0, ***)                                                           | 31.5(0.0, ***)  | -113.8(0.0, ***) | -1.4(0.43, )     | 34.3(0.0, ***) | -23.7(0.0, ***)  | 28.6(0.0, ***)  | 11.6(0.0, ***) | 18.2(0.0, ***)  |
| 8q     | 2.9(0.82, )                                                               | 46.1(0.0, ***)  | -191.1(0.0, ***) | -15.9(0.0, ***)  | 43.1(0.0, ***) | -14.8(0.09, *)   | 37.4(0.0, ***)  | 18.0(0.0, ***) | 53.0(0.0, ***)  |
| 12q    | -41.4(0.07, *)                                                            | 57.4(0.0, ***)  | -266.6(0.0, ***) | -12.0(0.02, **)  | 38.2(0.0, ***) | -31.0(0.02, **)  | 30.1(0.13, )    | 36.8(0.0, ***) | 56.9(0.0, ***)  |
| H2-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                  |                |                  |                 |                |                 |
| 1q     | -5.7(0.69, )                                                              | 22.4(0.01, ***) | -30.5(0.02, **)  | 6.2(0.5, )       | -16.1(0.39, )  | -76.1(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -13.3(0.0, ***)                                                           | 17.0(0.0, ***)  | -61.4(0.0, ***)  | 4.3(0.0, ***)    | 14.9(0.0, ***) | -60.3(0.0, ***)  | 9.5(0.0, ***)   | 2.5(0.19, )    | -21.8(0.0, ***) |
| 4q     | -40.4(0.0, ***)                                                           | 31.9(0.0, ***)  | -117.7(0.0, ***) | 0.2(0.88, )      | 33.4(0.0, ***) | -26.1(0.0, ***)  | 28.8(0.0, ***)  | 16.0(0.0, ***) | 18.7(0.0, ***)  |
| 8q     | 1.6(0.9, )                                                                | 47.7(0.0, ***)  | -210.0(0.0, ***) | -17.2(0.0, ***)  | 42.1(0.0, ***) | -18.8(0.03, **)  | 36.7(0.0, ***)  | 12.1(0.0, ***) | 54.1(0.0, ***)  |
| 12q    | -45.1(0.05, **)                                                           | 58.8(0.0, ***)  | -281.3(0.0, ***) | -13.3(0.03, **)  | 36.0(0.0, ***) | -35.8(0.01, ***) | 29.2(0.15, )    | 36.8(0.0, ***) | 58.0(0.0, ***)  |
| H3-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                  |                |                  |                 |                |                 |
| 1q     | -6.4(0.65, )                                                              | 22.4(0.01, ***) | 8.0(0.45, )      | 5.3(0.55, )      | -18.4(0.33, )  | -69.3(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -2.5(0.36, )                                                              | 18.5(0.0, ***)  | 12.0(0.0, ***)   | 5.6(0.0, ***)    | 14.2(0.0, ***) | -51.9(0.0, ***)  | 11.7(0.0, ***)  | 4.2(0.04, **)  | -22.0(0.0, ***) |
| 4q     | -14.4(0.0, ***)                                                           | 31.7(0.0, ***)  | -19.4(0.0, ***)  | 0.4(0.82, )      | 33.0(0.0, ***) | -21.8(0.0, ***)  | 30.1(0.0, ***)  | 12.1(0.0, ***) | 19.6(0.0, ***)  |
| 8q     | 32.2(0.01, ***)                                                           | 47.7(0.0, ***)  | -28.8(0.0, ***)  | -12.3(0.01, ***) | 41.5(0.0, ***) | -9.7(0.23, )     | 37.5(0.0, ***)  | 21.7(0.0, ***) | 53.0(0.0, ***)  |
| 12q    | 30.9(0.13, )                                                              | 57.1(0.0, ***)  | -17.7(0.04, **)  | -14.4(0.03, **)  | 35.4(0.0, ***) | -21.1(0.09, *)   | 30.0(0.14, )    | 39.1(0.0, ***) | 57.7(0.0, ***)  |
| H4-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                  |                |                  |                 |                |                 |
| 1q     | -6.4(0.66, )                                                              | 22.4(0.01, ***) | 8.4(0.42, )      | 5.6(0.49, )      | -18.4(0.33, )  | -68.9(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | 0.3(0.92, )                                                               | 20.8(0.0, ***)  | 10.8(0.0, ***)   | 5.1(0.0, ***)    | 14.2(0.0, ***) | -54.2(0.0, ***)  | 8.8(0.0, ***)   | 1.0(0.58, )    | -22.9(0.0, ***) |
| 4q     | -13.0(0.01, ***)                                                          | 32.1(0.0, ***)  | -21.0(0.0, ***)  | -0.4(0.84, )     | 32.7(0.0, ***) | -21.5(0.0, ***)  | 30.3(0.0, ***)  | 13.6(0.0, ***) | 18.3(0.0, ***)  |
| 8q     | 27.6(0.02, **)                                                            | 47.5(0.0, ***)  | -35.2(0.0, ***)  | -14.3(0.0, ***)  | 39.8(0.0, ***) | -12.0(0.15, )    | 37.0(0.0, ***)  | 22.0(0.0, ***) | 53.7(0.0, ***)  |
| 12q    | 26.0(0.2, )                                                               | 58.2(0.0, ***)  | -28.1(0.0, ***)  | -13.4(0.04, **)  | 33.7(0.0, ***) | -24.1(0.07, *)   | 29.2(0.15, )    | 38.7(0.0, ***) | 57.9(0.0, ***)  |
| H5-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                  |                |                  |                 |                |                 |
| 1q     | -6.5(0.65, )                                                              | 22.4(0.01, ***) | 2.2(0.81, )      | 6.2(0.47, )      | -17.0(0.37, )  | -73.1(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -7.6(0.0, ***)                                                            | 18.1(0.0, ***)  | -6.4(0.0, ***)   | -0.1(0.93, )     | 14.2(0.0, ***) | -54.9(0.0, ***)  | 10.4(0.0, ***)  | 4.2(0.03, **)  | -22.0(0.0, ***) |
| 4q     | -24.8(0.0, ***)                                                           | 33.0(0.0, ***)  | -49.6(0.0, ***)  | -4.3(0.03, **)   | 33.9(0.0, ***) | -22.6(0.0, ***)  | 30.1(0.0, ***)  | 17.0(0.0, ***) | 19.3(0.0, ***)  |
| 8q     | 23.7(0.04, **)                                                            | 46.4(0.0, ***)  | -25.0(0.0, ***)  | -18.2(0.0, ***)  | 41.7(0.0, ***) | -8.1(0.33, )     | 38.7(0.0, ***)  | 19.2(0.0, ***) | 53.3(0.0, ***)  |
| 12q    | 23.4(0.23, )                                                              | 59.1(0.0, ***)  | -18.7(0.25, )    | -18.5(0.02, **)  | 35.3(0.0, ***) | -17.6(0.15, )    | 31.1(0.13, )    | 38.0(0.0, ***) | 58.1(0.0, ***)  |
| H6-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                  |                |                  |                 |                |                 |
| 1q     | -7.2(0.62, )                                                              | 22.4(0.01, ***) | 2.8(0.82, )      | 3.3(0.73, )      | -18.6(0.33, )  | -69.1(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -0.9(0.72, )                                                              | 17.2(0.0, ***)  | 12.1(0.0, ***)   | 2.4(0.04, **)    | 12.6(0.0, ***) | -55.0(0.0, ***)  | 9.5(0.0, ***)   | 3.6(0.04, **)  | -21.8(0.0, ***) |
| 4q     | -6.7(0.15, )                                                              | 32.4(0.0, ***)  | 6.3(0.0, ***)    | -1.8(0.28, )     | 33.3(0.0, ***) | -21.2(0.0, ***)  | 28.9(0.0, ***)  | 13.7(0.0, ***) | 18.4(0.0, ***)  |
| 8q     | 35.8(0.0, ***)                                                            | 46.8(0.0, ***)  | 21.2(0.0, ***)   | -17.1(0.0, ***)  | 40.2(0.0, ***) | -10.6(0.2, )     | 36.7(0.0, ***)  | 22.3(0.0, ***) | 53.2(0.0, ***)  |
| 12q    | 38.9(0.04, **)                                                            | 58.7(0.0, ***)  | 36.1(0.0, ***)   | -18.4(0.01, **)  | 34.0(0.0, ***) | -20.7(0.1, )     | 29.2(0.15, )    | 38.6(0.0, ***) | 58.5(0.0, ***)  |
| H7-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                  |                |                  |                 |                |                 |
| 1q     | -7.8(0.59, )                                                              | 22.4(0.01, ***) | -32.9(0.07, *)   | 6.0(0.51, )      | -19.6(0.3, )   | -66.6(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | 3.1(0.19, )                                                               | 19.9(0.0, ***)  | -13.2(0.0, ***)  | 2.3(0.03, **)    | 12.2(0.0, ***) | -49.7(0.0, ***)  | 9.5(0.0, ***)   | 0.2(0.92, )    | -22.3(0.0, ***) |
| 4q     | -0.1(0.99, )                                                              | 32.6(0.0, ***)  | 2.8(0.32, )      | -1.9(0.28, )     | 31.7(0.0, ***) | -18.2(0.0, ***)  | 29.0(0.0, ***)  | 14.6(0.0, ***) | 18.6(0.0, ***)  |
| 8q     | 34.8(0.0, ***)                                                            | 48.7(0.0, ***)  | 11.1(0.25, )     | -16.0(0.0, ***)  | 40.0(0.0, ***) | -9.5(0.26, )     | 36.6(0.0, ***)  | 17.1(0.0, ***) | 53.0(0.0, ***)  |
| 12q    | 34.5(0.04, **)                                                            | 57.9(0.0, ***)  | 19.3(0.0, ***)   | -14.3(0.03, **)  | 34.2(0.0, ***) | -21.6(0.1, )     | 29.7(0.14, )    | 37.6(0.0, ***) | 57.8(0.0, ***)  |

*Note:* The table depicts the RMSFE of the forecast period 2010Q1 to 2019Q4. The VAR(1) is estimated starting in 1997Q1 and is re-estimated each quarter. The ABM is calibrated to 40 different reference quarters from 2010Q1 to 2019Q4 and the ABM results are obtained as an average of 100 Monte Carlo simulations. We report the p-values of the Diebold-Mariano tests (Harvey et al., 1997). We test the null hypothesis whether the tested models have the same accuracy as the VAR(1) model. We denote 10 percent, 5 percent and 1 percent significance by \*, \*\*, and \*\*\* respectively.

Table 16: Heuristic ABM forecast performance under ARX expectations, relative to ARX(1)

|        | GDP                                                                       | Inflation      | Household consumption | Investment      | Euribor         |
|--------|---------------------------------------------------------------------------|----------------|-----------------------|-----------------|-----------------|
| ARX(1) | <i>RMSFE for different forecast horizons</i>                              |                |                       |                 |                 |
| 1q     | 0.38                                                                      | 0.61           | 0.59                  | 3.46            | 0.04            |
| 2q     | 0.67                                                                      | 0.58           | 0.86                  | 4.54            | 0.06            |
| 4q     | 1.29                                                                      | 0.56           | 1.33                  | 6.09            | 0.1             |
| 8q     | 2.6                                                                       | 0.58           | 2.44                  | 9.67            | 0.16            |
| 12q    | 3.51                                                                      | 0.6            | 3.27                  | 13.41           | 0.21            |
| ABM    | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -7.7(0.49, )                                                              | 1.4(0.78, )    | 9.1(0.27, )           | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -21.8(0.0, ***)                                                           | 1.9(0.0, ***)  | 4.2(0.07, *)          | 0.2(0.73, )     | -42.4(0.0, ***) |
| 4q     | -37.9(0.0, ***)                                                           | 5.3(0.0, ***)  | -20.6(0.0, ***)       | -8.5(0.0, ***)  | -28.4(0.0, ***) |
| 8q     | 2.2(0.81, )                                                               | -4.8(0.0, ***) | -8.1(0.02, **)        | 2.6(0.0, ***)   | -8.3(0.02, **)  |
| 12q    | 1.5(0.91, )                                                               | -2.0(0.0, ***) | -7.6(0.11, )          | 4.7(0.08, *)    | 1.6(0.73, )     |
| H1-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -6.4(0.54, )                                                              | 1.4(0.78, )    | -29.4(0.01, ***)      | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -32.5(0.0, ***)                                                           | 3.2(0.0, ***)  | -56.5(0.0, ***)       | -0.0(0.95, )    | -38.9(0.0, ***) |
| 4q     | -64.2(0.0, ***)                                                           | 3.5(0.0, ***)  | -112.1(0.0, ***)      | -11.0(0.0, ***) | -26.6(0.0, ***) |
| 8q     | -24.9(0.05, *)                                                            | 0.7(0.14, )    | -213.9(0.0, ***)      | -0.4(0.84, )    | -5.8(0.15, )    |
| 12q    | -82.8(0.0, ***)                                                           | 5.9(0.04, **)  | -318.4(0.0, ***)      | -11.2(0.04, **) | 2.0(0.68, )     |
| H2-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -6.4(0.54, )                                                              | 1.4(0.78, )    | -32.0(0.01, ***)      | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -33.7(0.0, ***)                                                           | 3.3(0.0, ***)  | -59.4(0.0, ***)       | 0.2(0.79, )     | -39.8(0.0, ***) |
| 4q     | -65.8(0.0, ***)                                                           | 6.4(0.0, ***)  | -116.8(0.0, ***)      | -11.3(0.0, ***) | -26.4(0.0, ***) |
| 8q     | -33.4(0.01, **)                                                           | 1.7(0.34, )    | -239.7(0.0, ***)      | -2.3(0.26, )    | -7.2(0.05, **)  |
| 12q    | -93.4(0.0, ***)                                                           | 5.7(0.05, *)   | -338.0(0.0, ***)      | -14.2(0.01, **) | 1.3(0.76, )     |
| H3-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -6.4(0.56, )                                                              | 1.4(0.78, )    | 5.6(0.65, )           | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -14.0(0.0, ***)                                                           | 2.9(0.0, ***)  | 10.9(0.0, ***)        | 1.8(0.01, ***)  | -40.2(0.0, ***) |
| 4q     | -33.3(0.0, ***)                                                           | 6.6(0.0, ***)  | -15.9(0.0, ***)       | -7.0(0.0, ***)  | -25.5(0.0, ***) |
| 8q     | 6.5(0.54, )                                                               | 1.5(0.0, ***)  | -33.2(0.0, ***)       | 6.1(0.0, ***)   | -7.1(0.06, *)   |
| 12q    | 2.8(0.87, )                                                               | 3.3(0.0, ***)  | -35.6(0.0, ***)       | 6.7(0.0, ***)   | 1.6(0.74, )     |
| H4-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -6.3(0.57, )                                                              | 1.4(0.78, )    | 5.9(0.63, )           | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -14.4(0.0, ***)                                                           | 1.4(0.0, ***)  | 10.1(0.0, ***)        | 1.8(0.01, ***)  | -38.2(0.0, ***) |
| 4q     | -34.2(0.0, ***)                                                           | 5.2(0.0, ***)  | -18.5(0.0, ***)       | -7.4(0.0, ***)  | -26.1(0.0, ***) |
| 8q     | 5.6(0.61, )                                                               | 0.6(0.08, *)   | -41.0(0.0, ***)       | 6.5(0.0, ***)   | -5.5(0.16, )    |
| 12q    | -2.9(0.87, )                                                              | 1.0(0.71, )    | -44.8(0.0, ***)       | 5.8(0.0, ***)   | 4.5(0.36, )     |
| H5-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -7.1(0.51, )                                                              | 1.4(0.78, )    | 0.3(0.97, )           | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -24.9(0.0, ***)                                                           | 4.7(0.0, ***)  | -9.6(0.0, ***)        | 0.6(0.36, )     | -38.8(0.0, ***) |
| 4q     | -51.6(0.0, ***)                                                           | 3.9(0.0, ***)  | -55.9(0.0, ***)       | -10.0(0.0, ***) | -22.3(0.0, ***) |
| 8q     | -10.1(0.35, )                                                             | 0.8(0.17, )    | -43.3(0.0, ***)       | 0.5(0.61, )     | -8.3(0.02, **)  |
| 12q    | -11.2(0.53, )                                                             | 0.0(1.0, )     | -47.4(0.03, **)       | 2.3(0.04, **)   | 3.7(0.47, )     |
| H6-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -7.3(0.52, )                                                              | 1.4(0.78, )    | 1.6(0.91, )           | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -17.0(0.0, ***)                                                           | 1.7(0.0, ***)  | 11.1(0.0, ***)        | 1.4(0.04, **)   | -39.8(0.0, ***) |
| 4q     | -33.5(0.0, ***)                                                           | 4.0(0.0, ***)  | -1.9(0.31, )          | -7.7(0.0, ***)  | -24.9(0.0, ***) |
| 8q     | 9.4(0.34, )                                                               | -2.5(0.0, ***) | 9.0(0.2, )            | 4.5(0.0, ***)   | -6.3(0.08, *)   |
| 12q    | 9.3(0.56, )                                                               | -1.1(0.0, ***) | 15.4(0.27, )          | 7.3(0.0, ***)   | 3.1(0.52, )     |
| H7-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -7.9(0.49, )                                                              | 1.4(0.78, )    | -37.8(0.06, *)        | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -10.7(0.0, ***)                                                           | 2.9(0.0, ***)  | -16.0(0.0, ***)       | 1.9(0.0, ***)   | -38.9(0.0, ***) |
| 4q     | -22.6(0.0, ***)                                                           | 4.9(0.0, ***)  | -10.6(0.0, ***)       | -6.9(0.0, ***)  | -24.3(0.0, ***) |
| 8q     | 9.5(0.23, )                                                               | -3.9(0.1, *)   | -6.7(0.39, )          | 5.6(0.08, *)    | -6.0(0.11, )    |
| 12q    | 7.2(0.6, )                                                                | 2.8(0.12, )    | -0.9(0.9, )           | 6.5(0.02, **)   | 2.2(0.63, )     |

Note: ABMs under ARX expectations vs. ARX(1). The ARX(1) is estimated from 1997Q1 and re-estimated each quarter. ABM results average 100 Monte Carlo simulations per reference quarter (2010Q1–2019Q4). Diebold–Mariano  $p$ -values in parentheses; \*, \*\*, \*\*\*: 10%, 5%, 1% significance.

Table 17: Heuristic ABM forecast performance under ARX expectations, relative to VARX(1)

|         | GDP                                                                        | Inflation      | Household cons.  | Gov. cons.       | Exports        | Imports          | GDP EA          | Infl. EA       | Euribor         |
|---------|----------------------------------------------------------------------------|----------------|------------------|------------------|----------------|------------------|-----------------|----------------|-----------------|
| VARX(1) | <i>RMSFE for different forecast horizons</i>                               |                |                  |                  |                |                  |                 |                |                 |
| 1q      | 0.39                                                                       | 0.75           | 0.6              | 0.57             | 1.26           | 1.38             | 0.36            | 0.15           | 0.04            |
| 2q      | 0.65                                                                       | 0.71           | 0.83             | 0.94             | 2.44           | 2.23             | 0.71            | 0.16           | 0.07            |
| 4q      | 1.3                                                                        | 0.87           | 1.39             | 1.68             | 4.64           | 4.23             | 1.46            | 0.18           | 0.12            |
| 8q      | 2.87                                                                       | 1.17           | 2.76             | 3.11             | 8.34           | 7.66             | 3.19            | 0.19           | 0.23            |
| 12q     | 4.04                                                                       | 1.64           | 4.07             | 4.37             | 11.91          | 10.74            | 4.5             | 0.25           | 0.31            |
| H1-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                |                 |
| 1q      | -5.6(0.64, )                                                               | 19.3(0.01, **) | -28.4(0.03, **)  | 4.9(0.61, )      | -15.9(0.4, )   | -75.9(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q      | -35.2(0.0, ***)                                                            | 21.6(0.0, ***) | -61.2(0.0, ***)  | 0.6(0.65, )      | 9.1(0.01, **)  | -70.0(0.0, ***)  | 3.6(0.15, )     | 9.7(0.0, ***)  | -35.6(0.0, ***) |
| 4q      | -62.2(0.0, ***)                                                            | 37.3(0.0, ***) | -102.9(0.0, ***) | 0.4(0.81, )      | 23.3(0.0, ***) | -42.5(0.0, ***)  | 20.2(0.0, ***)  | 15.7(0.0, ***) | -11.5(0.0, ***) |
| 8q      | -12.8(0.33, )                                                              | 50.4(0.0, ***) | -176.6(0.0, ***) | -12.4(0.02, **)  | 31.9(0.0, ***) | -34.2(0.0, ***)  | 27.3(0.02, **)  | 15.7(0.0, ***) | 25.0(0.0, ***)  |
| 12q     | -58.6(0.02, **)                                                            | 65.5(0.0, ***) | -236.5(0.0, ***) | -12.9(0.12, )    | 29.5(0.0, ***) | -41.3(0.01, ***) | 22.6(0.3, )     | 33.1(0.03, **) | 33.7(0.0, ***)  |
| H2-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                |                 |
| 1q      | -5.6(0.64, )                                                               | 19.3(0.01, **) | -31.1(0.02, **)  | 5.1(0.61, )      | -15.9(0.4, )   | -75.5(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q      | -36.5(0.0, ***)                                                            | 21.7(0.0, ***) | -64.2(0.0, ***)  | 1.0(0.49, )      | 9.0(0.01, **)  | -70.9(0.0, ***)  | 1.2(0.63, )     | -0.3(0.88, )   | -36.4(0.0, ***) |
| 4q      | -63.7(0.0, ***)                                                            | 39.2(0.0, ***) | -107.3(0.0, ***) | -0.5(0.75, )     | 22.0(0.0, ***) | -45.0(0.0, ***)  | 16.0(0.0, ***)  | 10.6(0.0, ***) | -11.4(0.0, ***) |
| 8q      | -20.5(0.12, )                                                              | 50.9(0.0, ***) | -199.3(0.0, ***) | -13.7(0.01, ***) | 30.0(0.0, ***) | -37.4(0.0, ***)  | 25.8(0.02, **)  | 12.3(0.01, **) | 24.0(0.0, ***)  |
| 12q     | -67.7(0.01, ***)                                                           | 65.4(0.0, ***) | -252.2(0.0, ***) | -12.8(0.12, )    | 29.4(0.0, ***) | -43.0(0.01, ***) | 21.5(0.32, )    | 30.8(0.04, **) | 33.3(0.0, ***)  |
| H3-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                |                 |
| 1q      | -5.5(0.65, )                                                               | 19.3(0.01, **) | 6.3(0.58, )      | 0.2(0.98, )      | -18.2(0.34, )  | -68.5(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q      | -16.3(0.0, ***)                                                            | 21.4(0.0, ***) | 8.2(0.01, ***)   | 0.5(0.69, )      | 10.7(0.0, ***) | -57.2(0.0, ***)  | 4.2(0.08, *)    | 6.9(0.0, ***)  | -36.9(0.0, ***) |
| 4q      | -31.7(0.0, ***)                                                            | 39.3(0.0, ***) | -10.9(0.0, ***)  | -0.9(0.56, )     | 23.9(0.0, ***) | -34.5(0.0, ***)  | 20.2(0.0, ***)  | 9.2(0.0, ***)  | -10.6(0.0, ***) |
| 8q      | 15.6(0.17, )                                                               | 50.7(0.0, ***) | -17.4(0.0, ***)  | -12.6(0.01, **)  | 31.2(0.0, ***) | -25.0(0.01, ***) | 28.3(0.01, **)  | 13.9(0.0, ***) | 24.1(0.0, ***)  |
| 12q     | 15.7(0.42, )                                                               | 64.6(0.0, ***) | -9.1(0.03, **)   | -14.1(0.11, )    | 28.9(0.0, ***) | -25.7(0.08, *)   | 24.0(0.27, )    | 30.3(0.0, ***) | 33.4(0.0, ***)  |
| H4-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                |                 |
| 1q      | -5.5(0.65, )                                                               | 19.3(0.01, **) | 6.6(0.56, )      | 6.3(0.48, )      | -18.2(0.34, )  | -69.3(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q      | -16.8(0.0, ***)                                                            | 20.2(0.0, ***) | 7.4(0.01, **)    | 1.8(0.12, )      | 8.0(0.03, **)  | -61.7(0.0, ***)  | 2.2(0.36, )     | 1.7(0.36, )    | -34.9(0.0, ***) |
| 4q      | -32.6(0.0, ***)                                                            | 38.4(0.0, ***) | -13.3(0.0, ***)  | -2.9(0.12, )     | 22.5(0.0, ***) | -37.4(0.0, ***)  | 18.6(0.0, ***)  | 15.9(0.0, ***) | -11.1(0.0, ***) |
| 8q      | 14.7(0.2, )                                                                | 50.3(0.0, ***) | -24.2(0.0, ***)  | -11.6(0.02, **)  | 30.6(0.0, ***) | -27.2(0.0, ***)  | 27.0(0.02, **)  | 13.2(0.0, ***) | 25.2(0.0, ***)  |
| 12q     | 10.7(0.59, )                                                               | 63.7(0.0, ***) | -16.4(0.0, ***)  | -14.4(0.12, )    | 27.1(0.0, ***) | -29.5(0.05, **)  | 21.4(0.33, )    | 30.6(0.06, *)  | 35.4(0.0, ***)  |
| H5-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                |                 |
| 1q      | -6.2(0.61, )                                                               | 19.3(0.01, **) | 1.0(0.92, )      | 5.2(0.57, )      | -16.7(0.38, )  | -72.7(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q      | -27.5(0.0, ***)                                                            | 22.8(0.0, ***) | -12.9(0.0, ***)  | 1.9(0.13, )      | 8.1(0.03, **)  | -66.2(0.0, ***)  | 2.5(0.31, )     | 7.9(0.0, ***)  | -35.4(0.0, ***) |
| 4q      | -49.8(0.0, ***)                                                            | 37.6(0.0, ***) | -49.1(0.0, ***)  | 0.4(0.8, )       | 21.7(0.0, ***) | -42.3(0.0, ***)  | 17.7(0.0, ***)  | 12.1(0.0, ***) | -7.7(0.0, ***)  |
| 8q      | 0.5(0.96, )                                                                | 50.4(0.0, ***) | -26.2(0.0, ***)  | -12.7(0.01, **)  | 28.2(0.0, ***) | -26.2(0.01, ***) | 25.7(0.02, **)  | 14.0(0.0, ***) | 23.2(0.0, ***)  |
| 12q     | 3.6(0.85, )                                                                | 63.3(0.0, ***) | -18.6(0.1, *)    | -14.5(0.11, )    | 26.9(0.0, ***) | -27.7(0.07, *)   | 21.8(0.32, )    | 35.5(0.03, **) | 34.9(0.0, ***)  |
| H6-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                |                 |
| 1q      | -6.4(0.6, )                                                                | 19.3(0.01, **) | 2.3(0.86, )      | 5.6(0.55, )      | -18.4(0.33, )  | -68.9(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q      | -19.4(0.0, ***)                                                            | 20.4(0.0, ***) | 8.4(0.01, ***)   | 2.1(0.06, *)     | 7.5(0.04, **)  | -59.0(0.0, ***)  | 2.8(0.26, )     | 4.2(0.04, **)  | -36.4(0.0, ***) |
| 4q      | -31.9(0.0, ***)                                                            | 37.6(0.0, ***) | 2.5(0.06, *)     | -0.2(0.88, )     | 22.1(0.0, ***) | -35.0(0.0, ***)  | 17.4(0.0, ***)  | 12.9(0.0, ***) | -10.1(0.0, ***) |
| 8q      | 18.1(0.09, *)                                                              | 48.8(0.0, ***) | 19.8(0.0, ***)   | -11.5(0.02, **)  | 28.2(0.0, ***) | -26.0(0.01, ***) | 25.9(0.02, **)  | 14.4(0.0, ***) | 24.7(0.0, ***)  |
| 12q     | 21.3(0.24, )                                                               | 62.9(0.0, ***) | 32.0(0.0, ***)   | -13.6(0.11, )    | 26.3(0.0, ***) | -27.5(0.07, *)   | 21.4(0.32, )    | 28.5(0.06, *)  | 34.4(0.0, ***)  |
| H7-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                |                 |
| 1q      | -7.0(0.57, )                                                               | 19.3(0.01, **) | -36.8(0.06, *)   | 3.4(0.73, )      | -19.3(0.31, )  | -65.9(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q      | -12.9(0.0, ***)                                                            | 21.4(0.0, ***) | -19.5(0.0, ***)  | 3.5(0.02, **)    | 7.0(0.06, *)   | -58.8(0.0, ***)  | 3.8(0.12, )     | 3.3(0.06, *)   | -35.6(0.0, ***) |
| 4q      | -21.1(0.0, ***)                                                            | 38.3(0.0, ***) | -5.8(0.17, )     | -2.8(0.11, )     | 20.3(0.0, ***) | -35.2(0.0, ***)  | 17.8(0.0, ***)  | 13.0(0.0, ***) | -9.5(0.0, ***)  |
| 8q      | 18.2(0.04, **)                                                             | 48.1(0.0, ***) | 6.0(0.01, **)    | -12.5(0.02, **)  | 29.6(0.0, ***) | -23.0(0.02, **)  | 26.4(0.02, **)  | 11.3(0.01, **) | 24.8(0.0, ***)  |
| 12q     | 19.5(0.22, )                                                               | 64.4(0.0, ***) | 18.9(0.04, **)   | -14.5(0.1, )     | 27.2(0.0, ***) | -25.4(0.1, )     | 21.5(0.32, )    | 33.9(0.03, **) | 33.9(0.0, ***)  |

*Note:* The table depicts the RMSFE of the forecast period 2010Q1 to 2019Q4. The VARX(1) is estimated starting in 1997Q1 and is re-estimated each quarter. The ABM with ARX expectations is calibrated to 40 different reference quarters from 2010Q1 to 2019Q4 and the ABM results are obtained as an average of 100 Monte Carlo simulations. We report the p-values of the Diebold-Mariano tests (Harvey et al., 1997). We test the null hypothesis whether the tested models have the same accuracy as the VARX(1) model. We denote 10 percent, 5 percent and 1 percent significance by \*, \*\*, and \*\*\* respectively.

## *E.2 VAR expectations*

Table 18: Heuristic ABM forecast performance under VAR expectations, relative to AR(1)

|        | GDP                                                                      | Inflation       | Household consumption | Investment       | Euribor         |
|--------|--------------------------------------------------------------------------|-----------------|-----------------------|------------------|-----------------|
| AR(1)  | <i>RMSFE for different forecast horizons</i>                             |                 |                       |                  |                 |
| 1q     | 0.41                                                                     | 0.57            | 0.56                  | 3.3              | 0.03            |
| 2q     | 0.67                                                                     | 0.57            | 0.82                  | 4.48             | 0.06            |
| 4q     | 1.32                                                                     | 0.56            | 1.47                  | 6.21             | 0.11            |
| 8q     | 2.59                                                                     | 0.58            | 2.72                  | 9.61             | 0.16            |
| 12q    | 3.47                                                                     | 0.6             | 3.68                  | 13.27            | 0.2             |
| ABM    | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -3.8(0.69, )                                                             | -3.7(0.34, )    | 4.2(0.49, )           | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -10.2(0.0, ***)                                                          | 0.6(0.0, ***)   | 0.6(0.68, )           | 0.9(0.24, )      | -39.6(0.0, ***) |
| 4q     | -27.8(0.0, ***)                                                          | 3.4(0.0, ***)   | -1.5(0.75, )          | -5.6(0.0, ***)   | -19.0(0.0, ***) |
| 8q     | 4.1(0.66, )                                                              | -1.0(0.01, ***) | 5.8(0.37, )           | 2.6(0.0, ***)    | -8.7(0.13, )    |
| 12q    | 6.7(0.57, )                                                              | 1.2(0.5, )      | 6.6(0.32, )           | 6.4(0.02, **)    | -2.5(0.74, )    |
| H1-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -2.1(0.82, )                                                             | -3.7(0.34, )    | -35.2(0.01, ***)      | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -22.4(0.0, ***)                                                          | 1.3(0.0, ***)   | -60.9(0.0, ***)       | 0.2(0.81, )      | -40.6(0.0, ***) |
| 4q     | -60.7(0.0, ***)                                                          | 4.7(0.0, ***)   | -92.0(0.0, ***)       | -9.8(0.0, ***)   | -17.5(0.0, ***) |
| 8q     | -26.3(0.05, *)                                                           | -2.4(0.08, *)   | -181.4(0.0, ***)      | -1.4(0.51, )     | -8.0(0.18, )    |
| 12q    | -91.8(0.0, ***)                                                          | 3.6(0.12, )     | -274.0(0.0, ***)      | -13.9(0.01, **)  | -0.9(0.91, )    |
| H2-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -2.1(0.82, )                                                             | -3.7(0.34, )    | -38.2(0.01, ***)      | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -21.6(0.0, ***)                                                          | 3.4(0.0, ***)   | -64.2(0.0, ***)       | 0.2(0.78, )      | -38.5(0.0, ***) |
| 4q     | -61.2(0.0, ***)                                                          | 4.1(0.0, ***)   | -96.1(0.0, ***)       | -9.5(0.0, ***)   | -18.1(0.0, ***) |
| 8q     | -26.6(0.05, *)                                                           | 0.2(0.58, )     | -201.8(0.0, ***)      | -1.7(0.44, )     | -9.0(0.13, )    |
| 12q    | -93.3(0.0, ***)                                                          | 6.7(0.05, **)   | -287.8(0.0, ***)      | -14.4(0.01, ***) | -1.3(0.86, )    |
| H3-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -2.8(0.76, )                                                             | -3.7(0.34, )    | 1.2(0.89, )           | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -6.7(0.03, **)                                                           | 0.4(0.15, )     | 8.3(0.0, ***)         | 1.6(0.02, **)    | -39.6(0.0, ***) |
| 4q     | -29.0(0.0, ***)                                                          | 3.9(0.0, ***)   | -5.7(0.26, )          | -5.2(0.0, ***)   | -16.9(0.0, ***) |
| 8q     | 10.0(0.36, )                                                             | 0.3(0.58, )     | -18.8(0.01, ***)      | 6.9(0.0, ***)    | -10.2(0.08, *)  |
| 12q    | 3.0(0.87, )                                                              | 4.4(0.0, ***)   | -17.8(0.23, )         | 6.0(0.0, ***)    | -2.9(0.69, )    |
| H4-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -2.7(0.77, )                                                             | -3.7(0.34, )    | 1.7(0.85, )           | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -7.4(0.01, **)                                                           | 1.1(0.01, ***)  | 7.2(0.0, ***)         | 1.1(0.13, )      | -39.6(0.0, ***) |
| 4q     | -25.4(0.0, ***)                                                          | 4.4(0.0, ***)   | -6.8(0.18, )          | -4.4(0.0, ***)   | -16.9(0.0, ***) |
| 8q     | 11.3(0.29, )                                                             | 3.3(0.1, )      | -25.1(0.0, ***)       | 7.3(0.0, ***)    | -8.4(0.16, )    |
| 12q    | 5.9(0.74, )                                                              | 2.4(0.2, )      | -24.5(0.09, *)        | 5.9(0.0, ***)    | -3.7(0.62, )    |
| H5-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -3.0(0.75, )                                                             | -3.7(0.34, )    | -3.8(0.65, )          | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -16.0(0.0, ***)                                                          | 1.6(0.0, ***)   | -11.4(0.0, ***)       | 0.5(0.53, )      | -38.9(0.0, ***) |
| 4q     | -43.5(0.0, ***)                                                          | 4.9(0.0, ***)   | -39.8(0.0, ***)       | -7.2(0.0, ***)   | -17.7(0.0, ***) |
| 8q     | -1.0(0.93, )                                                             | -1.5(0.35, )    | -22.6(0.08, *)        | 2.5(0.03, **)    | -6.7(0.29, )    |
| 12q    | -3.0(0.86, )                                                             | -0.3(0.87, )    | -23.0(0.3, )          | 2.9(0.0, ***)    | -1.0(0.9, )     |
| H6-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -3.6(0.7, )                                                              | -3.7(0.34, )    | -4.6(0.58, )          | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -5.1(0.05, *)                                                            | 2.1(0.0, ***)   | 7.8(0.0, ***)         | 1.7(0.01, **)    | -40.0(0.0, ***) |
| 4q     | -22.0(0.0, ***)                                                          | 2.1(0.0, ***)   | 12.3(0.0, ***)        | -4.7(0.0, ***)   | -18.6(0.0, ***) |
| 8q     | 14.3(0.15, )                                                             | 0.6(0.31, )     | 21.9(0.02, **)        | 5.3(0.0, ***)    | -9.2(0.12, )    |
| 12q    | 17.8(0.26, )                                                             | 4.4(0.0, ***)   | 32.9(0.05, *)         | 9.0(0.0, ***)    | -2.7(0.7, )     |
| H7-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                  |                 |
| 1q     | -4.2(0.65, )                                                             | -3.7(0.34, )    | -45.1(0.0, ***)       | 8.8(0.26, )      | -80.9(0.0, ***) |
| 2q     | -2.6(0.26, )                                                             | 1.8(0.0, ***)   | -21.4(0.0, ***)       | 1.7(0.01, **)    | -42.3(0.0, ***) |
| 4q     | -11.3(0.05, **)                                                          | 3.8(0.0, ***)   | 0.6(0.58, )           | -3.7(0.0, ***)   | -18.4(0.0, ***) |
| 8q     | 14.8(0.04, **)                                                           | 2.3(0.0, ***)   | 4.4(0.3, )            | 6.0(0.01, ***)   | -8.7(0.14, )    |
| 12q    | 10.2(0.4, )                                                              | 2.3(0.01, **)   | 9.8(0.32, )           | 6.7(0.01, ***)   | -2.4(0.76, )    |

Note: ABMs under VAR expectations vs. AR(1). The AR(1) is estimated from 1997Q1 and re-estimated each quarter. ABM results average 100 Monte Carlo simulations per reference quarter (2010Q1–2019Q4). Diebold–Mariano  $p$ -values in parentheses; \*, \*\*, \*\*\*: 10%, 5%, 1% significance.

Table 19: Heuristic ABM forecast performance under VAR expectations, relative to VAR(1)

|        | GDP                                                                       | Inflation       | Household cons.  | Gov. cons.      | Exports        | Imports         | GDP EA          | Infl. EA       | Euribor         |
|--------|---------------------------------------------------------------------------|-----------------|------------------|-----------------|----------------|-----------------|-----------------|----------------|-----------------|
| VAR(1) | <i>RMSFE for different forecast horizons</i>                              |                 |                  |                 |                |                 |                 |                |                 |
| 1q     | 0.39                                                                      | 0.75            | 0.6              | 0.57            | 1.26           | 1.38            | 0.36            | 0.15           | 0.04            |
| 2q     | 0.7                                                                       | 0.7             | 0.84             | 0.94            | 2.61           | 2.35            | 0.76            | 0.16           | 0.07            |
| 4q     | 1.52                                                                      | 0.85            | 1.4              | 1.66            | 5.42           | 4.85            | 1.69            | 0.18           | 0.16            |
| 8q     | 3.45                                                                      | 1.11            | 2.76             | 3.04            | 9.88           | 8.85            | 3.71            | 0.2            | 0.37            |
| 12q    | 4.71                                                                      | 1.44            | 3.94             | 4.38            | 13.08          | 11.45           | 4.95            | 0.28           | 0.5             |
| ABM    | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                 |                |                 |                 |                |                 |
| 1q     | -9.1(0.55, )                                                              | 20.6(0.01, ***) | 9.5(0.3, )       | 10.1(0.26, )    | -17.8(0.35, )  | -72.2(0.0, ***) | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -5.1(0.04, **)                                                            | 18.5(0.0, ***)  | 2.7(0.41, )      | 5.9(0.0, ***)   | 14.5(0.0, ***) | -56.5(0.0, ***) | 8.0(0.0, ***)   | 5.4(0.0, ***)  | -22.1(0.0, ***) |
| 4q     | -11.5(0.02, **)                                                           | 35.7(0.0, ***)  | -6.7(0.0, ***)   | 0.7(0.66, )     | 33.8(0.0, ***) | -22.3(0.0, ***) | 28.1(0.0, ***)  | 11.9(0.0, ***) | 17.3(0.0, ***)  |
| 8q     | 28.2(0.01, ***)                                                           | 46.9(0.0, ***)  | 7.2(0.45, )      | -15.6(0.0, ***) | 40.0(0.0, ***) | -9.4(0.28, )    | 36.8(0.0, ***)  | 21.0(0.0, ***) | 53.0(0.0, ***)  |
| 12q    | 31.3(0.06, *)                                                             | 59.0(0.0, ***)  | 12.7(0.14, )     | -17.1(0.02, **) | 33.4(0.0, ***) | -19.0(0.16, )   | 29.0(0.15, )    | 39.3(0.0, ***) | 58.1(0.0, ***)  |
| H1-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                 |                |                 |                 |                |                 |
| 1q     | -7.3(0.62, )                                                              | 20.6(0.01, ***) | -27.7(0.03, **)  | 2.6(0.76, )     | -16.2(0.39, )  | -75.4(0.0, ***) | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -16.7(0.0, ***)                                                           | 19.1(0.0, ***)  | -57.4(0.0, ***)  | 1.9(0.1, )      | 15.4(0.0, ***) | -62.2(0.0, ***) | 9.0(0.0, ***)   | 6.7(0.0, ***)  | -23.0(0.0, ***) |
| 4q     | -40.1(0.0, ***)                                                           | 36.6(0.0, ***)  | -101.8(0.0, ***) | -0.8(0.6, )     | 33.9(0.0, ***) | -26.1(0.0, ***) | 29.4(0.0, ***)  | 12.2(0.0, ***) | 18.3(0.0, ***)  |
| 8q     | 5.4(0.67, )                                                               | 46.2(0.0, ***)  | -177.3(0.0, ***) | -16.0(0.0, ***) | 40.8(0.0, ***) | -17.9(0.04, **) | 36.7(0.0, ***)  | 17.3(0.0, ***) | 53.4(0.0, ***)  |
| 12q    | -41.4(0.07, *)                                                            | 60.0(0.0, ***)  | -249.6(0.0, ***) | -10.5(0.05, *)  | 34.0(0.0, ***) | -35.5(0.01, **) | 28.4(0.16, )    | 37.2(0.0, ***) | 58.7(0.0, ***)  |
| H2-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                 |                |                 |                 |                |                 |
| 1q     | -7.2(0.62, )                                                              | 20.6(0.01, ***) | -30.5(0.02, **)  | 2.9(0.72, )     | -16.1(0.39, )  | -75.6(0.0, ***) | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -16.0(0.0, ***)                                                           | 20.8(0.0, ***)  | -60.6(0.0, ***)  | 1.2(0.34, )     | 16.7(0.0, ***) | -58.9(0.0, ***) | 9.8(0.0, ***)   | 7.1(0.0, ***)  | -21.2(0.0, ***) |
| 4q     | -40.6(0.0, ***)                                                           | 36.2(0.0, ***)  | -106.1(0.0, ***) | -2.1(0.23, )    | 34.6(0.0, ***) | -24.2(0.0, ***) | 29.6(0.0, ***)  | 16.8(0.0, ***) | 17.9(0.0, ***)  |
| 8q     | 5.2(0.69, )                                                               | 47.6(0.0, ***)  | -197.3(0.0, ***) | -18.3(0.0, ***) | 42.9(0.0, ***) | -16.8(0.06, *)  | 37.0(0.0, ***)  | 16.1(0.0, ***) | 52.9(0.0, ***)  |
| 12q    | -42.4(0.07, *)                                                            | 61.3(0.0, ***)  | -262.5(0.0, ***) | -12.7(0.03, **) | 36.4(0.0, ***) | -34.0(0.01, **) | 29.0(0.15, )    | 38.4(0.0, ***) | 58.6(0.0, ***)  |
| H3-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                 |                |                 |                 |                |                 |
| 1q     | -8.0(0.59, )                                                              | 20.6(0.01, ***) | 6.7(0.56, )      | 5.3(0.59, )     | -18.5(0.33, )  | -69.2(0.0, ***) | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -1.7(0.53, )                                                              | 18.3(0.0, ***)  | 10.3(0.0, ***)   | 5.2(0.0, ***)   | 13.7(0.0, ***) | -52.9(0.0, ***) | 8.0(0.0, ***)   | -0.1(0.96, )   | -22.1(0.0, ***) |
| 4q     | -12.5(0.01, ***)                                                          | 36.0(0.0, ***)  | -11.1(0.0, ***)  | 1.2(0.46, )     | 33.5(0.0, ***) | -19.2(0.0, ***) | 29.0(0.0, ***)  | 15.8(0.0, ***) | 18.8(0.0, ***)  |
| 8q     | 32.6(0.01, ***)                                                           | 47.6(0.0, ***)  | -17.1(0.0, ***)  | -13.8(0.0, ***) | 41.4(0.0, ***) | -8.4(0.3, )     | 36.5(0.0, ***)  | 23.3(0.0, ***) | 52.4(0.0, ***)  |
| 12q    | 28.6(0.16, )                                                              | 60.3(0.0, ***)  | -10.1(0.3, )     | -14.5(0.03, **) | 34.6(0.0, ***) | -21.0(0.11, )   | 28.6(0.15, )    | 38.4(0.0, ***) | 57.9(0.0, ***)  |
| H4-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                 |                |                 |                 |                |                 |
| 1q     | -7.9(0.59, )                                                              | 20.6(0.01, ***) | 7.1(0.53, )      | 2.4(0.8, )      | -18.4(0.33, )  | -69.4(0.0, ***) | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -2.4(0.36, )                                                              | 19.0(0.0, ***)  | 9.2(0.0, ***)    | 0.2(0.88, )     | 13.8(0.0, ***) | -50.7(0.0, ***) | 10.5(0.0, ***)  | -1.5(0.43, )   | -22.2(0.0, ***) |
| 4q     | -9.4(0.05, **)                                                            | 36.3(0.0, ***)  | -12.2(0.0, ***)  | -4.1(0.03, **)  | 34.0(0.0, ***) | -19.3(0.0, ***) | 31.9(0.0, ***)  | 12.6(0.0, ***) | 18.8(0.0, ***)  |
| 8q     | 33.5(0.0, ***)                                                            | 49.2(0.0, ***)  | -23.2(0.0, ***)  | -15.8(0.0, ***) | 41.2(0.0, ***) | -9.4(0.25, )    | 37.7(0.0, ***)  | 20.1(0.0, ***) | 53.2(0.0, ***)  |
| 12q    | 30.7(0.13, )                                                              | 59.4(0.0, ***)  | -16.4(0.09, *)   | -16.1(0.03, **) | 35.2(0.0, ***) | -21.3(0.09, *)  | 29.7(0.14, )    | 37.8(0.0, ***) | 57.6(0.0, ***)  |
| H5-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                 |                |                 |                 |                |                 |
| 1q     | -8.2(0.59, )                                                              | 20.6(0.01, ***) | 2.0(0.84, )      | 5.7(0.51, )     | -17.0(0.37, )  | -72.9(0.0, ***) | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -10.6(0.0, ***)                                                           | 19.4(0.0, ***)  | -9.0(0.0, ***)   | 1.5(0.26, )     | 13.6(0.0, ***) | -58.2(0.0, ***) | 11.0(0.0, ***)  | 3.0(0.06, *)   | -21.5(0.0, ***) |
| 4q     | -25.2(0.0, ***)                                                           | 36.7(0.0, ***)  | -46.9(0.0, ***)  | -1.7(0.39, )    | 33.4(0.0, ***) | -24.8(0.0, ***) | 30.9(0.0, ***)  | 14.0(0.0, ***) | 18.2(0.0, ***)  |
| 8q     | 24.4(0.04, **)                                                            | 46.7(0.0, ***)  | -20.7(0.02, **)  | -17.1(0.0, ***) | 40.3(0.0, ***) | -9.9(0.24, )    | 38.2(0.0, ***)  | 21.5(0.0, ***) | 53.9(0.0, ***)  |
| 12q    | 24.1(0.21, )                                                              | 58.3(0.0, ***)  | -15.0(0.35, )    | -17.1(0.02, **) | 34.2(0.0, ***) | -21.4(0.09, *)  | 29.9(0.14, )    | 36.2(0.02, **) | 58.7(0.0, ***)  |
| H6-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                 |                |                 |                 |                |                 |
| 1q     | -8.8(0.56, )                                                              | 20.6(0.01, ***) | 1.2(0.93, )      | 6.8(0.48, )     | -18.6(0.33, )  | -69.2(0.0, ***) | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -0.3(0.91, )                                                              | 19.8(0.0, ***)  | 9.8(0.0, ***)    | 3.8(0.0, ***)   | 12.9(0.0, ***) | -50.0(0.0, ***) | 8.5(0.0, ***)   | 6.4(0.0, ***)  | -22.5(0.0, ***) |
| 4q     | -6.4(0.16, )                                                              | 34.8(0.0, ***)  | 7.8(0.0, ***)    | -0.7(0.68, )    | 33.1(0.0, ***) | -18.5(0.0, ***) | 30.5(0.0, ***)  | 14.4(0.0, ***) | 17.6(0.0, ***)  |
| 8q     | 35.8(0.0, ***)                                                            | 47.8(0.0, ***)  | 23.0(0.0, ***)   | -16.8(0.0, ***) | 40.7(0.0, ***) | -8.1(0.34, )    | 38.1(0.0, ***)  | 19.9(0.0, ***) | 52.9(0.0, ***)  |
| 12q    | 39.5(0.04, **)                                                            | 60.3(0.0, ***)  | 37.2(0.0, ***)   | -15.5(0.02, **) | 34.6(0.0, ***) | -18.4(0.16, )   | 29.5(0.15, )    | 37.2(0.0, ***) | 58.0(0.0, ***)  |
| H7-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                  |                 |                |                 |                 |                |                 |
| 1q     | -9.5(0.53, )                                                              | 20.6(0.01, ***) | -37.1(0.06, *)   | 6.6(0.5, )      | -19.6(0.3, )   | -66.5(0.0, ***) | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | 2.1(0.34, )                                                               | 19.5(0.0, ***)  | -18.7(0.0, ***)  | 4.9(0.0, ***)   | 13.9(0.0, ***) | -53.6(0.0, ***) | 7.6(0.0, ***)   | 1.4(0.48, )    | -24.5(0.0, ***) |
| 4q     | 2.9(0.49, )                                                               | 35.9(0.0, ***)  | -4.4(0.27, )     | -1.6(0.28, )    | 32.7(0.0, ***) | -19.2(0.0, ***) | 30.3(0.0, ***)  | 13.4(0.0, ***) | 17.7(0.0, ***)  |
| 8q     | 36.2(0.0, ***)                                                            | 47.7(0.0, ***)  | 5.8(0.56, )      | -16.4(0.0, ***) | 41.0(0.0, ***) | -10.1(0.24, )   | 36.8(0.0, ***)  | 22.9(0.0, ***) | 53.1(0.0, ***)  |
| 12q    | 33.8(0.04, **)                                                            | 59.4(0.0, ***)  | 15.7(0.0, ***)   | -16.9(0.02, **) | 34.5(0.0, ***) | -20.4(0.12, )   | 29.3(0.15, )    | 39.0(0.0, ***) | 58.1(0.0, ***)  |

Note: ABMs under VAR expectations vs. VAR(1). The VAR(1) is estimated from 1997Q1 and re-estimated each quarter. ABM results average 100 Monte Carlo simulations per reference quarter (2010Q1–2019Q4). Diebold–Mariano  $p$ -values in parentheses; \*, \*\*, \*\*\*: 10%, 5%, 1% significance.

### *E.3 VARX expectations*

Table 20: Heuristic ABM forecast performance under VARX expectations, relative to ARX(1)

|        | GDP                                                                       | Inflation      | Household consumption | Investment      | Euribor         |
|--------|---------------------------------------------------------------------------|----------------|-----------------------|-----------------|-----------------|
| ARX(1) | <i>RMSFE for different forecast horizons</i>                              |                |                       |                 |                 |
| 1q     | 0.38                                                                      | 0.61           | 0.59                  | 3.46            | 0.04            |
| 2q     | 0.67                                                                      | 0.58           | 0.86                  | 4.54            | 0.06            |
| 4q     | 1.29                                                                      | 0.56           | 1.33                  | 6.09            | 0.1             |
| 8q     | 2.6                                                                       | 0.58           | 2.44                  | 9.67            | 0.16            |
| 12q    | 3.51                                                                      | 0.6            | 3.27                  | 13.41           | 0.21            |
| ABM    | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -7.7(0.49, )                                                              | 1.4(0.78, )    | 9.3(0.25, )           | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -20.1(0.0, ***)                                                           | 2.8(0.0, ***)  | 4.8(0.04, **)         | 1.1(0.11, )     | -38.0(0.0, ***) |
| 4q     | -41.7(0.0, ***)                                                           | 3.9(0.0, ***)  | -20.7(0.0, ***)       | -8.0(0.0, ***)  | -26.8(0.0, ***) |
| 8q     | 1.4(0.88, )                                                               | 1.3(0.0, ***)  | -7.3(0.09, *)         | 3.1(0.0, ***)   | -8.1(0.04, **)  |
| 12q    | 2.3(0.87, )                                                               | -3.8(0.0, ***) | -6.1(0.31, )          | 5.3(0.07, *)    | 1.6(0.73, )     |
| H1-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -6.4(0.54, )                                                              | 1.4(0.78, )    | -29.4(0.01, ***)      | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -31.3(0.0, ***)                                                           | 4.1(0.0, ***)  | -56.2(0.0, ***)       | 0.3(0.62, )     | -38.9(0.0, ***) |
| 4q     | -64.1(0.0, ***)                                                           | 5.1(0.0, ***)  | -113.6(0.0, ***)      | -11.6(0.0, ***) | -25.7(0.0, ***) |
| 8q     | -25.8(0.05, **)                                                           | -0.2(0.32, )   | -216.4(0.0, ***)      | -0.9(0.65, )    | -8.7(0.02, **)  |
| 12q    | -86.0(0.0, ***)                                                           | 4.5(0.25, )    | -321.6(0.0, ***)      | -12.7(0.02, **) | 1.6(0.74, )     |
| H2-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -6.4(0.54, )                                                              | 1.4(0.78, )    | -32.1(0.01, ***)      | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -31.4(0.0, ***)                                                           | 4.0(0.0, ***)  | -59.3(0.0, ***)       | 0.2(0.73, )     | -40.0(0.0, ***) |
| 4q     | -66.9(0.0, ***)                                                           | 5.1(0.0, ***)  | -118.1(0.0, ***)      | -12.1(0.0, ***) | -27.5(0.0, ***) |
| 8q     | -28.6(0.03, **)                                                           | -0.1(0.77, )   | -239.2(0.0, ***)      | -1.2(0.54, )    | -8.0(0.02, **)  |
| 12q    | -90.1(0.0, ***)                                                           | 6.4(0.06, *)   | -337.6(0.0, ***)      | -13.1(0.02, **) | 2.8(0.57, )     |
| H3-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -6.4(0.56, )                                                              | 1.4(0.78, )    | 5.6(0.65, )           | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -14.0(0.0, ***)                                                           | 2.5(0.0, ***)  | 10.8(0.0, ***)        | 1.8(0.01, ***)  | -39.8(0.0, ***) |
| 4q     | -34.4(0.0, ***)                                                           | 4.4(0.0, ***)  | -15.2(0.0, ***)       | -7.5(0.0, ***)  | -27.1(0.0, ***) |
| 8q     | 4.5(0.67, )                                                               | -1.5(0.0, ***) | -33.3(0.0, ***)       | 5.4(0.0, ***)   | -6.3(0.09, *)   |
| 12q    | -2.0(0.91, )                                                              | 3.4(0.0, ***)  | -37.1(0.0, ***)       | 5.8(0.0, ***)   | 2.4(0.6, )      |
| H4-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -6.3(0.57, )                                                              | 1.4(0.78, )    | 5.9(0.63, )           | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -18.3(0.0, ***)                                                           | 1.9(0.0, ***)  | 10.7(0.0, ***)        | 1.4(0.05, **)   | -38.4(0.0, ***) |
| 4q     | -38.4(0.0, ***)                                                           | 4.6(0.0, ***)  | -19.3(0.0, ***)       | -7.5(0.0, ***)  | -25.0(0.0, ***) |
| 8q     | 5.1(0.63, )                                                               | 3.6(0.0, ***)  | -40.8(0.0, ***)       | 6.0(0.0, ***)   | -5.0(0.2, )     |
| 12q    | -0.8(0.96, )                                                              | 1.7(0.41, )    | -45.7(0.0, ***)       | 6.6(0.0, ***)   | 2.9(0.56, )     |
| H5-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -7.1(0.51, )                                                              | 1.4(0.78, )    | 0.3(0.97, )           | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -28.1(0.0, ***)                                                           | 2.6(0.0, ***)  | -10.2(0.0, ***)       | 0.6(0.35, )     | -40.5(0.0, ***) |
| 4q     | -53.3(0.0, ***)                                                           | 4.8(0.0, ***)  | -56.5(0.0, ***)       | -9.6(0.0, ***)  | -26.4(0.0, ***) |
| 8q     | -9.1(0.43, )                                                              | 0.5(0.24, )    | -41.8(0.0, ***)       | 0.9(0.48, )     | -6.3(0.07, *)   |
| 12q    | -13.3(0.48, )                                                             | -1.4(0.01, **) | -47.1(0.03, **)       | 1.5(0.38, )     | 3.8(0.44, )     |
| H6-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -7.3(0.52, )                                                              | 1.4(0.78, )    | 1.6(0.9, )            | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -16.8(0.0, ***)                                                           | 4.2(0.0, ***)  | 11.6(0.0, ***)        | 1.3(0.06, *)    | -39.8(0.0, ***) |
| 4q     | -31.4(0.0, ***)                                                           | 5.2(0.0, ***)  | -0.2(0.89, )          | -7.3(0.0, ***)  | -23.7(0.0, ***) |
| 8q     | 6.1(0.55, )                                                               | 1.3(0.31, )    | 9.0(0.19, )           | 4.6(0.0, ***)   | -7.9(0.03, **)  |
| 12q    | 7.0(0.68, )                                                               | -3.3(0.06, *)  | 16.1(0.25, )          | 6.7(0.0, ***)   | 1.5(0.76, )     |
| H7-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                 |                 |
| 1q     | -7.9(0.49, )                                                              | 1.4(0.78, )    | -37.8(0.06, *)        | 12.2(0.18, )    | -60.5(0.0, ***) |
| 2q     | -12.9(0.0, ***)                                                           | 1.2(0.0, ***)  | -15.7(0.0, ***)       | 1.4(0.03, **)   | -36.8(0.0, ***) |
| 4q     | -24.2(0.0, ***)                                                           | 3.6(0.0, ***)  | -9.0(0.01, **)        | -6.5(0.0, ***)  | -25.5(0.0, ***) |
| 8q     | 4.0(0.61, )                                                               | -4.0(0.0, ***) | -4.0(0.0, ***)        | 4.3(0.16, )     | -7.7(0.04, **)  |
| 12q    | 3.6(0.79, )                                                               | 1.0(0.0, ***)  | -0.3(0.97, )          | 5.0(0.05, *)    | 1.5(0.75, )     |

Note: ABMs under VARX expectations vs. ARX(1). The ARX(1) is estimated from 1997Q1 and re-estimated each quarter. ABM results average 100 Monte Carlo simulations per reference quarter (2010Q1–2019Q4). Diebold–Mariano  $p$ -values in parentheses; \*, \*\*, \*\*\*: 10%, 5%, 1% significance.

Table 21: Heuristic ABM forecast performance under VARX expectations, relative to VARX(1)

|         | GDP                                                                        | Inflation      | Household cons.  | Gov. cons.       | Exports        | Imports          | GDP EA          | Infl. EA        | Euribor         |
|---------|----------------------------------------------------------------------------|----------------|------------------|------------------|----------------|------------------|-----------------|-----------------|-----------------|
| VARX(1) | <i>RMSFE for different forecast horizons</i>                               |                |                  |                  |                |                  |                 |                 |                 |
| 1q      | 0.39                                                                       | 0.75           | 0.6              | 0.57             | 1.26           | 1.38             | 0.36            | 0.15            | 0.04            |
| 2q      | 0.65                                                                       | 0.71           | 0.83             | 0.94             | 2.44           | 2.23             | 0.71            | 0.16            | 0.07            |
| 4q      | 1.3                                                                        | 0.87           | 1.39             | 1.68             | 4.64           | 4.23             | 1.46            | 0.18            | 0.12            |
| 8q      | 2.87                                                                       | 1.17           | 2.76             | 3.11             | 8.34           | 7.66             | 3.19            | 0.19            | 0.23            |
| 12q     | 4.04                                                                       | 1.64           | 4.07             | 4.37             | 11.91          | 10.74            | 4.5             | 0.25            | 0.31            |
| ABM     | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                 |                 |
| 1q      | -6.9(0.57, )                                                               | 19.3(0.01, **) | 9.9(0.26, )      | 3.9(0.68, )      | -17.6(0.35, )  | -75.0(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )     | -63.3(0.0, ***) |
| 2q      | -22.5(0.0, ***)                                                            | 21.3(0.0, ***) | 1.9(0.53, )      | 1.9(0.11, )      | 6.4(0.08, *)   | -64.8(0.0, ***)  | 4.4(0.07, *)    | 10.0(0.0, ***)  | -34.7(0.0, ***) |
| 4q      | -40.0(0.0, ***)                                                            | 37.6(0.0, ***) | -15.4(0.0, ***)  | -0.7(0.64, )     | 19.6(0.0, ***) | -42.0(0.0, ***)  | 16.8(0.0, ***)  | 9.6(0.0, ***)   | -11.7(0.0, ***) |
| 8q      | 11.0(0.3, )                                                                | 50.7(0.0, ***) | 5.5(0.5, )       | -12.1(0.01, ***) | 27.4(0.0, ***) | -25.5(0.01, ***) | 26.8(0.02, **)  | 16.0(0.0, ***)  | 23.4(0.0, ***)  |
| 12q     | 15.3(0.35, )                                                               | 62.0(0.0, ***) | 14.7(0.11, )     | -14.8(0.08, *)   | 25.9(0.0, ***) | -24.1(0.11, )    | 22.1(0.3, )     | 35.9(0.03, **)  | 33.5(0.0, ***)  |
| H1-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                 |                 |
| 1q      | -5.6(0.64, )                                                               | 19.3(0.01, **) | -28.4(0.02, **)  | 4.0(0.66, )      | -15.9(0.4, )   | -75.2(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )     | -63.3(0.0, ***) |
| 2q      | -34.0(0.0, ***)                                                            | 22.3(0.0, ***) | -61.0(0.0, ***)  | 0.6(0.63, )      | 9.2(0.01, **)  | -69.5(0.0, ***)  | 3.1(0.2, )      | 4.7(0.01, ***)  | -35.5(0.0, ***) |
| 4q      | -62.1(0.0, ***)                                                            | 38.4(0.0, ***) | -104.3(0.0, ***) | -1.0(0.49, )     | 22.9(0.0, ***) | -43.9(0.0, ***)  | 17.6(0.0, ***)  | 15.7(0.0, ***)  | -10.7(0.0, ***) |
| 8q      | -13.6(0.3, )                                                               | 49.9(0.0, ***) | -178.7(0.0, ***) | -12.2(0.01, **)  | 32.0(0.0, ***) | -34.1(0.0, ***)  | 26.6(0.02, **)  | 12.2(0.01, ***) | 23.0(0.0, ***)  |
| 12q     | -61.3(0.01, **)                                                            | 65.0(0.0, ***) | -239.0(0.0, ***) | -12.2(0.13, )    | 29.7(0.0, ***) | -42.1(0.01, ***) | 21.9(0.31, )    | 29.8(0.05, *)   | 33.4(0.0, ***)  |
| H2-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                 |                 |
| 1q      | -5.6(0.64, )                                                               | 19.3(0.01, **) | -31.1(0.02, **)  | 7.7(0.4, )       | -15.9(0.4, )   | -75.1(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )     | -63.3(0.0, ***) |
| 2q      | -34.1(0.0, ***)                                                            | 22.3(0.0, ***) | -64.2(0.0, ***)  | 5.6(0.0, ***)    | 9.5(0.01, ***) | -69.3(0.0, ***)  | 0.7(0.79, )     | 2.5(0.19, )     | -36.7(0.0, ***) |
| 4q      | -64.8(0.0, ***)                                                            | 38.4(0.0, ***) | -108.6(0.0, ***) | 0.5(0.76, )      | 24.8(0.0, ***) | -42.6(0.0, ***)  | 17.8(0.0, ***)  | 16.2(0.0, ***)  | -12.3(0.0, ***) |
| 8q      | -16.2(0.22, )                                                              | 50.0(0.0, ***) | -198.8(0.0, ***) | -13.2(0.01, **)  | 32.1(0.0, ***) | -36.4(0.0, ***)  | 26.3(0.02, **)  | 14.9(0.0, ***)  | 23.4(0.0, ***)  |
| 12q     | -64.9(0.01, ***)                                                           | 65.7(0.0, ***) | -251.9(0.0, ***) | -11.4(0.16, )    | 29.5(0.0, ***) | -43.1(0.01, ***) | 22.5(0.3, )     | 31.7(0.05, **)  | 34.2(0.0, ***)  |
| H3-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                 |                 |
| 1q      | -5.5(0.65, )                                                               | 19.3(0.01, **) | 6.2(0.58, )      | 3.6(0.71, )      | -18.2(0.34, )  | -68.5(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )     | -63.3(0.0, ***) |
| 2q      | -16.3(0.0, ***)                                                            | 21.1(0.0, ***) | 8.1(0.01, ***)   | 3.0(0.0, ***)    | 9.6(0.01, ***) | -59.5(0.0, ***)  | 2.1(0.42, )     | 6.7(0.0, ***)   | -36.4(0.0, ***) |
| 4q      | -32.8(0.0, ***)                                                            | 37.9(0.0, ***) | -10.2(0.01, ***) | -2.1(0.15, )     | 22.7(0.0, ***) | -36.7(0.0, ***)  | 17.1(0.0, ***)  | 12.9(0.0, ***)  | -12.0(0.0, ***) |
| 8q      | 13.7(0.22, )                                                               | 49.3(0.0, ***) | -17.4(0.0, ***)  | -10.0(0.03, **)  | 28.5(0.0, ***) | -30.0(0.0, ***)  | 26.3(0.02, **)  | 15.1(0.0, ***)  | 24.7(0.0, ***)  |
| 12q     | 11.5(0.55, )                                                               | 64.6(0.0, ***) | -10.3(0.01, ***) | -12.1(0.14, )    | 27.1(0.0, ***) | -27.9(0.06, *)   | 21.8(0.31, )    | 31.0(0.04, **)  | 34.0(0.0, ***)  |
| H4-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                 |                 |
| 1q      | -5.5(0.65, )                                                               | 19.3(0.01, **) | 6.6(0.56, )      | -0.9(0.93, )     | -18.2(0.34, )  | -68.8(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )     | -63.3(0.0, ***) |
| 2q      | -20.7(0.0, ***)                                                            | 20.5(0.0, ***) | 7.9(0.01, ***)   | -2.3(0.07, *)    | 9.1(0.01, **)  | -55.6(0.0, ***)  | 3.2(0.18, )     | 7.4(0.0, ***)   | -35.1(0.0, ***) |
| 4q      | -36.7(0.0, ***)                                                            | 38.1(0.0, ***) | -14.1(0.0, ***)  | -3.3(0.03, **)   | 23.4(0.0, ***) | -33.5(0.0, ***)  | 20.5(0.0, ***)  | 11.0(0.0, ***)  | -10.1(0.0, ***) |
| 8q      | 14.3(0.21, )                                                               | 51.8(0.0, ***) | -24.1(0.0, ***)  | -11.7(0.01, **)  | 29.6(0.0, ***) | -26.6(0.01, ***) | 27.5(0.02, **)  | 13.8(0.0, ***)  | 25.5(0.0, ***)  |
| 12q     | 12.6(0.52, )                                                               | 64.0(0.0, ***) | -17.2(0.0, ***)  | -12.3(0.13, )    | 28.3(0.0, ***) | -27.1(0.07, *)   | 23.1(0.29, )    | 33.2(0.03, **)  | 34.3(0.0, ***)  |
| H5-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                 |                 |
| 1q      | -6.2(0.61, )                                                               | 19.3(0.01, **) | 1.0(0.92, )      | 4.2(0.64, )      | -16.8(0.38, )  | -73.3(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )     | -63.3(0.0, ***) |
| 2q      | -30.7(0.0, ***)                                                            | 21.2(0.0, ***) | -13.6(0.0, ***)  | 2.7(0.01, **)    | 6.7(0.07, *)   | -68.3(0.0, ***)  | 0.4(0.88, )     | 5.6(0.0, ***)   | -37.2(0.0, ***) |
| 4q      | -51.5(0.0, ***)                                                            | 38.2(0.0, ***) | -49.7(0.0, ***)  | -1.2(0.44, )     | 21.7(0.0, ***) | -42.2(0.0, ***)  | 17.0(0.0, ***)  | 13.7(0.0, ***)  | -11.4(0.0, ***) |
| 8q      | 1.5(0.9, )                                                                 | 50.3(0.0, ***) | -24.9(0.0, ***)  | -12.8(0.01, ***) | 27.0(0.0, ***) | -29.3(0.0, ***)  | 25.1(0.03, **)  | 13.6(0.0, ***)  | 24.6(0.0, ***)  |
| 12q     | 1.7(0.93, )                                                                | 62.8(0.0, ***) | -18.3(0.11, )    | -15.3(0.09, *)   | 25.8(0.0, ***) | -28.7(0.06, *)   | 21.4(0.33, )    | 32.1(0.04, **)  | 34.9(0.0, ***)  |
| H6-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                 |                 |
| 1q      | -6.4(0.6, )                                                                | 19.3(0.01, **) | 2.3(0.85, )      | 4.3(0.66, )      | -18.4(0.33, )  | -68.2(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )     | -63.3(0.0, ***) |
| 2q      | -19.2(0.0, ***)                                                            | 22.4(0.0, ***) | 8.9(0.0, ***)    | -0.6(0.6, )      | 7.9(0.03, **)  | -61.5(0.0, ***)  | 4.7(0.03, **)   | 5.5(0.0, ***)   | -36.4(0.0, ***) |
| 4q      | -29.8(0.0, ***)                                                            | 38.4(0.0, ***) | 4.1(0.0, ***)    | -1.2(0.48, )     | 21.1(0.0, ***) | -36.4(0.0, ***)  | 18.7(0.0, ***)  | 16.2(0.0, ***)  | -9.0(0.0, ***)  |
| 8q      | 15.2(0.17, )                                                               | 50.7(0.0, ***) | 19.8(0.0, ***)   | -12.7(0.01, **)  | 27.4(0.0, ***) | -26.8(0.01, ***) | 26.1(0.02, **)  | 14.9(0.0, ***)  | 23.5(0.0, ***)  |
| 12q     | 19.3(0.3, )                                                                | 62.1(0.0, ***) | 32.6(0.0, ***)   | -14.8(0.09, *)   | 26.2(0.0, ***) | -25.0(0.1, *)    | 22.6(0.3, )     | 31.8(0.04, **)  | 33.3(0.0, ***)  |
| H7-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                  |                  |                |                  |                 |                 |                 |
| 1q      | -7.0(0.57, )                                                               | 19.3(0.01, **) | -36.8(0.06, *)   | 1.5(0.89, )      | -19.3(0.31, )  | -66.3(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )     | -63.3(0.0, ***) |
| 2q      | -15.2(0.0, ***)                                                            | 20.0(0.0, ***) | -19.3(0.0, ***)  | 1.8(0.12, )      | 5.2(0.16, )    | -61.5(0.0, ***)  | 5.1(0.02, **)   | 6.6(0.0, ***)   | -33.5(0.0, ***) |
| 4q      | -22.7(0.0, ***)                                                            | 37.4(0.0, ***) | -4.3(0.31, )     | -1.0(0.5, )      | 20.8(0.0, ***) | -31.6(0.0, ***)  | 18.0(0.0, ***)  | 14.2(0.0, ***)  | -10.6(0.0, ***) |
| 8q      | 13.3(0.14, )                                                               | 48.0(0.0, ***) | 8.4(0.0, ***)    | -12.4(0.01, ***) | 29.2(0.0, ***) | -20.9(0.03, **)  | 27.0(0.02, **)  | 12.5(0.0, ***)  | 23.6(0.0, ***)  |
| 12q     | 16.4(0.31, )                                                               | 63.7(0.0, ***) | 19.3(0.03, **)   | -14.9(0.1, *)    | 27.8(0.0, ***) | -23.0(0.12, )    | 22.9(0.29, )    | 32.1(0.04, **)  | 33.4(0.0, ***)  |

Note: ABMs under VARX expectations vs. VARX(1). The VARX(1) is estimated from 1997Q1 and re-estimated each quarter. ABM results average 100 Monte Carlo simulations per reference quarter (2010Q1–2019Q4). Diebold–Mariano  $p$ -values in parentheses; \*, \*\*, \*\*\*: 10%, 5%, 1% significance.

## F Forecast Results under ARX, VAR, and VARX Expectations

This appendix reports the per-mechanism heatmaps and horizon-by-horizon RMSFE tables for the OC-ABM, the base ABM, and the best-performing heuristic under ARX, VAR, and VARX expectations. Section 5 summarises the cross-mechanism pattern via Figure 11 and Section 5.2; the AR-expectations results are presented in detail in Section 5.1.

### F.1 ARX expectations

ARX expectations augment the AR specification with exogenous predictors (Euro area variables and the Euribor rate), providing the ABM with additional information about the external environment. The ARX(1) benchmark is correspondingly harder to beat because it, too, conditions on these predictors.

Figure 24 displays the 12-quarter RMSFE relative to the ARX(1) benchmark. The OC-ABM retains a statistically significant consumption advantage of 10.7% ( $p < 0.01$ ) and an investment improvement of 8.7% ( $p < 0.01$ ), but the margin over the dynamic heuristics narrows under ARX expectations: the exogenous predictors allow the ARX(1) to capture some of the medium-term dynamics that the AR(1) missed, reducing the scope for structural improvement. The habit rule H6 produces a +15.4% consumption point estimate (not significant), the buffer-stock H7 a  $-0.9\%$  point estimate (not significant), and the static rules H1–H5 show consumption losses ranging from  $-35.6\%$  to  $-338\%$ .

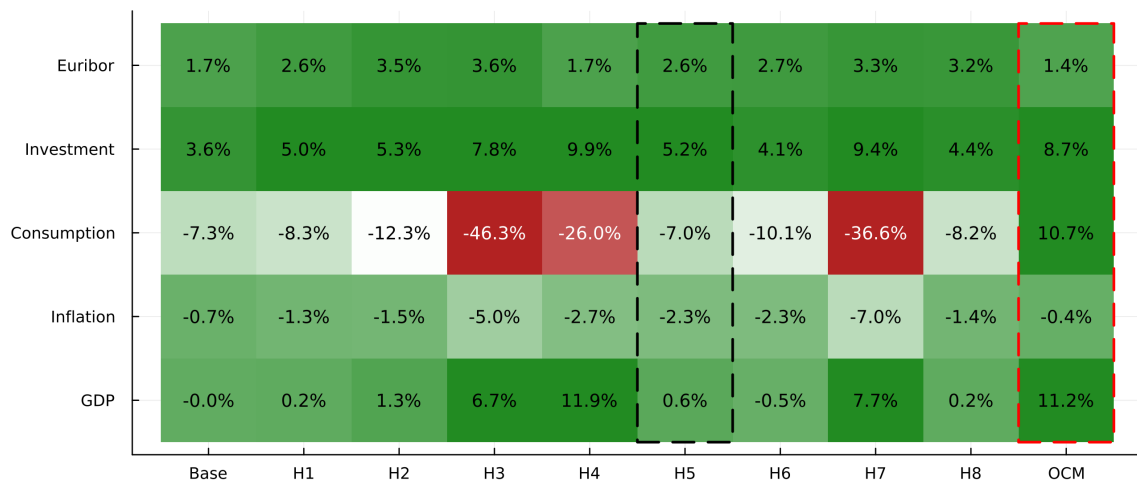


Figure 24: Twelve-quarter relative RMSFE of heuristic ABMs (H1–H7) and the OC-ABM under ARX expectations, relative to the ARX(1) benchmark. Evaluation period: 2010Q1–2019Q4, 100 Monte Carlo simulations.

Table 22 benchmarks against the VARX(1). The familiar horizon-dependent pattern holds. At 12 quarters, the OC-ABM achieves 23.0% for GDP, 63.2% for inflation ( $p < 0.01$ ), and 28.2% for household consumption ( $p < 0.01$ ). Government consumption remains difficult for all ABM variants, with losses around 16% at 12 quarters, a pattern that persists across all expectation mechanisms and likely reflects the exogeneity of fiscal policy in the model.

Table 22: Forecast performance under ARX expectations, relative to VARX(1)

|         | GDP                                                                        | Inflation      | Household cons. | Gov. cons.       | Exports        | Imports          | GDP EA          | Infl. EA       | Euribor         |
|---------|----------------------------------------------------------------------------|----------------|-----------------|------------------|----------------|------------------|-----------------|----------------|-----------------|
| VARX(1) | <i>RMSFE for different forecast horizons</i>                               |                |                 |                  |                |                  |                 |                |                 |
| 1q      | 0.39                                                                       | 0.75           | 0.6             | 0.57             | 1.26           | 1.38             | 0.36            | 0.15           | 0.04            |
| 2q      | 0.65                                                                       | 0.71           | 0.83            | 0.94             | 2.44           | 2.23             | 0.71            | 0.16           | 0.07            |
| 4q      | 1.3                                                                        | 0.87           | 1.39            | 1.68             | 4.64           | 4.23             | 1.46            | 0.18           | 0.12            |
| 8q      | 2.87                                                                       | 1.17           | 2.76            | 3.11             | 8.34           | 7.66             | 3.19            | 0.19           | 0.23            |
| 12q     | 4.04                                                                       | 1.64           | 4.07            | 4.37             | 11.91          | 10.74            | 4.5             | 0.25           | 0.31            |
| ABM     | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                 |                  |                |                  |                 |                |                 |
| 1q      | -6.9(0.57, )                                                               | 19.3(0.01, **) | 9.7(0.27, )     | 2.6(0.79, )      | -17.6(0.35, )  | -68.9(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q      | -24.3(0.0, ***)                                                            | 20.6(0.0, ***) | 1.3(0.66, )     | 0.5(0.67, )      | 6.7(0.07, *)   | -61.2(0.0, ***)  | 2.3(0.37, )     | 1.8(0.4, )     | -38.9(0.0, ***) |
| 4q      | -36.2(0.0, ***)                                                            | 38.5(0.0, ***) | -15.3(0.0, ***) | -4.0(0.01, ***)  | 22.1(0.0, ***) | -40.1(0.0, ***)  | 18.1(0.0, ***)  | 11.4(0.0, ***) | -13.1(0.0, ***) |
| 8q      | 11.6(0.25, )                                                               | 47.6(0.0, ***) | 4.8(0.58, )     | -14.1(0.01, ***) | 29.8(0.0, ***) | -23.9(0.02, **)  | 26.0(0.02, **)  | 15.1(0.0, ***) | 23.2(0.0, ***)  |
| 12q     | 14.6(0.35, )                                                               | 62.6(0.0, ***) | 13.4(0.16, )    | -15.5(0.09, *)   | 27.9(0.0, ***) | -22.4(0.15, )    | 22.0(0.31, )    | 30.9(0.04, **) | 33.4(0.0, ***)  |
| H6-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                 |                  |                |                  |                 |                |                 |
| 1q      | -6.4(0.6, )                                                                | 19.3(0.01, **) | 2.3(0.86, )     | 5.6(0.55, )      | -18.4(0.33, )  | -68.9(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q      | -19.4(0.0, ***)                                                            | 20.4(0.0, ***) | 8.4(0.01, ***)  | 2.1(0.06, *)     | 7.5(0.04, **)  | -59.0(0.0, ***)  | 2.8(0.26, )     | 4.2(0.04, **)  | -36.4(0.0, ***) |
| 4q      | -31.9(0.0, ***)                                                            | 37.6(0.0, ***) | 2.5(0.06, *)    | -0.2(0.88, )     | 22.1(0.0, ***) | -35.0(0.0, ***)  | 17.4(0.0, ***)  | 12.9(0.0, ***) | -10.1(0.0, ***) |
| 8q      | 18.1(0.09, *)                                                              | 48.8(0.0, ***) | 19.8(0.0, ***)  | -11.5(0.02, **)  | 28.2(0.0, ***) | -26.0(0.01, ***) | 25.9(0.02, **)  | 14.4(0.0, ***) | 24.7(0.0, ***)  |
| 12q     | 21.3(0.24, )                                                               | 62.9(0.0, ***) | 32.0(0.0, ***)  | -13.6(0.11, )    | 26.3(0.0, ***) | -27.5(0.07, *)   | 21.4(0.32, )    | 28.5(0.06, *)  | 34.4(0.0, ***)  |
| OC-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                 |                  |                |                  |                 |                |                 |
| 1q      | -4.8(0.68, )                                                               | 19.3(0.01, **) | -51.4(0.0, ***) | 3.7(0.7, )       | -15.4(0.42, )  | -75.6(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q      | -37.7(0.0, ***)                                                            | 21.0(0.0, ***) | -22.9(0.0, ***) | -0.4(0.76, )     | 9.5(0.01, ***) | -64.1(0.0, ***)  | 2.1(0.4, )      | 3.7(0.04, **)  | -37.1(0.0, ***) |
| 4q      | -41.4(0.0, ***)                                                            | 39.1(0.0, ***) | -3.6(0.2, )     | -3.0(0.08, *)    | 20.6(0.0, ***) | -33.8(0.0, ***)  | 18.1(0.0, ***)  | 11.8(0.0, ***) | -11.9(0.0, ***) |
| 8q      | 17.9(0.05, **)                                                             | 47.8(0.0, ***) | 13.9(0.0, ***)  | -15.2(0.0, ***)  | 28.8(0.0, ***) | -19.5(0.05, **)  | 27.0(0.02, **)  | 11.9(0.02, **) | 23.6(0.0, ***)  |
| 12q     | 23.0(0.14, )                                                               | 63.2(0.0, ***) | 28.2(0.0, ***)  | -16.2(0.09, *)   | 27.6(0.0, ***) | -19.1(0.19, )    | 23.1(0.28, )    | 30.4(0.05, **) | 33.3(0.0, ***)  |

ABMs under ARX expectations vs. VARX(1). See Section 5 for estimation details. Diebold–Mariano  $p$ -values in parentheses; \*, \*\*, \*\*\*: 10%, 5%, 1% significance.

Table 23 isolates the OC-ABM's contribution relative to the univariate ARX(1). The household consumption gain of 10.7% ( $p < 0.01$ ) at 12 quarters is smaller than the 21.8% under AR expectations, consistent with the ARX(1) absorbing some predictive content through exogenous variables. The GDP improvement of 11.2% is not statistically significant ( $p = 0.37$ ), paralleling the AR result and reinforcing the finding that GDP gains, while consistent in sign, are imprecisely estimated. Investment shows a robust 8.7% improvement ( $p < 0.01$ ).

Table 23: Forecast performance under ARX expectations, relative to ARX(1)

|        | GDP                                                                       | Inflation      | Household consumption | Investment     | Euribor         |
|--------|---------------------------------------------------------------------------|----------------|-----------------------|----------------|-----------------|
| ARX(1) | <i>RMSFE for different forecast horizons</i>                              |                |                       |                |                 |
| 1q     | 0.38                                                                      | 0.61           | 0.59                  | 3.46           | 0.04            |
| 2q     | 0.67                                                                      | 0.58           | 0.86                  | 4.54           | 0.06            |
| 4q     | 1.29                                                                      | 0.56           | 1.33                  | 6.09           | 0.1             |
| 8q     | 2.6                                                                       | 0.58           | 2.44                  | 9.67           | 0.16            |
| 12q    | 3.51                                                                      | 0.6            | 3.27                  | 13.41          | 0.21            |
| ABM    | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                |                 |
| 1q     | -7.7(0.49, )                                                              | 1.4(0.78, )    | 9.1(0.27, )           | 12.2(0.18, )   | -60.5(0.0, ***) |
| 2q     | -21.8(0.0, ***)                                                           | 1.9(0.0, ***)  | 4.2(0.07, *)          | 0.2(0.73, )    | -42.4(0.0, ***) |
| 4q     | -37.9(0.0, ***)                                                           | 5.3(0.0, ***)  | -20.6(0.0, ***)       | -8.5(0.0, ***) | -28.4(0.0, ***) |
| 8q     | 2.2(0.81, )                                                               | -4.8(0.0, ***) | -8.1(0.02, **)        | 2.6(0.0, ***)  | -8.3(0.02, **)  |
| 12q    | 1.5(0.91, )                                                               | -2.0(0.0, ***) | -7.6(0.11, )          | 4.7(0.08, *)   | 1.6(0.73, )     |
| H6-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                |                 |
| 1q     | -7.3(0.52, )                                                              | 1.4(0.78, )    | 1.6(0.91, )           | 12.2(0.18, )   | -60.5(0.0, ***) |
| 2q     | -17.0(0.0, ***)                                                           | 1.7(0.0, ***)  | 11.1(0.0, ***)        | 1.4(0.04, **)  | -39.8(0.0, ***) |
| 4q     | -33.5(0.0, ***)                                                           | 4.0(0.0, ***)  | -1.9(0.31, )          | -7.7(0.0, ***) | -24.9(0.0, ***) |
| 8q     | 9.4(0.34, )                                                               | -2.5(0.0, ***) | 9.0(0.2, )            | 4.5(0.0, ***)  | -6.3(0.08, *)   |
| 12q    | 9.3(0.56, )                                                               | -1.1(0.0, ***) | 15.4(0.27, )          | 7.3(0.0, ***)  | 3.1(0.52, )     |
| OC-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                |                 |
| 1q     | -5.7(0.58, )                                                              | 1.4(0.78, )    | -52.5(0.0, ***)       | 12.2(0.18, )   | -60.5(0.0, ***) |
| 2q     | -34.9(0.0, ***)                                                           | 2.4(0.0, ***)  | -19.3(0.0, ***)       | -0.2(0.79, )   | -40.4(0.0, ***) |
| 4q     | -43.1(0.0, ***)                                                           | 6.3(0.0, ***)  | -8.3(0.0, ***)        | -8.5(0.0, ***) | -27.0(0.0, ***) |
| 8q     | 9.1(0.25, )                                                               | -4.4(0.0, ***) | 2.2(0.79, )           | 5.0(0.12, )    | -7.8(0.03, **)  |
| 12q    | 11.2(0.37, )                                                              | -0.4(0.67, )   | 10.7(0.0, ***)        | 8.7(0.0, ***)  | 1.4(0.78, )     |

ABMs under ARX expectations vs. ARX(1). See Section 5 for estimation details. Diebold–Mariano  $p$ -values in parentheses; \*, \*\*, \*\*\*: 10%, 5%, 1% significance.

## F.2 VAR expectations

Under VAR expectations, the ABM’s aggregate forecasting module exploits cross-variable correlations, the same information the VAR(1) benchmark uses.

Figure 25 displays the 12-quarter RMSFE under VAR expectations, measured relative to the AR(1) benchmark. The OC-ABM achieves 18.6% for household consumption ( $p < 0.01$ ), 15.2% for GDP, and 8.2% for investment ( $p < 0.01$ ). Among heuristic models, H6-ABM (habit on the income-plus-wealth specification) delivers the largest consumption point estimate at +32.9% (marginally significant at  $p = 0.05$ ) and the largest GDP point estimate at +17.8%, while H7-ABM (Deaton buffer-stock) yields a +9.8% consumption point estimate that is not statistically distinguishable from zero. Static rules (H1–H5) show consumption losses ranging from  $-10.1\%$  to  $-287.8\%$ . The OC-ABM is the only specification whose consumption gain is statistically significant under VAR expectations, reflecting both the structural shape constraints imposed by the Euler equation and the lower forecast variance relative to the noisier H6 estimates.

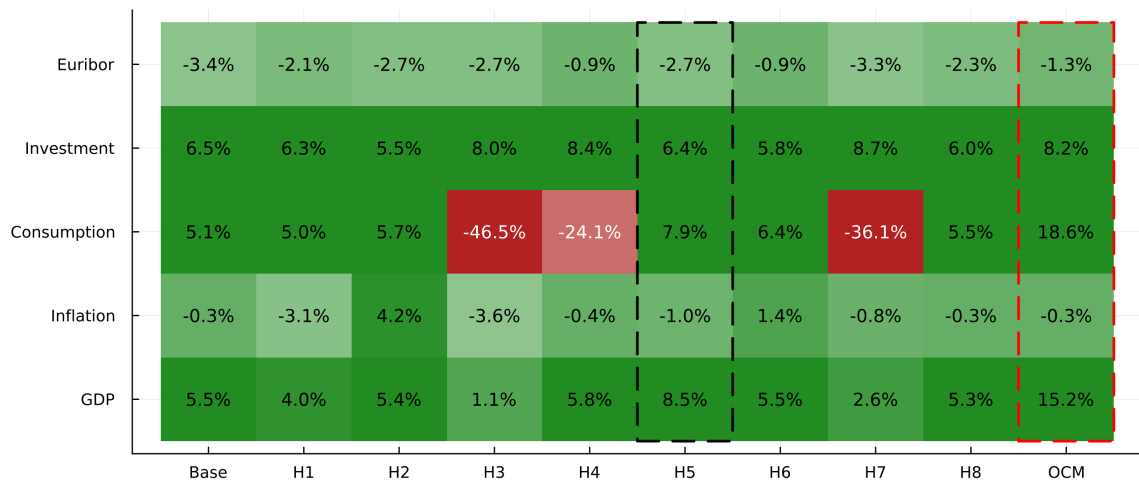


Figure 25: Twelve-quarter relative RMSFE of heuristic ABMs (H1–H7) and the OC-ABM under VAR expectations, relative to the AR(1) benchmark. Evaluation period: 2010Q1–2019Q4, 100 Monte Carlo simulations.

Table 24 provides horizon-by-horizon detail for the VAR(1) benchmark. At 12 quarters, the OC-ABM reaches 37.5% for GDP ( $p < 0.05$ ) and 23.9% for household consumption ( $p < 0.05$ ). The H6-ABM follows at 39.5% for GDP ( $p < 0.05$ ) and 37.2% for consumption ( $p < 0.01$ ), the largest point estimates among heuristics. The inflation gains of around 58% ( $p < 0.01$ ) are shared across all ABM variants, indicating they arise from the model’s structural inflation dynamics rather than from the household block.

Table 24: Forecast performance under VAR expectations, relative to VAR(1)

|        | GDP                                                                       | Inflation       | Household cons. | Gov. cons.      | Exports        | Imports         | GDP EA          | Infl. EA       | Euribor         |
|--------|---------------------------------------------------------------------------|-----------------|-----------------|-----------------|----------------|-----------------|-----------------|----------------|-----------------|
| VAR(1) | <i>RMSFE for different forecast horizons</i>                              |                 |                 |                 |                |                 |                 |                |                 |
| 1q     | 0.39                                                                      | 0.75            | 0.6             | 0.57            | 1.26           | 1.38            | 0.36            | 0.15           | 0.04            |
| 2q     | 0.7                                                                       | 0.7             | 0.84            | 0.94            | 2.61           | 2.35            | 0.76            | 0.16           | 0.07            |
| 4q     | 1.52                                                                      | 0.85            | 1.4             | 1.66            | 5.42           | 4.85            | 1.69            | 0.18           | 0.16            |
| 8q     | 3.45                                                                      | 1.11            | 2.76            | 3.04            | 9.88           | 8.85            | 3.71            | 0.2            | 0.37            |
| 12q    | 4.71                                                                      | 1.44            | 3.94            | 4.38            | 13.08          | 11.45           | 4.95            | 0.28           | 0.5             |
| ABM    | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                 |                 |                |                 |                 |                |                 |
| 1q     | -9.1(0.55, )                                                              | 20.6(0.01, ***) | 9.5(0.3, )      | 10.1(0.26, )    | -17.8(0.35, )  | -72.2(0.0, ***) | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -5.1(0.04, **)                                                            | 18.5(0.0, ***)  | 2.7(0.41, )     | 5.9(0.0, ***)   | 14.5(0.0, ***) | -56.5(0.0, ***) | 8.0(0.0, ***)   | 5.4(0.0, ***)  | -22.1(0.0, ***) |
| 4q     | -11.5(0.02, **)                                                           | 35.7(0.0, ***)  | -6.7(0.0, ***)  | 0.7(0.66, )     | 33.8(0.0, ***) | -22.3(0.0, ***) | 28.1(0.0, ***)  | 11.9(0.0, ***) | 17.3(0.0, ***)  |
| 8q     | 28.2(0.01, ***)                                                           | 46.9(0.0, ***)  | 7.2(0.45, )     | -15.6(0.0, ***) | 40.0(0.0, ***) | -9.4(0.28, )    | 36.8(0.0, ***)  | 21.0(0.0, ***) | 53.0(0.0, ***)  |
| 12q    | 31.3(0.06, *)                                                             | 59.0(0.0, ***)  | 12.7(0.14, )    | -17.1(0.02, **) | 33.4(0.0, ***) | -19.0(0.16, )   | 29.0(0.15, )    | 39.3(0.0, ***) | 58.1(0.0, ***)  |
| H6-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                 |                 |                |                 |                 |                |                 |
| 1q     | -8.8(0.56, )                                                              | 20.6(0.01, ***) | 1.2(0.93, )     | 6.8(0.48, )     | -18.6(0.33, )  | -69.2(0.0, ***) | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -0.3(0.91, )                                                              | 19.8(0.0, ***)  | 9.8(0.0, ***)   | 3.8(0.0, ***)   | 12.9(0.0, ***) | -50.0(0.0, ***) | 8.5(0.0, ***)   | 6.4(0.0, ***)  | -22.5(0.0, ***) |
| 4q     | -6.4(0.16, )                                                              | 34.8(0.0, ***)  | 7.8(0.0, ***)   | -0.7(0.68, )    | 33.1(0.0, ***) | -18.5(0.0, ***) | 30.5(0.0, ***)  | 14.4(0.0, ***) | 17.6(0.0, ***)  |
| 8q     | 35.8(0.0, ***)                                                            | 47.8(0.0, ***)  | 23.0(0.0, ***)  | -16.8(0.0, ***) | 40.7(0.0, ***) | -8.1(0.34, )    | 38.1(0.0, ***)  | 19.9(0.0, ***) | 52.9(0.0, ***)  |
| 12q    | 39.5(0.04, **)                                                            | 60.3(0.0, ***)  | 37.2(0.0, ***)  | -15.5(0.02, **) | 34.6(0.0, ***) | -18.4(0.16, )   | 29.5(0.15, )    | 37.2(0.0, ***) | 58.0(0.0, ***)  |
| OC-ABM | <i>Percentage improvements (+) or losses (-) relative to VAR(1) model</i> |                 |                 |                 |                |                 |                 |                |                 |
| 1q     | -6.3(0.67, )                                                              | 20.6(0.01, ***) | -51.5(0.0, ***) | 4.9(0.6, )      | -15.7(0.41, )  | -77.6(0.0, ***) | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q     | -14.9(0.0, ***)                                                           | 19.6(0.0, ***)  | -25.1(0.0, ***) | 1.5(0.24, )     | 15.6(0.0, ***) | -57.7(0.0, ***) | 10.1(0.0, ***)  | 0.8(0.69, )    | -22.9(0.0, ***) |
| 4q     | -21.9(0.0, ***)                                                           | 35.7(0.0, ***)  | -2.9(0.24, )    | -2.0(0.19, )    | 32.8(0.0, ***) | -17.7(0.0, ***) | 30.2(0.0, ***)  | 13.1(0.0, ***) | 18.5(0.0, ***)  |
| 8q     | 34.3(0.0, ***)                                                            | 44.7(0.0, ***)  | 12.8(0.2, )     | -17.2(0.0, ***) | 40.9(0.0, ***) | -5.4(0.51, )    | 37.4(0.0, ***)  | 21.3(0.0, ***) | 53.0(0.0, ***)  |
| 12q    | 37.5(0.02, **)                                                            | 58.3(0.0, ***)  | 23.9(0.01, **)  | -15.1(0.03, **) | 33.9(0.0, ***) | -17.4(0.18, )   | 29.7(0.14, )    | 36.8(0.0, ***) | 58.6(0.0, ***)  |

ABMs under VAR expectations vs. VAR(1). See Section 5 for estimation details. Diebold–Mariano  $p$ -values in parentheses; \*, \*\*, \*\*\*: 10%, 5%, 1% significance.

Table 25 provides the horizon-by-horizon detail underlying Figure 25. The OC-ABM's household consumption improvement of 18.6% ( $p < 0.01$ ) and GDP gain of 15.2% at 12 quarters confirm the pattern visible in the heatmap. H6-ABM's gains of 17.8% for GDP (not significant) and 32.9% for consumption (marginally significant at  $p = 0.05$ ) represent the best heuristic performance under VAR expectations.

Table 25: Forecast performance under VAR expectations, relative to AR(1)

|        | GDP                                                                      | Inflation       | Household consumption | Investment     | Euribor         |
|--------|--------------------------------------------------------------------------|-----------------|-----------------------|----------------|-----------------|
| AR(1)  | <i>RMSFE for different forecast horizons</i>                             |                 |                       |                |                 |
| 1q     | 0.41                                                                     | 0.57            | 0.56                  | 3.3            | 0.03            |
| 2q     | 0.67                                                                     | 0.57            | 0.82                  | 4.48           | 0.06            |
| 4q     | 1.32                                                                     | 0.56            | 1.47                  | 6.21           | 0.11            |
| 8q     | 2.59                                                                     | 0.58            | 2.72                  | 9.61           | 0.16            |
| 12q    | 3.47                                                                     | 0.6             | 3.68                  | 13.27          | 0.2             |
| ABM    | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                |                 |
| 1q     | -3.8(0.69, )                                                             | -3.7(0.34, )    | 4.2(0.49, )           | 8.8(0.26, )    | -80.9(0.0, ***) |
| 2q     | -10.2(0.0, ***)                                                          | 0.6(0.0, ***)   | 0.6(0.68, )           | 0.9(0.24, )    | -39.6(0.0, ***) |
| 4q     | -27.8(0.0, ***)                                                          | 3.4(0.0, ***)   | -1.5(0.75, )          | -5.6(0.0, ***) | -19.0(0.0, ***) |
| 8q     | 4.1(0.66, )                                                              | -1.0(0.01, ***) | 5.8(0.37, )           | 2.6(0.0, ***)  | -8.7(0.13, )    |
| 12q    | 6.7(0.57, )                                                              | 1.2(0.5, )      | 6.6(0.32, )           | 6.4(0.02, **)  | -2.5(0.74, )    |
| H6-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                |                 |
| 1q     | -3.6(0.7, )                                                              | -3.7(0.34, )    | -4.6(0.58, )          | 8.8(0.26, )    | -80.9(0.0, ***) |
| 2q     | -5.1(0.05, *)                                                            | 2.1(0.0, ***)   | 7.8(0.0, ***)         | 1.7(0.01, **)  | -40.0(0.0, ***) |
| 4q     | -22.0(0.0, ***)                                                          | 2.1(0.0, ***)   | 12.3(0.0, ***)        | -4.7(0.0, ***) | -18.6(0.0, ***) |
| 8q     | 14.3(0.15, )                                                             | 0.6(0.31, )     | 21.9(0.02, **)        | 5.3(0.0, ***)  | -9.2(0.12, )    |
| 12q    | 17.8(0.26, )                                                             | 4.4(0.0, ***)   | 32.9(0.05, *)         | 9.0(0.0, ***)  | -2.7(0.7, )     |
| OC-ABM | <i>Percentage improvements (+) or losses (-) relative to AR(1) model</i> |                 |                       |                |                 |
| 1q     | -1.2(0.9, )                                                              | -3.7(0.34, )    | -60.4(0.0, ***)       | 8.8(0.26, )    | -80.9(0.0, ***) |
| 2q     | -20.4(0.0, ***)                                                          | 1.9(0.0, ***)   | -27.9(0.0, ***)       | 0.1(0.86, )    | -40.5(0.0, ***) |
| 4q     | -39.8(0.0, ***)                                                          | 3.5(0.0, ***)   | 2.1(0.47, )           | -6.6(0.0, ***) | -17.3(0.0, ***) |
| 8q     | 12.3(0.09, *)                                                            | -5.2(0.0, ***)  | 11.5(0.02, **)        | 5.2(0.0, ***)  | -8.8(0.13, )    |
| 12q    | 15.2(0.19, )                                                             | -0.3(0.88, )    | 18.6(0.01, ***)       | 8.2(0.0, ***)  | -1.3(0.87, )    |

ABMs under VAR expectations vs. AR(1). See Section 5 for estimation details. Diebold–Mariano  $p$ -values in parentheses; \*, \*\*, \*\*\*: 10%, 5%, 1% significance.

### F.3 VARX expectations

VARX expectations combine multivariate dynamics with exogenous predictors, giving the ABM the richest information set. Under this specification, H6-ABM remains the best heuristic for consumption forecasting, consistent with its performance under the other three mechanisms.

Figure 26 shows the 12-quarter results relative to the ARX(1) benchmark. The OC-ABM improves GDP by 13.4%, household consumption by 9.9% ( $p < 0.01$ ), and investment by 9.9% ( $p < 0.01$ ). H6-ABM achieves a +16.1% consumption point estimate (not significant) and a +7.0% GDP point estimate. H7-ABM yields a  $-0.3\%$  consumption point estimate (not significant). Static heuristics (H1–H5) deliver consumption losses ranging from  $-37.1\%$  to  $-337.6\%$ . Only the OC-ABM produces a statistically significant consumption gain under this mechanism.

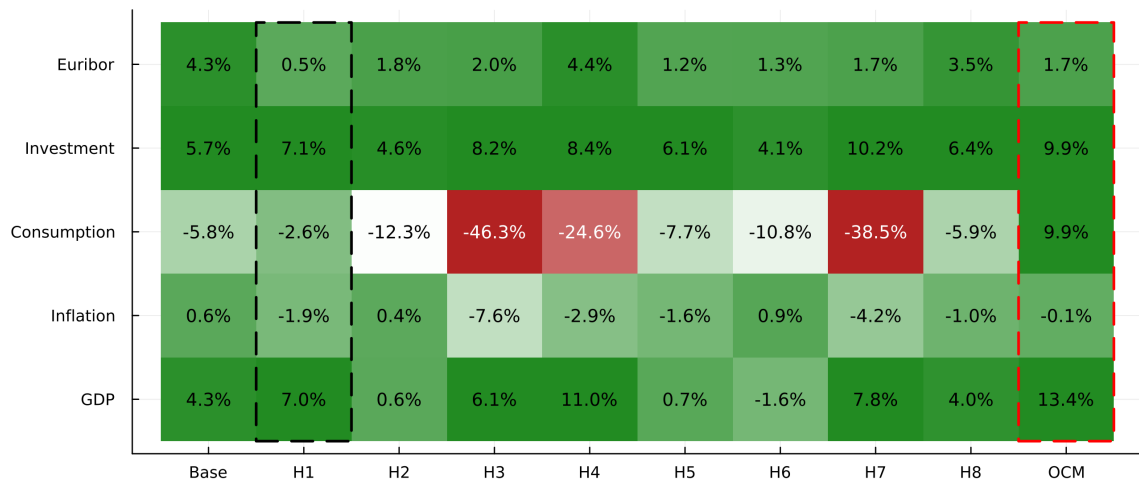


Figure 26: Twelve-quarter relative RMSFE of heuristic ABMs (H1–H7) and the OC-ABM under VARX expectations, relative to the ARX(1) benchmark. Evaluation period: 2010Q1–2019Q4, 100 Monte Carlo simulations.

Table 26 provides horizon-by-horizon detail for the VARX(1) benchmark. At 12 quarters, the OC-ABM delivers 24.9% for GDP, 63.3% for inflation ( $p < 0.01$ ), and 27.5% for household consumption ( $p < 0.01$ ). The inflation gains, like those under VAR expectations, are shared with the base ABM (63.6%), confirming they originate from the model’s price-setting module rather than the household block.

Table 26: Forecast performance under VARX expectations, relative to VARX(1)

|         | GDP                                                                        | Inflation      | Household cons. | Gov. cons.       | Exports         | Imports          | GDP EA          | Infl. EA       | Euribor         |
|---------|----------------------------------------------------------------------------|----------------|-----------------|------------------|-----------------|------------------|-----------------|----------------|-----------------|
| VARX(1) | <i>RMSFE for different forecast horizons</i>                               |                |                 |                  |                 |                  |                 |                |                 |
| 1q      | 0.39                                                                       | 0.75           | 0.6             | 0.57             | 1.26            | 1.38             | 0.36            | 0.15           | 0.04            |
| 2q      | 0.65                                                                       | 0.71           | 0.83            | 0.94             | 2.44            | 2.23             | 0.71            | 0.16           | 0.07            |
| 4q      | 1.3                                                                        | 0.87           | 1.39            | 1.68             | 4.64            | 4.23             | 1.46            | 0.18           | 0.12            |
| 8q      | 2.87                                                                       | 1.17           | 2.76            | 3.11             | 8.34            | 7.66             | 3.19            | 0.19           | 0.23            |
| 12q     | 4.04                                                                       | 1.64           | 4.07            | 4.37             | 11.91           | 10.74            | 4.5             | 0.25           | 0.31            |
| ABM     | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                 |                  |                 |                  |                 |                |                 |
| 1q      | -6.9(0.57, )                                                               | 19.3(0.01, **) | 9.9(0.26, )     | 3.9(0.68, )      | -17.6(0.35, )   | -75.0(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q      | -22.5(0.0, ***)                                                            | 21.3(0.0, ***) | 1.9(0.53, )     | 1.9(0.11, )      | 6.4(0.08, *)    | -64.8(0.0, ***)  | 4.4(0.07, *)    | 10.0(0.0, ***) | -34.7(0.0, ***) |
| 4q      | -40.0(0.0, ***)                                                            | 37.6(0.0, ***) | -15.4(0.0, ***) | -0.7(0.64, )     | 19.6(0.0, ***)  | -42.0(0.0, ***)  | 16.8(0.0, ***)  | 9.6(0.0, ***)  | -11.7(0.0, ***) |
| 8q      | 11.0(0.3, )                                                                | 50.7(0.0, ***) | 5.5(0.5, )      | -12.1(0.01, ***) | 27.4(0.0, ***)  | -25.5(0.01, ***) | 26.8(0.02, **)  | 16.0(0.0, ***) | 23.4(0.0, ***)  |
| 12q     | 15.3(0.35, )                                                               | 62.0(0.0, ***) | 14.7(0.11, )    | -14.8(0.08, *)   | 25.9(0.0, ***)  | -24.1(0.11, )    | 22.1(0.3, )     | 35.9(0.03, **) | 33.5(0.0, ***)  |
| H6-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                 |                  |                 |                  |                 |                |                 |
| 1q      | -6.4(0.6, )                                                                | 19.3(0.01, **) | 2.3(0.85, )     | 4.3(0.66, )      | -18.4(0.33, )   | -68.2(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q      | -19.2(0.0, ***)                                                            | 22.4(0.0, ***) | 8.9(0.0, ***)   | -0.6(0.6, )      | 7.9(0.03, **)   | -61.5(0.0, ***)  | 4.7(0.03, **)   | 5.5(0.0, ***)  | -36.4(0.0, ***) |
| 4q      | -29.8(0.0, ***)                                                            | 38.4(0.0, ***) | 4.1(0.0, ***)   | -1.2(0.48, )     | 21.1(0.0, ***)  | -36.4(0.0, ***)  | 18.7(0.0, ***)  | 16.2(0.0, ***) | -9.0(0.0, ***)  |
| 8q      | 15.2(0.17, )                                                               | 50.7(0.0, ***) | 19.8(0.0, ***)  | -12.7(0.01, **)  | 27.4(0.0, ***)  | -26.8(0.01, ***) | 26.1(0.02, **)  | 14.9(0.0, ***) | 23.5(0.0, ***)  |
| 12q     | 19.3(0.3, )                                                                | 62.1(0.0, ***) | 32.6(0.0, ***)  | -14.8(0.09, *)   | 26.2(0.0, ***)  | -25.0(0.1, *)    | 22.6(0.3, )     | 31.8(0.04, **) | 33.3(0.0, ***)  |
| OC-ABM  | <i>Percentage improvements (+) or losses (-) relative to VARX(1) model</i> |                |                 |                  |                 |                  |                 |                |                 |
| 1q      | -4.8(0.68, )                                                               | 19.3(0.01, **) | -51.2(0.0, ***) | 5.0(0.6, )       | -15.4(0.42, )   | -75.3(0.0, ***)  | -67.7(0.0, ***) | 4.7(0.71, )    | -63.3(0.0, ***) |
| 2q      | -38.2(0.0, ***)                                                            | 20.1(0.0, ***) | -25.3(0.0, ***) | 1.7(0.23, )      | 10.2(0.01, ***) | -62.2(0.0, ***)  | 3.1(0.21, )     | 9.5(0.0, ***)  | -37.6(0.0, ***) |
| 4q      | -42.7(0.0, ***)                                                            | 39.0(0.0, ***) | -4.0(0.16, )    | 0.3(0.78, )      | 23.2(0.0, ***)  | -34.5(0.0, ***)  | 17.7(0.0, ***)  | 17.7(0.0, ***) | -11.2(0.0, ***) |
| 8q      | 20.5(0.02, **)                                                             | 47.6(0.0, ***) | 13.7(0.0, ***)  | -13.3(0.01, ***) | 31.0(0.0, ***)  | -19.5(0.05, *)   | 27.1(0.02, **)  | 9.2(0.04, **)  | 24.2(0.0, ***)  |
| 12q     | 24.9(0.1, )                                                                | 63.3(0.0, ***) | 27.5(0.0, ***)  | -13.4(0.13, )    | 28.4(0.0, ***)  | -20.2(0.18, )    | 22.8(0.29, )    | 30.4(0.05, *)  | 33.5(0.0, ***)  |

ABMs under VARX expectations vs. VARX(1). See Section 5 for estimation details. Diebold–Mariano  $p$ -values in parentheses; \*, \*\*, \*\*\*: 10%, 5%, 1% significance.

Table 27 provides the horizon-by-horizon detail underlying Figure 26. The OC-ABM achieves 13.4% for GDP and 9.9% for household consumption ( $p < 0.01$ ) at 12 quarters, while the base ABM *deteriorates* by  $-5.8\%$  for consumption. H6-ABM achieves a  $+16.1\%$  consumption point estimate (not significant) and a  $+7.0\%$  GDP point estimate. The divergence between the OC-ABM and the base ABM is consistent with the microfounded household block contributing to the forecasting gain. Investment benefits across all variants, with the OC-ABM reaching  $9.9\%$  ( $p < 0.01$ ).

Table 27: Forecast performance under VARX expectations, relative to ARX(1)

|        | GDP                                                                       | Inflation      | Household consumption | Investment     | Euribor         |
|--------|---------------------------------------------------------------------------|----------------|-----------------------|----------------|-----------------|
| ARX(1) | <i>RMSFE for different forecast horizons</i>                              |                |                       |                |                 |
| 1q     | 0.38                                                                      | 0.61           | 0.59                  | 3.46           | 0.04            |
| 2q     | 0.67                                                                      | 0.58           | 0.86                  | 4.54           | 0.06            |
| 4q     | 1.29                                                                      | 0.56           | 1.33                  | 6.09           | 0.1             |
| 8q     | 2.6                                                                       | 0.58           | 2.44                  | 9.67           | 0.16            |
| 12q    | 3.51                                                                      | 0.6            | 3.27                  | 13.41          | 0.21            |
| ABM    | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                |                 |
| 1q     | -7.7(0.49, )                                                              | 1.4(0.78, )    | 9.3(0.25, )           | 12.2(0.18, )   | -60.5(0.0, ***) |
| 2q     | -20.1(0.0, ***)                                                           | 2.8(0.0, ***)  | 4.8(0.04, **)         | 1.1(0.11, )    | -38.0(0.0, ***) |
| 4q     | -41.7(0.0, ***)                                                           | 3.9(0.0, ***)  | -20.7(0.0, ***)       | -8.0(0.0, ***) | -26.8(0.0, ***) |
| 8q     | 1.4(0.88, )                                                               | 1.3(0.0, ***)  | -7.3(0.09, *)         | 3.1(0.0, ***)  | -8.1(0.04, **)  |
| 12q    | 2.3(0.87, )                                                               | -3.8(0.0, ***) | -6.1(0.31, )          | 5.3(0.07, *)   | 1.6(0.73, )     |
| H6-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                |                 |
| 1q     | -7.3(0.52, )                                                              | 1.4(0.78, )    | 1.6(0.9, )            | 12.2(0.18, )   | -60.5(0.0, ***) |
| 2q     | -16.8(0.0, ***)                                                           | 4.2(0.0, ***)  | 11.6(0.0, ***)        | 1.3(0.06, *)   | -39.8(0.0, ***) |
| 4q     | -31.4(0.0, ***)                                                           | 5.2(0.0, ***)  | -0.2(0.89, )          | -7.3(0.0, ***) | -23.7(0.0, ***) |
| 8q     | 6.1(0.55, )                                                               | 1.3(0.31, )    | 9.0(0.19, )           | 4.6(0.0, ***)  | -7.9(0.03, **)  |
| 12q    | 7.0(0.68, )                                                               | -3.3(0.06, *)  | 16.1(0.25, )          | 6.7(0.0, ***)  | 1.5(0.76, )     |
| OC-ABM | <i>Percentage improvements (+) or losses (-) relative to ARX(1) model</i> |                |                       |                |                 |
| 1q     | -5.7(0.58, )                                                              | 1.4(0.78, )    | -52.3(0.0, ***)       | 12.2(0.18, )   | -60.5(0.0, ***) |
| 2q     | -35.4(0.0, ***)                                                           | 1.3(0.0, ***)  | -21.6(0.0, ***)       | -0.3(0.68, )   | -41.0(0.0, ***) |
| 4q     | -44.4(0.0, ***)                                                           | 6.1(0.0, ***)  | -8.7(0.0, ***)        | -9.4(0.0, ***) | -26.2(0.0, ***) |
| 8q     | 12.0(0.12, )                                                              | -4.9(0.0, ***) | 2.1(0.8, )            | 5.9(0.06, *)   | -7.0(0.07, *)   |
| 12q    | 13.4(0.28, )                                                              | -0.1(0.69, )   | 9.9(0.0, ***)         | 9.9(0.0, ***)  | 1.7(0.73, )     |

ABMs under VARX expectations vs. ARX(1). See Section 5 for estimation details. Diebold–Mariano  $p$ -values in parentheses; \*, \*\*, \*\*\*: 10%, 5%, 1% significance.